

THE ORGANIZATION AND CONSEQUENCES OF SPECIES INTERACTIONS ACROSS
ECOLOGICAL SCALES

by

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ABSTRACT

Species interactions structure ecological communities, but predicting when interactions occur, how species are positioned within interaction networks, and how interactions scale up to community-level outcomes remain central challenges in ecology. The architecture of species interactions emerges from processes operating across multiple scales, from species-level traits to community structure to biogeographic contexts. In this dissertation, I examine how interspecific interactions are organized across these scales in mutualistic, antagonistic, and experimental systems. At the species level, I show that low-dimensional representations of species positions within avian frugivory networks can be used in concert with species traits and phylogenetic information to improve predictions of missing links. I also show that, for some taxa, both the geographic range size and the climatic niche breadth of a species are associated with its role in local and regional interaction webs. At broader scales, I use a global set of insular host-helminth interaction networks to show how island characteristics like isolation and area are associated with differences in species assemblages and network structure. Additionally, I review the current state of knowledge on how major classes of anthropogenic impacts affect the structural properties of plant-animal transport networks globally. Finally, I show how interspecific interactions scale to community-level outcomes under alternative environmental regimes in experimental microbial communities. By linking predictive models, network analyses, spatial approaches, synthesis, and experiments, this dissertation contributes to a multiscale view of how ecological interactions are organized across systems.

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CHAPTER 1

BRIEF INTRODUCTION

Species interactions structure ecological communities, but predicting when interactions occur and how interactions scale up to community-level outcomes remain central challenges in ecology. Natural systems are characterized by a diverse set of interacting organisms (Janzen, 1974); organizing these interaction sets into ecological networks can help us gain insight by linking pairwise interactions to community- and ecosystem-level processes (Lever et al., 2014; Rohr et al., 2014). In the framework of interspecific interaction networks, individual species represent nodes, while the interactions between species represent links. Network approaches have furthered our understanding of a variety of systems reliant on interspecific interactions, including consumer-resource, host-parasite, and mutualistic interactions, and network perspectives have yielded insights on features as specific as the dynamics of singular focal species all the way up to broad-scale patterns of community assembly and stability (Guimaraes Jr, 2020; Saravia et al., 2022).

One reason network approaches are useful is that they provide a set of descriptive and mathematical tools for characterizing the architecture of species interactions. At the level of individual species, node-level properties can describe the number of interactions a species participates in (degree), the total weight of those interactions (strength), or the position of a species relative to the rest of the network. Centrality, for example, describes how connected a species is to other members of a network and is often used to identify species whose loss may have disproportionate effects on associated communities (Maia et al., 2019; Martins et al., 2024; Traveset et al., 2017). Understanding the ecological and evolutionary processes that give rise to variation in node-level properties is therefore important for understanding how interaction networks are organized and how they may respond to disturbance (Martins et al., 2025; Moulatlet et al., 2023; Zhang et al., 2025).

In addition to node-level characterizations, network approaches can also summarize the overlying architecture of species interactions within a community. Network-level properties include summaries of how links are distributed among species, such as degree distributions and connectance, as well as properties describing the arrangement of links, such as modularity and nestedness (Bascompte et al., 2003; Bascompte et al., 2006). These structural properties are of interest because they can influence emergent features of ecological systems, including stability, resilience, robustness to species loss, and the propagation of perturbations through communities (Bascompte & Scheffer, 2022; Martins et al., 2024; Suweis et al., 2013; Thébault & Fontaine, 2010). As a result, understanding the factors that shape network structure can provide insight into both the assembly of ecological communities and the potential responses of those communities to environmental change.

The architecture of species interactions emerges from processes operating across multiple scales. Species-level traits (Rafferty & Ives, 2013; Ramos-Robles, Dáttilo, et al., 2018), phylogenetic relationships (Eriksson, 2016), geographic distributions (Moulatlet et al., 2023), environmental tolerances (Slatyer et al., 2013), local community composition (Saravia et al., 2022), biogeographic context (Dalsgaard et al., 2013; Nogales et al., 2024; Sugiura, 2010; Traveset et al., 2016), and anthropogenic impacts (Birkenbach et al., 2024; Cruz et al., 2023; Teixido et al., 2022) can all influence which interactions occur and how they are organized. However, the relationship between these factors and interaction structure is not always straightforward. Processes that shape species assemblages may not translate directly into the structure of interactions among those species, and the meaning of network properties may shift between local communities, regional metawebs, and broader macroecological contexts. In this dissertation, I examine how interspecific interactions are organized across these scales in mutualistic, antagonistic, and competitive systems.

Like any data collection process, the sampling of ecological networks is imperfect. Even intensive sampling may fail to capture rare interactions or, in some cases, rare interactors (Young et al., 2021). Because biased errors in network construction can alter inferred network structure, incomplete sampling creates challenges for understanding how interactions are organized (Blüthgen et al., 2008; Young et al., 2021). In chapter two, I address this problem by comparing random-forest regression models informed by combinations of phylogenetic structure, species traits, and latent network structure in their ability to fill in potentially missing interactions in a diverse network of fruiting plants and frugivorous birds in Brazil’s Atlantic forest. This chapter highlights the potential importance of latent structural features for predicting mutualistic interactions, and encourages a clear conceptual link between prediction performance metrics and the overall goal of predicting cryptic links.

Species’ roles within interaction networks may also depend on processes operating across spatial scales. Highly central species may have disproportionate effects on their associated communities (Maia et al., 2019; Martins et al., 2024; Traveset et al., 2017), but the factors contributing to species’ position within interaction networks may differ across biogeographic scales. Identifying the eco-evolutionary processes that give rise to highly central species in local and regional-scale interaction networks is essential for understanding how ecological networks are organized and how they may respond to disturbances (Martins et al., 2025; Moulatlet et al., 2023; Zhang et al., 2025). In chapter three, I combine a globally distributed dataset of floral visitation networks comprising 30,330 interactions among 6,857 species with large-scale occurrence data to evaluate how species’ geographic and environmental range sizes impact their role in networks across scales. These results suggest that the apparent impacts of geographic range size on species centrality in large-scale metawebs are primarily driven by interaction turnover and the scaling of interaction specificity from local

to global networks. When variation in interaction turnover is incorporated, niche breadth exerts the strongest positive effect on plant centrality within both local and global floral visitation networks.

The structure of species interaction networks is shaped by both biotic and abiotic contexts, and the increasing availability of ecological data has permitted examinations into how spatial and environmental gradients influence the structure of these species interaction networks (Tylianakis & Morris, 2017). In chapter four, I take a macroecological approach to investigate how bipartite interaction networks between free-living hosts and helminth parasites vary across gradients of island area and isolation. I find that relationships between island characteristics and the species richness of both hosts and helminth parasites are broadly consistent with predictions from classical island biogeography theory for free-living species. In contrast, network nestedness showed a more limited response, increasing weakly with island area while failing to respond to either measure of island connectivity. These results suggest that island characteristics shape the richness of host-parasite assemblages more predictably than the structure of interactions within them.

Global ecosystems are subject to anthropogenically driven changes at increasing rates, and as a result, so are the mutualistic interaction networks that occur within them. Structural changes in mutualistic networks can be driven by a diverse set of drivers, including shifting climates, urbanization and land use changes, and invasion of non-native species (Cruz et al., 2023; Teixido et al., 2022; Traveset & Richardson, 2014). In chapter five, I characterize the current state of understanding of how these drivers affect the structural properties of two major classes of plant-animal transport mutualisms: seed dispersal and pollination networks (Bronstein, 2015). I also examine how knowledge of these patterns can be used to better understand, predict, and conserve these critical natural systems.

Finally, in chapter six, I investigate how species-level performance and interspecific interactions scale up to community assembly outcomes. Single-species vital rates are one possible source of simplified demographic information that may be useful in predicting community outcomes (Nestor et al., 2023; Ram et al., 2019). Using experimental microbial communities, I test whether demographic parameters derived from monoculture growth experiments can predict multi-strain community outcomes across nutrient environments. Together, these results show that single-species growth parameters can provide useful predictions of community assembly, but that their predictive value depends on environmental contexts.

Collectively, these chapters show that the structure and consequences of interspecific interactions are shaped by processes operating across multiple scales. Species traits, phylogenetic relationships, latent network structure, geographic range size, climatic niche breadth, island biogeography, anthropogenic change, and environmental contexts can provide insight into how ecological interactions are organized, but their importance varies across systems and scales. Together, this work contributes to a multiscale view of ecological interactions in which community structure depends not only on whether species interact, but also on how species are embedded within networks, where those networks occur, and how interactions scale up to community-level outcomes.

CHAPTER 2

COMPARING THE POWER OF PHYLOGENETIC, TRAIT, AND NETWORK STRUCTURE INFORMATION TO PREDICT PLANT-FRUGIVORE INTERACTIONS¹

¹Published as: Foster, G., & T. Dallas. Comparing the power of phylogenetic, trait, and network structure information to predict plant-frugivore interactions. *Oikos* 3, e11156. 2026. Reproduced here under the CC BY 4.0 license terms. See Appendix E.

Abstract

Due to the constraints of limited effort and sampling error, observed species interaction networks are an imperfect representation of the “true” underlying community. Link prediction methods allow us to construct a potentially more complete representation of a given empirical network by guiding targeted sampling of predicted links, as well as offer insight into potential interactions that may occur as species’ ranges shift. Various data types can predict interactions; understanding how different kinds of information compare in their ability to predict links between different types of nodes is important. To this end, we compare random-forest regression models informed by combinations of phylogenetic structure, species traits, and latent network structure-hidden features inferred from the observed network topology-in their ability to predict interactions in a diverse network of fruiting plants and frugivorous birds in Brazil’s Atlantic forest. We found that for our dataset, latent structure derived through a single-value decomposition approach was the most important determinant of model predictive performance. While incorporating trait or phylogenetic information alongside latent features had little effect on discriminatory power, they did meaningfully increase overall model accuracy. Our results highlight the potential importance of latent structural features for predicting mutualistic interactions, and encourage a clear conceptual link between prediction performance metrics and the overall goal of predicting cryptic links.

Introduction

Natural systems are characterized by a diverse set of interacting organisms; organizing these interaction sets into ecological networks can help us gain insight into community and ecosystem-level processes. Whether consumer-resource, host-parasite, or mutualistic networks, understanding which species interact and why can give us insight into features as specific as the dynamics of one focal species all the way up to broad-scale patterns of community assembly and stability (Guimaraes Jr, 2020; Saravia et al., 2022). As such, ecologists have put a great deal of effort into observing and quantifying the interactions found in nature. However, like any data collection process, the sampling of ecological networks is imperfect. No matter how much effort we put into characterizing a network, we may often miss observing interactions (links), or even interactors themselves (nodes) (Young et al., 2021). Moreover, the likelihood of observing a given interaction is related to factors such as the abundances of both interactors (Canard et al., 2014), with the total interaction set of rare species less likely to be represented accurately (Cirtwill et al., 2019; Olesen, Bascompte, et al., 2011). Errors in the construction of ecological networks, especially when biased, can lead to dramatic changes in network structure (Blüthgen et al., 2008; Young et al., 2021). Ecological networks are also temporally dynamic. Local species turnover can add or subtract links, while regional-scale extinctions, evolution, or range shifts may rewire existing connections (Olesen, Stefanescu, et al., 2011a).

Increasing sampling intensity or duration could solve these problems; as sampling effort increases it is less likely for rare links to go unobserved, allowing for a more accurate representation of the true network structure. However, the resources that can be devoted to sampling are limited. While the increasing availability of passive monitoring technologies (Quintero et al., 2022), as well efforts to extract interaction data from existing large-scale occurrence datasets (Putman et al., 2021) may increase

the availability of interaction data, the problem of incompletely sampling interactions will never fully disappear. A potential way to address this shortfall is to utilize predictive approaches to decide which particular species or links should be targeted for additional sampling. To this end, the goal of predicting cryptic links between nodes allows us to more productively focus limited sampling resources (Terry & Lewis, 2020). Predictive approaches also allow for us to predict network organization arising from novel complements of species. Identifying likely or unlikely possible linkages between invading species and already present interactors allows us to get a more complete picture of biotic constraints on invasibility (Minoarivelo & Hui, 2016), and ultimately a better understanding of how species interaction networks may be altered by anthropogenic forcing.

Mutualistic interactions are dependent on a complex set of factors including life history (Ramos-Robles, Dáttilo, et al., 2018), phenotypic traits (Rafferty & Ives, 2013), co-evolutionary history (Eriksson, 2016), spatial and temporal distributions (Fricke & Svenning, 2020; Laurindo et al., 2020; Menke et al., 2012), as well as other biotic interactions (Carreira et al., 2020). Incorporating information that either directly influences, or conveys information about these factors into link prediction frameworks should hopefully improve prediction performance (Dallas et al., 2017). When applying link prediction in practice, we often have a variety of species and interaction-level properties at our disposal, many of which may perform better or worse at predicting in certain types of networks or interactions. Understanding which types of information may be better or worse at predicting interactions in different contexts, however, is still an open avenue of research.

For plant-frugivore seed dispersal systems, information on species morphological traits (gape-width, fruit characteristics, etc), may be at the root of why many taxa do or do not interact (Bender et al., 2018; González-Castro et al., 2015; Moran &

Catterall, 2010). Trait matching link prediction approaches are able to use suites of continuous or discrete traits that either directly or indirectly influence interaction probability across a large number of species. These approaches perform well in situations when a small number of traits are linked to interaction probabilities across many species (e.g. flower and beak morphometric traits in flower-hummingbird pollinator systems (Pichler et al., 2020a)). However, while trait data may give us insight into biological underpinnings of network connections we observe, they may not always be the most appropriate for link prediction methods. Interactions may be determined by a large suite of traits that individually contribute small amounts. Alternatively, traits relevant for one taxonomic group may not be relevant for another (gape size may be predictive for birds, but not for primates). Or interaction propensities may be controlled by traits difficult to quantify, such as microhabitat usage or foraging strategies. In all of these cases, trait-matching approaches may perform relatively poorly for some or all types of nodes in a given network. In these cases, alternative information sources such as phylogenetic information or latent network structure (discussed below) may be useful for link prediction.

Using phylogenetic relationships between species may be useful for predicting links due to a number of factors. Firstly, traits that mediate species interactions may be phylogenetically conserved, allowing us to treat information about phylogenetic relationships between organisms as a proxy for traits. This assumption may be especially useful when traits governing interactions are numerous, hard to measure, or hard to identify. However, phylogenetic conservatism may break down at fine evolutionary scales due to local selection pressures on traits governing interactions (Pérez et al., 2007), making phylogenetic information an imperfect proxy in some contexts (Rafferty & Ives, 2013). Additionally, while post-hoc correlative investigation may be able to suggest potential traits of interest, this approach ultimately further abstracts from

the mechanisms governing interactions even if prediction performance is quite good. Despite these shortcomings, in the use-case of targeting potential links for surveillance (in essence a prediction problem), a phylogenetic approach may still be quite useful. In addition to proxying traits, as coevolutionary history between organisms can often be an important factor governing the presence of an interaction, incorporating phylogenetic information may be vital to represent this linkage. While plant-frugivore networks may generally be less coevolutionarily linked and more asymmetric (species with few interactions tend to interact with partners with many interactions) than some other classes of ecological networks (Jordano, 1987; Maglianesi et al., 2024; Wheelwright & Orians, 1982), capturing nodes' shared evolutionary history may still be important for prediction. By definition phylogenetic methods require a phylogeny that includes all the species you intend to predict interactions for, causing potential limitations for predicting in systems where phylogenetic information may be incomplete. However, as the cost and time required to sequence non-model genomes has and continues to decline, the problem of data availability continues to become a more and more surmountable barrier to prediction. To-date, phylogeny-based interaction predictions have been successfully implemented in a number of both mutualistic (Braga et al., 2021) and antagonistic (Pearse & Altermatt, 2013) plant-animal network systems.

Independent of the phylogenetic relationships and traits of individual nodes in a network, latent structural features of networks themselves may actually be useful for predicting missing links. In this context, the term "latent features" refers broadly to hidden properties derived from the topology of the interaction network, which can be approximated through any number of dimensionality-reduction approaches (Poisot et al., 2021). While there are a multitude of different methods used to create latent network features for interaction prediction (single-value decomposition, random dot

product graphs (Strydom et al., 2022), graph motif distributions, matrix factorization (Seo & Hutchinson, 2018), etc), we focus on single-value decomposition, an eigendecomposition approach already used with some success in describing the structure of and assisting in the prediction of empirical food webs, (Banville et al., 2023) host-virus associations (Poisot et al., 2021; Poisot et al., 2023), and seed-dispersal interactions (Nunes Martinez & Mistretta Pires, 2024). Once an existing network is known, calculating latent structural features is computationally tractable. While arguably even more agnostic to the mechanisms driving species interactions, this structural information may capture particular types of real linkages difficult to discover based on trait or phylogenetic information alone.

Prevailing interaction prediction approaches have often centered on a single information type, but recent work in host-parasite systems have emphasized the power of machine learning approaches to synthesize multiple data types to predict interactions (Strydom et al., 2021). Despite the use of this variety of information streams for link prediction problems, studies directly comparing the predictive capacity of phylogenetic, trait, and latent features are few, and have not been applied in plant-frugivore systems. Beyond looking at the performance of these feature classes independently however, knowing how they may be used together is another important avenue of research. By looking at the performance of models trained on combinations of different data types we are better able to understand whether different methods overlap in the types of links they predict well, and perhaps more importantly whether we can use these information streams in concert to create models that are more than the sum of their parts. To this end, we apply random forest prediction algorithms to predict plant-frugivore interactions in Atlantic forests in Brazil using combinations of species trait data, phylogenetic information, and latent structural features.

Methods

Trait and Interaction Data

The full interaction network was published by Bello et al. (Bello et al., 2017), and includes a total of 5,226 unique species interactions between 787 plant species and 342 frugivore species. This network effectively represents a regional metaweb of observed interactions, with the realized set of interactions at a given site being a subset, the exact composition of which is determined by local assembly processes. By taking a metaweb approach—predicting “possible” interactions independent of local geographic or temporal variation in species abundance—we aim to better understand the factors associated with the potential for interaction. This general approach is shared by a number of other studies predicting potential interactions across large spatial scales. (Dansereau et al., 2024; Hao et al., 2025; Strydom et al., 2022) Due to the availability of phylogenetic and trait data, we restricted our analyses to avian-plant interactions, representing the most species-rich set of frugivores in this dataset (3,856 unique interactions between 394 and 242 plant and bird species, respectively). Of our avian-frugivore species (hereafter frugivores), the number of unique plant interactions per frugivore (frugivore degree) ranged from 1 (53 species) to 120 unique plant interactions recorded for *Turdus rufiventris*; median frugivore degree was 5. The number of unique frugivore interactions per plant (plant degree) was generally lower, ranging from 1 (128 species) to 80 unique frugivore interactions recorded for *Myrsine coriacea*; median plant degree was 2. Overall this interaction network is characterized by a relatively high degree of phylogenetic generalism. Out of 187 frugivore species that interacted with more than one plant for which phylogenetic information is available, only 9 interacted with a group of plants more phylogenetically related than random chance (Kembel et al., 2010), see supplement).

The Bello et al. dataset includes a number of informative traits for both plant and frugivore nodes that could be potentially useful for prediction, many of which we used in our trait-based models. These include frugivore allometric measures such as body mass and gape size, and ecological traits such as degree of frugivory (scored 1-3). Frugivore mean body mass was unavailable for 3 species, while mean gape size was unavailable for 68 species. Plant traits include fruit diameter, fruit color, growth form, and fruit lipid concentration (scored 1-3). For trait-based models, species were filtered to only include those with complete sets of trait data (174/242 frugivore species). As part of our research goal was to compare the power of phylogenetic and trait data, we chose to omit these species for trait based models rather than phylogenetically impute missing trait information. Categorical variables (such as plant growth form) were transformed into a series of binary variables through one-hot encoding. Continuous data on fruit size was available for 280/394 plant species; plants without trait data were again excluded from trait-based models. Details on plant and frugivore traits used in the analysis are available in Table 2.1).

Phylogenetic Data

For plant species, phylogenetic data was retrieved from the BIEN database, accessed using the *r* package *BIEN* (Maitner et al., 2018). Out of 787 plant species in our dataset, 646 occurred in the BIEN phylogeny. 61 plant species with a recorded avian frugivore interaction did not occur in this phylogeny. For absent species with a congeneric in our phylogeny, we added those species as a polytomy at the parent node of that genera, allowing us to add an additional 46 species to our analysis. 16 species did not have congenics present in the network, and were discarded from the analysis. Results were qualitatively the same if these polytomies are omitted from the analysis (see supplement). Bird phylogenies were retrieved from VertLife (Jetz et al., 2012); for our analysis, we used 100 trees sampled from the Bayesian posterior

distribution. Phylogenetic information was unavailable for two bird species in our network. As with plants, both species were added as polytomies at their respective parent genera. Each sampled tree was used in one of one hundred replicates for each model incorporating phylogenetic data. For both the plant and bird phylogenies we performed eigenvalue decomposition, and trained phylogenetic models on the first 4 dominant eigenvectors for plants and the first 3 dominant eigenvectors for birds. The incorporated eigenvectors explained 37.2% of variation for plants. While the exact amount of variation explained by the incorporated eigenvectors differed among sampled trees for birds due to variation in posterior samples, on average the incorporated eigenvectors explained $46.02 \pm 2.29\%$ of variation for birds.

Latent Methods

We used singular value decomposition of each training interaction matrix to create continuous latent features for prediction using the `svd()` function in base R (R Core Team, 2021). For each training iteration, we fixed the dimensions of the training matrix to be equal to the full network, adding interactions (1's) of only the training set before creating latent features. As such, in any individual training matrix, some species with few interactions may have no recorded interactions. Latent features of training subsets were highly correlated with those derived for imputing based on the full interaction matrix (see supplementary Figures A.11 and A.12). We used the first three axes of variation for both plants and birds as continuous predictors. For imputation of missing links across the entire network, we utilized the entire interaction matrix to construct latent features.

Model Structure and Comparison

We used random forest regression models informed by different feature types to make comparisons across types of information, as implemented in the R package *randomForest* (Liaw & Wiener, 2002). Random forest models are effectively an ensembled

series of classification trees, each of which is applied to bootstrapped subsamples of training data and a portion of the potential predictor variables. By incorporating the information of many decision trees together, random forest techniques can often boast high predictive accuracy across a variety of ecological settings, the ability to uncover complex and nonlinear interactions between predictor variables, and internally cross-validate by repeatedly testing on "out of bag samples" (portions of the data not included in a given bootstrap subsample) (Breiman, 2001; Cutler et al., 2007). In the context of ecological networks, random forest models have been successfully applied to predict species interactions across a variety of systems. (Desjardins-Proulx et al., 2017; Pichler et al., 2020a; Sydenham et al., 2022) We tested a total of 7 models; three using predictors from only one type of information (traits, phylogeny, or latent features), three pairwise combinations of each predictor class, and one model incorporating all possible predictors. In order to validate model performance, we trained 100 iterations of each model. Like most empirically observed ecological networks, our training network was very sparse (Vázquez et al., 2009). Our total interaction matrix included 3,643 observed interactions and 90,917 unobserved interactions. To address this class imbalance, for each iteration we first randomly selected 80% of our data for training, and then randomly removed unobserved interactions from the edgelist (i.e. rows of the edgelist where the interaction value is equal to 0) from the training data until we achieved a 1:3 ratio of observed:unobserved interactions (see supplementary Figures A.6-A.8 for the results of alternative prevalence values). After model training, performance was evaluated on the remaining 20% test set. We analyze model performance both through area under the receiver-operator curve (AUC), which measures model discriminatory power, and root mean squared error (RMSE), which measures prediction accuracy. These two measures are commonly used to evaluate the effectiveness of predictive approaches (Norberg et al., 2019), but the choice of which metric to optimize may be different depending on the use case in question (see

discussion). For each model iteration we also recorded the optimal suitability threshold for classification as the value that maximized Youden’s J statistic (the sum of specificity and sensitivity minus one, a common diagnostic statistic for dichotomous tests (Youden, 1950)). For comparisons of model accuracy metrics across models, we performed post hoc two-sided pairwise t -tests adjusted for multiple comparisons (Holm, 1979).

After validating model predictive performance using this test-train split, we then re-ran 100 iterations of each model using the full network without internal class balancing. Using the full suite of interaction information allows us to better predict potential unobserved links. We present the pairwise suitability correlations of each of these full model outputs, as well as present a list of links predicted to be highly suitable by one or more of our models but that are not observed to occur in this data set. We then used the average optimal threshold value for each model as model-specific thresholds for classification. For each model, variable importance was quantified by the mean decrease in Gini coefficient (MDG) after permutation of each variable as implemented in the R package *randomForest* (Liaw & Wiener, 2002). A common variable importance metric for classification problems, MDG is calculated by normalizing the sum of all decreases in node purity (the ability of a given tree to correctly distinguish links from nonlinks) given permutation by the total amount of decision trees in the ensemble (Calle & Urrea, 2011; Khalilia et al., 2011). A higher MDG value for a given variable indicates that variable is more important to model performance.

Results

Model Performance

Using an 80-20 split of testing and training data, all models were able to predict

testing interactions with reasonable accuracy and discriminatory power (Figure 2.1). AUC was lowest for trait only models ($0.84 \pm .009$) followed by phylogeny only models ($0.84 \pm .008$). All models including latent information outperformed latent-agnostic models in terms of discriminatory power, with the greatest AUC achieved by trio ($0.91 \pm .007$) and phylogeny-latent models ($0.91 \pm .006$); pairwise t -tests indicated no significant difference between them ($p = 0.14$). However, prediction accuracy did not follow the same trends as discriminatory power. Latent and phylogeny-only models had the lowest overall accuracy ($57.7, \pm 0.40$ and 56.1 ± 0.44 , respectively), while the most accurate models were those combining trait information with another information type. The trait-phylogeny (RMSE: $49.3 \pm .42$) and trio (RMSE: $49.4 \pm .44$) models were the most accurate (pairwise t -tests indicating no difference, $p = 0.8$), closely followed by the traits-latent model (RMSE: $49.8 \pm .37$). The trait-only model had only middling accuracy performance however (RMSE: $52.7 \pm .54$).

Variable Importance on Entire Network

After validating model performance on novel test-data, we then retrain replicate models of each type on the full network in order to generate more robust predictions. Variable importance was quantified by mean decrease in Gini coefficient after permutation of each variable. In all models incorporating them, latent SVD features were unilaterally the most important variables for prediction (Figure 2.2); all models combining latent features with other data sources ranked plant SVD axis 1 followed by frugivore SVD axis 1 as the two most important variables, with some variation as the ordering of the rest of the SVD axis importance. The latent-only model generally followed similar patterns of variable importance as composite models, though frugivore SVD axis 1 had higher importance than the corresponding plant SVD axis. For the trio model, frugivore mass and fruit diameter were the next most important traits, which was consistent with the ordering of all other trait models. While the

next most important trait was frugivore gape size for the trio model, for other models incorporating trait information fruit length was the next most important trait.

In general, we found that continuous frugivore traits (body mass, gape width) were more informative to prediction than continuous plant traits. Similarly, phylogenetic information about frugivores was generally more important for prediction than phylogenetic information for plants. In the phylogenetic-only model, all three frugivore phylogenetic eigenvectors were more important than all plant eigenvectors, while for the trio model only plant eigenvalue axis 2 was more important than frugivore eigenvalue axis 2. Out of the remaining traits, the trio model was most informed by frugivory level, with the remaining categorical plant traits impacting Gini coefficient very little for all models.

Comparing models

Across the entire network, suitability values between all models were highly correlated (Figure 2.3); most models were able to correctly discriminate between observed and unobserved links. While the ability to reconstitute observed links is important for model performance, focusing on model agreement on the suitability of unobserved interactions can provide insight into which models may identify different potential interactions that are yet unobserved but may occur given the appropriate ecological context. Across all potential links, Spearman’s rank correlation of predicted suitability values showed low correlation between phylogenetic and trait models ($\rho = 0.643$), despite both exhibiting very similar overall performance. Phylogenetic and latent models showed the lowest agreement in suitability predictions of all pairwise combinations of models, with $\rho = 0.566$ when comparing across all links, which then dropped to $\rho = 0.032$ when only looking at unobserved links. This marked drop suggests that most of the agreement in suitability rankings between these models was in known interactions. The low agreement between the ranking of potential interac-

tions poses a barrier to meaningfully target potential sampling based on the results of these two models used independently, highlighting the importance of composite models that synthesize multiple information types to create predictions.

Similarly, suitability predictions of composite and individual component models tended to mirror relative information type importance. For example, suitability values from the trait-latent model were more tightly correlated with those from the latent model ($\rho = 0.882$) than the trait model ($\rho = 0.773$). Table 2.2 presents the confusion matrix for all models after classifying points according to each models' average optimal threshold value across all training iterations. We see that at their optimal threshold value, all models exhibit markedly low false-negative rates with the exception of the trait only model, which exhibited a false-negative rate of 4.80%.

Discussion

Models incorporating latent structural features consistently outperformed models that excluded them in terms of discriminatory power, but had generally lower accuracy. In contrast, trait based and phylogenetic models displayed lower AUC values, but a better RMSE. Overall performance of phylogenetic and trait-only models were remarkably similar in terms of both performance measures used. Incorporating latent features into each model once again yielded nearly identical discriminatory capacity, but an overall slightly higher accuracy for the trait-latent features model compared to the phylogeny-latent feature model. The discriminatory power of each individual model generally reflects the relative importance of variable information types (Figure 2.2).

The strength of latent features for predicting interactions may lie in their particular suitability for relatively generalist interaction networks. Frugivorous vertebrates, for instance, often exhibit high levels of generalism in their interactions (Richardson

et al., 2000), often more-so than their plant-partners (Salazar-Rivera et al., 2020). In systems characterized by low interaction specificity, latent features may be especially effective at capturing the neutral processes that underlie community assembly. Abundant, wide-ranging, or highly generalist species may drive a "rich get richer" dynamic, where hyper-generalists that already interact with multiple species are more likely to accumulate additional partners. This pattern is reflected in our results: the 20 links with the highest predicted suitability from our trio model were spread across 16 bird species, many of which already participated in a large number of observed interactions. While the median number of interactions per bird in the dataset was 5, the median degree for these 16 species was 56. The top predicted missing link connected the azure-shouldered tanager (*Thraupis cyanoptera*) and *Myrsine umbellata*, two relatively common species that have indeed been documented as interacting in other datasets (Fricke & Svenning, 2020; Silva et al., 2002). Host abundance, in particular, has been identified as a key driver of interaction probability in other Neotropical frugivory networks (Laurindo et al., 2020). Overall, latent features derived from network dimensionality reduction or motif completion may be especially powerful tools in the context of generalist interaction networks.

While comparing models in terms of discriminatory power highlights the importance of latent features, incorporating phylogenetic or trait information alongside latent features actually improves model accuracy measures (RMSE), despite slightly reducing discriminatory power. The question whether it is better to optimize one metric or the other, is largely dependent on the particular use case. As we've presented the problem of link prediction, primarily as a way to guide sampling efforts to more efficiently measure networks given limited resources, discriminatory power is likely a more apt metric. Beyond ranking different links, the relative confidence in each link is of less concern as the most likely links will be validated with additional

observation data. However, if the aim was something closer to filling in a partially sampled network and then analyzing its properties, a metric like RMSE or H-measure (Hand, 2009) may be more appropriate as it better allows you to directly incorporate uncertainty about particular linkages into your resultant analyses.

In models incorporating frugivore and plant phylogenetic information, the frugivore phylogeny tended to be more important to model performance. Ideally, this is likely because patterns of interactions are more phylogenetically conserved between frugivores than plants. On average, frugivores tend to interact with a more phylogenetically conserved group of plants than vice versa, with a given plant’s fruit being consumed with a more phylogenetically dispersed set of avian taxa. Some of this phenomenon may be due to the frugivore phylogeny being more fully summarized by a smaller number of eigenvectors. On average, the first three eigenvectors of the decomposed frugivore phylogeny explained 46.02% of total variation across the phylogeny on average, while the first four eigenvectors only explained 37.5% of variation for plants. The difference in variation and resultant effects on prediction may be due to overall structural features of the phylogeny, or differences in the total phylogenetic divergence characterized by the tree. Frugivory is a strategy that spans the avian tree, occurring in a diverse set of taxonomic orders (Daniel Kissling et al., 2009), though this dataset reflects a relatively narrower set of frugivorous taxa. Out of our 242 avian species investigated in this study, the majority belonged to the orders *Passeriformes*, *Piciformes*, and *Columbiformes*, comprising 193, 18, and 10 species, respectively. Ornithochory is also widely distributed across plant taxa in this system, a pattern reflected in other tropical systems (Kuhlmann & Ribeiro, 2016; Pizo et al., 2021).

Our modeling approach represents a useful framework for predicting species interactions in a variety of systems, but care must be taken when interpreting the results

in the particular context of each system. While we ultimately find that latent features dominate prediction in our system, the potential for information leakage between test and train sets means we must be conservative in our interpretation of comparisons across models. Due to the nature of matrix decomposition approaches, there is no clear way to utilize either latent or phylogenetic features to predict out of bounds. Even when creating latent features from training subsets of interaction data, the dimensions of the training matrix must be fixed to the size of the full network in order to generate predictions for all species within the network. As such, while useful for imputing potentially missing interactions between known species, latent features are unsuitable for predicting interactions of species that are not already represented within the existing species pool. The dominance of latent features using internal cross-validation metrics support their importance in our system. However, as latent features are derived from portions of or all of the observed interaction network, their predictive capacity may be sensitive to network characteristics such as network size, connectance, and the proportion of links that are still unsampled (see supplement). Additional work is needed to fully explore these relationships in order to explore in which contexts latent features might be more or less useful for prediction. Despite their caveats, latent features have already been successfully implemented to predict links in a variety of ecological network types, and can provide significant improvements in predictive performance in a variety of ecological (Banville et al., 2023; Nunes Martinez & Mistretta Pires, 2024; Poisot et al., 2021; Poisot et al., 2023) and non-ecological (Wu et al., 2019; Yeung et al., 2002; Zeng et al., 2020) contexts.

Future work investigating the utility of latent features to predict frugivorous interactions in a variety of empirical systems and the mechanisms through which they act represents an important avenue for future research. While there are currently no clear methods through which we can apply these methods for out-of-sample prediction, these methods are promising ways to target potentially missing links in existing

networks. Similarly, our presented interpretation of the efficacy of phylogenetic information for predicting missing links assumes that phylogenetic relatedness reflects conserved traits or shared evolutionary histories that govern interactions. However, while phylogenetic relationships do impact at least some of the species traits governing interactions, there may be other confounding information (such as a phylogenetic biases in research effort) within phylogenetic information as well. While not reducing overall link prediction performance, this does mean we should once again take a more conservative interpretation of our prediction results when comparing across models, as is the case with any correlative approach.

Link prediction is a promising method to deal with the realities of incompletely sampled natural systems, and may play an increasingly important role in helping to gain insight into how interactions networks may change over time. Anthropogenic effects on climate, landcover, and biodiversity all have the potential to fundamentally alter mutualistic networks, whether through changing mutualistic assemblages themselves, or the probability of interactions within a network (Memmott et al., 2007; Teixido et al., 2022; Tylianakis & Morris, 2017). Positive feedbacks within mutualistic communities may often make them more likely to exhibit alternative stable states, and changes in mutualistic communities may result in drastic changes to network structure and resulting assemblages (Bascompte & Scheffer, 2023; Lever et al., 2014). Link prediction methods provide an opportunity for us to better understand which species are interacting now, and which may interact in the future. Prediction approaches can also be useful when predicting the potential effects of invaders on mutualistic network structure (Fricke & Svenning, 2020; Traveset & Richardson, 2014). However, at its core, predicting which interactions are unsampled but present in a given ecological context is related to but distinct from the task of predicting how interactions may shift in response to changing biotic or abiotic conditions. If the eco-evolutionary processes

governing the formation of novel interactions over short timescales are sufficiently different from those governing the maintenance and structure of long-standing interactions across a community, then different modeling approaches may vary in their effectiveness at predicting missing links versus truly novel ones. To fully understand and predict the scope of mutualistic network change, future work applying predictive approaches to temporally sampled mutualistic networks is necessary. Understanding the spatial and temporal scales at which mutualistic interactions turn over or change strength, and how we can utilize predictive approaches to forecast these changes is a vital next step for understanding the dynamics of mutualistic networks in a changing world.

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Open Research statement

All *R* code and data necessary to reproduce this is available on figshare at <https://figshare.com/s/544021fc48732ca5bb13>. A version-controlled history of this project is also publicly available on GitHub at <https://github.com/GTFoster/Fruglink>.

Tables

Table 2.1 Frugivore and plant traits used for interaction prediction. Non-ordinal factors such as fruit color were transformed through one-hot encoding prior to inclusion in the model.

Species	Trait	Units	Range/Levels
Frugivore	Body Size	g	6.60 - 3500
	Gape Size	mm	2.80-36.29
	Degree of Frugivory	-	low, medium, high
Plant	Fruit Diameter	mm	1.2-325
	Fruit Color	-	yellow, red, black, brown, green, other
	Fruit Lipid Content	-	low, medium, high
	Plant Growth Form	-	tree, liana, palm, scrub, other

Table 2.2 True and false positive and negative rates for models trained on the entire interaction network, classified assuming the average threshold value across all training interactions that maximized Youden’s J statistic. Percentages represent proportions in each category out of all total links each model predicted; the absolute count of each category is represented in parentheses. Total number of predicted links differed across models according to availability of covariate data.

Model	True Positives	False Positives	True Negatives	False Negatives
Latent	3.85% (3,643)	0.28% (263)	95.87% (90,654)	0.00% (0)
Phy	3.87% (3,350)	0.27% (231)	95.66% (82,806)	0.29% (175)
Traits	1.86% (778)	0.22% (92)	93.20% (39,021)	4.73% (1,979)
PhyLatent	4.07% (3,525)	0.124% (107)	95.80% (82,930)	0.00% (0)
PhyTraits	6.63% (2,723)	0.35% (143)	93.02% (38,212)	0.01% (2)
TraitsLatent	6.59% (2,757)	0.44% (185)	92.97% (38,928)	0.00% (0)
Trio	6.863% (2,725)	0.15% (61)	93.22% (38,294)	0.00% (0)

Figures

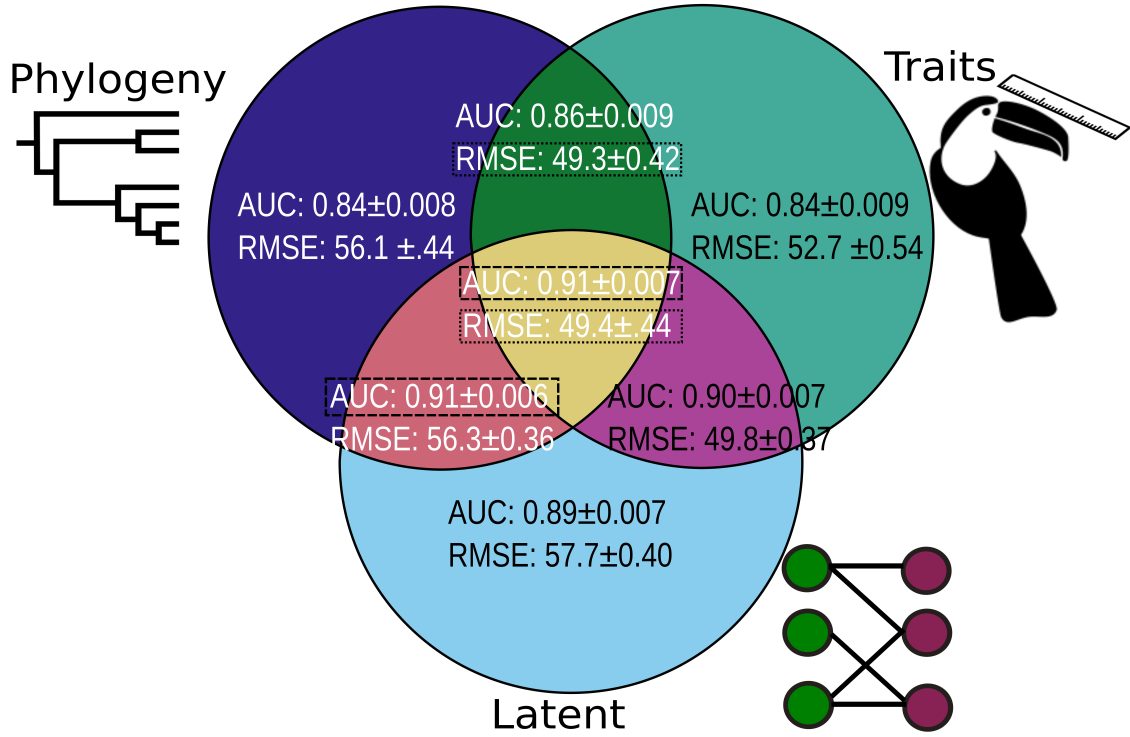


Figure 2.1 Summary performance metrics of all 7 models, as measured by area under the receiver operating characteristic curve (AUC) and root mean square error (RMSE); highest performing models for each metric are outlined in dashed (AUC) or dotted lines (RMSE). Mean metric values are presented from 100 replicates of each model structure alongside standard deviation. Model discriminatory power between links and non-links is maximized by including latent structural features, with the inclusion of trait, phylogenetic information, or both actually slightly decreasing discriminatory power. However, inclusion of trait and phylogenetic information, while not improving AUC, does increase overall model accuracy as measured by root mean squared error. Pairwise *t*-test adjusted for multiple comparisons showed no significant difference between the discriminatory power of the highest performing models (Trio, PhyLatent, $p = 0.13$), or between the RMSE of the two highest accuracy models (Trio, PhyTraits $p = 0.8$).

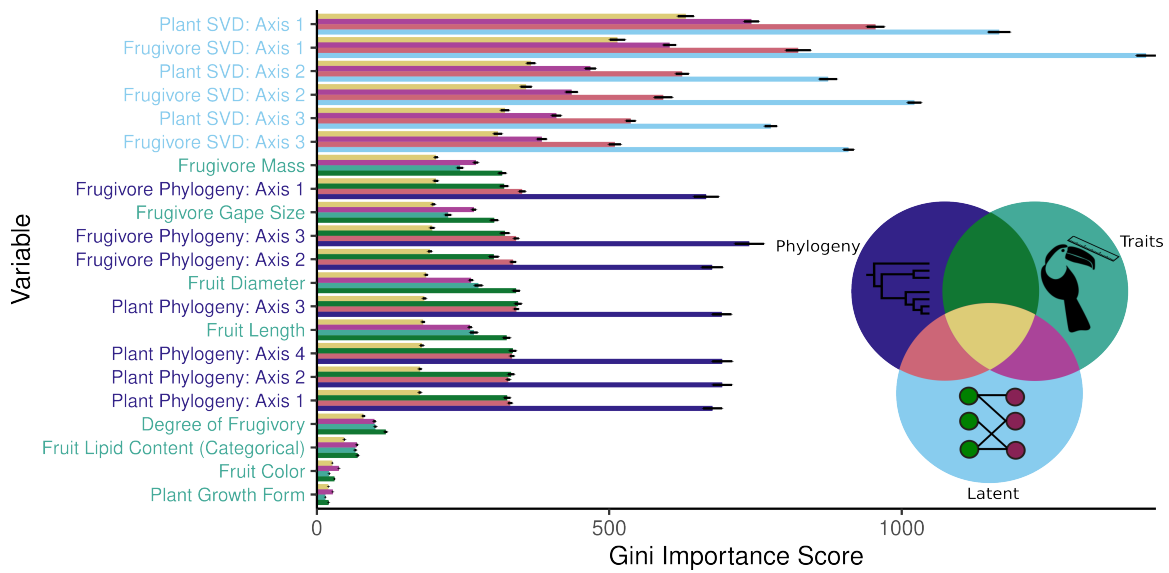


Figure 2.2 Variable importance across all models as measure by Gini importance score; color scheme is consistent with figure 1. In all models that include them, latent traits were consistently the most important variables for prediction. These were followed by continuous frugivore traits (body mass, gape size), and frugivore phylogenetic axes. Plant phylogenies and continuous trait information were generally less important for prediction than frugivore traits. Categorical plant traits (Lipid content, fruit color, growth form) were the least important variables for prediction.

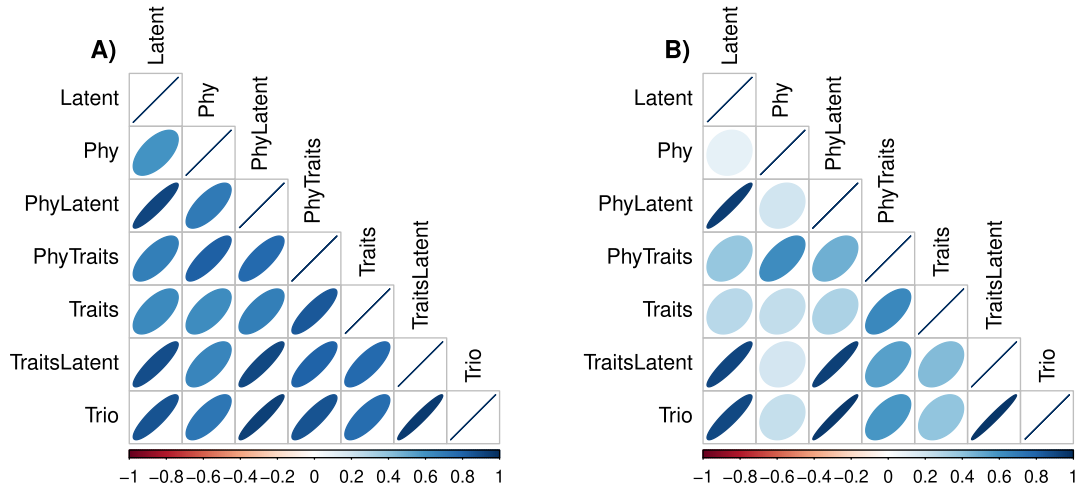


Figure 2.3 Pairwise Spearman's rank correlations of link suitabilities across models for all potential interaction (A), as well as only unobserved interactions (B). The latter set of links represents both true forbidden links, as well as other potential interactions not observed in our dataset.

CHAPTER 3

THE IMPACTS OF GEOGRAPHIC RANGE SIZE AND NICHE BREADTH ON SPECIES CENTRALITY IN FLORAL VISITATION NETWORKS¹

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Abstract

Understanding the factors that influence species' roles within ecological networks can provide insights into the rules governing the assembly of interaction networks. Centrality, a property describing how connected a given node is to the rest of a network, is often used to identify species whose loss may result in disproportionate effects on biotic communities. Understanding the drivers of species' centrality in floral visitation networks is crucial for understanding network responses to global change. Here, we test how geographic range size and niche breadth influence species centrality in floral visitation networks across spatial scales. We combine a globally distributed dataset of floral visitation networks comprising 30,330 interactions among 6,857 species with large-scale occurrence data to evaluate how species' geographic and environmental range sizes impact their role in networks across scales. In global metawebs we observe a strong positive effect of geographic range size on closeness centrality, but that effect largely disappeared after accounting for spatial turnover in interactions. Instead, niche breadth showed a positive relationship with the standardized centrality of flowering plant species in both network types. Site-level centrality was tightly linked to species ranges. Together, these results emphasize the need for a multi-scale perspective to understand the drivers and consequences of changes in ecological network properties across environments and scales.

Introduction

Originating in mathematics, graph theory perspectives have become increasingly popular for the study of ecological networks throughout recent years (Bascompte & Jordano, 2007; Guimaraes Jr, 2020; Landi et al., 2018; Valdovinos, 2019). Under this framework, interacting species can be represented as a series of nodes connected by links (interactions). What constitutes an "interaction" is variable, set by the context

of the system, and may include predation events, infections (Dallas et al., 2017), floral visits (Young et al., 2021), or any number of other biotic interactions. By quantifying the full set of links present in a given system, we have gained insights into the structural properties of species interaction networks that allow us to better predict their potential responses to change (Gilaranz et al., 2017). For example, foundational work has linked network properties such as nestedness (the tendency for species with few interactions to link to species with many other interactions) and modularity (the tendency for species to interact more within given groups or “modules” than between them) to features such as network stability in response to perturbations or species extinctions (Okuyama & Holland, 2008; Suweis et al., 2013; Thébault & Fontaine, 2010).

In addition to these network-level properties, we can also measure the contribution of an individual node to overall network structure considering node-level properties. One of the most commonly analyzed node-level properties is centrality: a suite of measures describing how connected a given node is to the rest of the network. Highly central species may represent keystone species in a network, and removal of these nodes is likely to have outsized impacts on their associated communities (Maia et al., 2019; Martins et al., 2024; Traveset et al., 2017). Identifying the eco-evolutionary processes that give rise to highly central species in local and regional-scale interaction networks is essential for understanding how ecological networks are organized and how they may respond to disturbances (Martins et al., 2025; Moulatlet et al., 2023; Zhang et al., 2025). Previous work has emphasized the importance of species traits (Lázaro et al., 2020; Maia et al., 2019; Watts et al., 2016) and evolutionary history (Burin et al., 2021; Martín González et al., 2015) on influencing both network and node-level properties. Investigating determinants of a species’ role within floral-visitation networks (defined here as its position and the distribution of interactions

relative to other members in a network, captured through node-level properties such as centrality), can provide insights into biological processes underpinning interaction network structure across scales. Understanding these drivers in the context of floral visitation networks, which are often characterized by interaction generalism (Stanton, 2003; Waser et al., 1996) and frequent facultative interactions (Geib & Galen, 2012), is of particular conservation importance as global declines in pollinator abundance and diversity threaten to potentially reduce pollination services in many ecosystems (Potts et al., 2010; Rodger et al., 2021).

In addition to morphological characteristics and phylogenetic history, the size of a species' geographic range may also impact its role in pollination networks at both local and global scales (Dáttilo et al., 2020; Moulatlet et al., 2023). As species ranges increase in size, so does the total number of potential interaction partners. As a consequence, in regional metawebs (the set of species interactions observed across multiple local webs (Poisot et al., 2012)) wide-ranging species may increase in their total number of interactions as both potential partners and interactions turn over across space (Fricke & Svenning, 2020; Martins et al., 2022). This accumulation of interactions across area, parallel to the classical species-area relationship (Burkle & Knight, 2012; Dallas & Jordano, 2021; Galiana et al., 2018; Sugiura & Taki, 2012), may thus induce positive associations between species range size and species centrality at regional or global scales. Additionally, species with broad geographic ranges may also be more central in local networks if geographic range size is associated with demographic or functional traits that may allow for generalist interaction strategies or high local abundance (Holt et al., 1997; Hurtado et al., 2024; Moulatlet et al., 2025). This potential confounder makes it difficult to differentiate the potential reasons geographic range size may impact species roles in mutualistic networks. The development of null modeling approaches, in which many random permutations of

network linkages are summarized into an observed distribution of potential network property values, has been instrumental in making meaningful comparisons across ecological networks of different size and connectance (Blüthgen & Staab, 2024; Dormann et al., 2017; Guimaraes Jr, 2020). Here, we utilize a null modeling approach in order to tease apart the otherwise confounding effects of turnover and range size *per se* on species centrality across scales.

Species’ geographic range size is driven by a number of biogeographic factors, including evolutionary history (Alzate et al., 2025; Webb & Gaston, 2000), dispersal limitation (Alzate & Onstein, 2022), biotic interactions (Galiana et al., 2023; Pigot & Tobias, 2013), and climatic constraints (Hausharter et al., 2023). As species increase in the geographic extent they occupy, they may geographically overlap with more potential partners, increasing their centrality. However, geographic range size might not be the strongest predictor of the number of potential partners. Instead, species that occur in more climatically diverse environments may co-occur with a larger set of potential partners (Batstone et al., 2018; Slatyer et al., 2013). Therefore, species with broad abiotic niche breadths may have higher centrality in the global metaweb as a function of greater niche overlap with potential partners. A positive relationship between niche breadth and local centrality may also be observed if climatic and interaction generalism are positively associated, a feature observed in some food web systems (Slatyer et al., 2013). Niche breadth and geographic range size are fundamentally linked, though the exact shape of the relationship is dependent on the geographic availability of environments across space (Dallas & Ten Caten, 2025).

Biologically, species centrality has a fundamentally different meaning in local networks and global metawebs, and understanding how centrality scales between these contexts is still an open question in network ecology. Most existing work investigating the determinants of species centrality within networks to date has focused

on local scales ((Maia et al., 2019), but see (Moulatlet et al., 2023)), as here node centrality has clear links to network properties (Cagua et al., 2019a; González et al., 2010; Maia et al., 2019). Species that occur in many local networks and are highly central in many or all of them may also exhibit high centrality in regional or global metawebs ((Estrada, 2007; Martins et al., 2025) though see (Dallas & Jordano, 2022)). However, even species with low local centrality can achieve high centrality in the global metaweb if they exhibit high interaction turnover across a wide geographic area (Carstensen et al., 2014; Simanonok & Burkle, 2014). At a global level, these central species may appear as relative generalists with many potential partners. However, at local scales, the realized set of interactions of any given individual or population may be relatively narrow (Arroyo-Correa et al., 2023; Bison et al., 2015). As such, understanding how centrality scales from local to global scales helps us better understand how population-level interaction patterns may vary across mutualistic species' ranges.

To address these questions, we quantified variation in species-level centrality relative to geographic and environmental range size across a global assemblage of avian-plant and insect-plant floral visitation networks. Specifically, we tested whether species with broader geographic ranges or environmental niches occupy more central positions within local networks or global metawebs. We predicted that centrality will increase with range size and niche breadth in both contexts (Figure 3.1). At the metaweb scale, we expected species with broad ranges to turnover in interactions across geographic or environmental space, allowing them to accumulate more interactions across sites. At the local scale, we predicted a similar positive relationship driven by the positive association between interaction generalism and climatic or habitat generalism. Finally, we investigated whether these relationships persist after using null-model approaches to account for the impact of species co-occurrence

on potential centrality measures. If range size or niche breadth continues to be positively associated with centrality under null models that control for co-occurrence, it would suggest a fundamental link between geographic or climate distributions and factors determining species role within mutualistic networks. Together, our results suggest that the impacts of range size on species centrality in large-scale metawebs are primarily driven by interaction turnover and the scaling of interaction specificity from local to global networks. When variation in interaction turnover is incorporated, niche breadth exerts the strongest positive effect on plant centrality within global networks. Consequently, species with large geographic ranges may appear to be relatively generalist in global metawebs due to high interaction turnover, but local-scale interaction patterns may exhibit more nuanced and variable relationships with species range limits.

Methods

Data

We queried floral visitation networks through the Web of Life dataset, resulting in 97 networks with a total of 30,330 unique interactions between 6,857 species. We combined these networks into two global metawebs: one describing avian-plant interactions and another describing insect-plant interactions. Most studies only characterized one of these types of interactions; for the three studies that reported both insect and avian floral visits, we divided the network as appropriate. The avian-plant metaweb is comprised of 2,033 unique species-level interactions between 605 plant and 120 bird species across 31 sites. The insect-plant metaweb is larger, consisting of 28,297 interactions between 1,960 plants and 4,182 insect species across 66 sites. While expert range maps are available for most birds and some well-studied plant taxa, the majority of species in our study do not have up-to-date expert species range estimates available. In order to investigate the role of range size in determining

species-level network traits, we estimated species ranges from available occurrence points.

Estimating Geographic Range Size

For each species, we queried global occurrence records from the Global Biotic Information Facility (GBIF) using the *gbifdb* R package. Occurrence points were first processed using the *coordinateCleaner* package; points were removed if they were present only in the fossil record, occurred within 2 km of country centroids, were not terrestrial, or were marked as an “introduced” individual. To avoid overestimating species range size through including spurious or transient records in GBIF, we used a quantiled minimum convex polygon (MCP) approach. Within each continent – based on a 7 continent model (Asia, Europe, Africa, Oceania, North America, Antarctica, and South America) – we created the smallest minimum convex polygon that contains the central 95% of the occurrence points within each continent. Range polygons were not created for species-continent combinations with less than 10 occurrence points. Total species’ geographic range size was calculated as the total terrestrial area occupied across all continents. The points included within these polygons were also used for the calculation of environmental range (below). 2,554 species within our network had insufficient occurrence data to create species ranges, leaving a total of 4,303 species within our analysis (110 birds, 2,126 insects, and 2,067 plants).

Estimating Niche Breadth

For each occurrence point, we extracted 56 climatic values from WorldClim at a 2.5 arc-minute resolution (Fick & Hijmans, 2017). These values represent a number of climatic variables potentially important for limiting species’ distributions, averaged across the years 1970-2000. As these variables represent a high-dimensional environmental space that is collinear in many aspects, we performed a principal component analysis in order to reduce the dimensionality of our niche space (Dallas & Ten Cat-

en, 2025; Kriticos et al., 2014). The first two principal components of this climatic space explain over 77% of the total climatic variation. For each species, we plotted each occurrence point that fell within the 95% quantiled geographic range polygon calculated above onto the Cartesian plane defined by these two PCA axes, combining points from all continents on which a species occurred. We then again employed a minimum convex polygon approach to draw an environmental range in this PCA space for each species (i.e., its niche breadth).

Centrality Measures

Insect and avian floral visitor metawebs contained 6142 and 725 total species, respectively. The vast majority of species in each network were connected to one major component (the largest set of interconnected nodes). As centrality measures become non-comparable in unconnected graphs, we dropped species that were not part of these major components. This resulted in the removal of 51 (0.83%) and 4 (0.55%) species from the insect and avian metawebs, respectively. Closeness centrality (which describes the mean proximity of one node to all other reachable nodes within the network, Figure 3.1) was calculated for nodes in the remaining simply connected networks using the *igraph* package. We chose closeness centrality as an intermediate between hyper-local measures of centrality (e.g., degree centrality) and hyper-global measures (e.g., eigenvalue centrality). At the global metaweb scale, the rank of alternative measures of centrality were positively correlated, though to varying degrees (Figure B.2).

Null model expectations of species centrality

While centrality measures are designed to be comparable between nodes of the same network, their absolute value is heavily influenced by network size and connectance, potentially hindering comparisons across networks. To address this when measuring species-level centrality in local networks, we used a null-model approach to standard-

ize centrality within each local network. For each local network, we first measured the closeness centrality of all component species using a similar approach to the global metaweb. We then used the **vegan** package to generate 1000 randomized versions of each local network by permuting interaction links while preserving each node’s degree (effectively preserving marginal totals of the interaction matrix). This allowed us to characterize the potential distribution of centrality values for each node, given that node’s degree and the network’s overall structure. We then z -transformed the observed centrality values using the mean and standard deviation of this null distribution to create a z -transformed measure of local closeness centrality for each species-site combination.

To extend this approach to the metaweb level, we pooled permuted local webs to construct null global metawebs. This allowed us to generate a null distribution of species-level closeness centrality across the entire metaweb, accounting for each species’ presence across multiple local networks. Observed metaweb centrality values were then z -transformed relative to this distribution. We investigated both measured and z -transformed centrality at the metaweb scale. At the local scale we only modeled z -transformed centrality, as raw measures of centrality are not directly comparable across differently sized networks. One insect-plant and three avian-plant networks were too small to create realistic null distributions, and were subsequently excluded for local centrality analyses.

Statistical Analyses

We examined the role of species range size and niche breadth on species centrality through Bayesian generalized linear models implemented in Stan (Carpenter et al., 2017) through the *R* package *brms* (Bürkner, 2017). In order to facilitate comparison and meaningful interpretation of interaction effects, both geographic range sizes and niche breadth values were log-transformed and then rescaled to a mean of 0

and standard deviation of 1. All models therefore shared the generalized form of predicting a given species' centrality as a function of geographic range size, niche breadth, and the interaction between these predictors. Niche breadth or geographic range size may exert fundamentally different effects on the centrality of flowering plant species compared to floral visitors. As such, we analyze node types (floral visitor or flowering plant) separately for each network or set of networks, effectively repeating our modeling approach for all factorial combinations of: scale (metaweb vs local webs), network type (insect-plant vs avian-plant), and interactor type (floral visitor vs flowering plant). Accounting for our two alternative centrality measures at the metaweb scale (measured and z -transformed closeness centrality), this resulted in 12 models total. We use vague priors centered on 0 for all slope and intercept coefficients. All models were fit using 5 Markov chain Monte-Carlo simulations, each with 5,000 iterations following a 1,000 iteration burn-in period; posterior samples were sampled without thinning (Link & Eaton, 2012). Simulation convergence was checked through visual inspection of traceplots. Gelman-Rubin statistics fell below 1.01 for all parameters across all models. For slope terms we sampled the resultant posterior distributions and approximated the two-sided posterior probability (POP_2) as twice the proportion of posterior samples falling above or below 0 (whichever is smallest), a Bayesian analog to traditional frequentist p-values (Shi & Yin, 2021). Relationships between fit slope values across models are presented as pairs plots in supplemental Figures B.7-B.12.

For the subset of species which occurred across multiple sites, we further investigated the role interaction turnover may play in determining species centrality in global metawebs. As both the degree of turnover and number of sites at which a species occurs may have nonlinear effects on metaweb centrality, we investigate these factors through a generalized additive modeling (GAM) approach. For each node that oc-

curred in multiple sites (1,166 species), we calculated the Simpson’s dissimilarity of interaction partner sets across all sites in which they occurred (Baselga & Leprieur, 2015), as implemented in the *R* package **betapart** (Baselga & Orme, 2012). We created GAM models predicting metaweb centrality (either raw values or *z*-transformed, scaled to a mean of 0 and standard deviation of 1 to facilitate comparisons across models) as a function of the tensor product of interaction turnover and number of local sites. We investigate the *p*-value of tensor terms, as well as compare estimated degrees of freedom to maximum allowable degrees of freedom (*k'*) and visually inspected model residuals to assess model fit.

Results

Species Occurrences and Interactions

Species’ geographic range sizes in insect-plant networks varied widely, ranging from 2,232 km² (*Chaetanthera lycopodioides*, a range-restricted montane Aster endemic to Chile and Argentina Arroyo et al., 2007) to over 119 million km² (*Sonchus oleraceus*, a cosmopolitan weed native to Eurasia), with a median value of 3,749,391 km². Geographic distributions of species in avian-plant visitation networks ranged from 2,518 km² (*Barbacenia longiflora*, a monocot forb restricted to Minas Gerais, Brazil) to over 85 million km² (*Prunella vulgaris*, a cosmopolitan invasive weed), with a median value of 1,536,647 km². Across sites, geographic and environmental range size were positively correlated ($r=0.667$, $p < 0.001$). Typical of large-scale interaction metawebs, realized connectance (the percentage of realized links out of the total possible number of interactions) was low (Jordano, 1987). At the metaweb scale, only 0.32% and 2.8% of potential links were realized in insect and avian interaction metawebs, respectively. At the site-level, average connectance of insect-plant networks was 0.194, whereas average avian-plant network connectance was 0.356.

Global Metawebs

At the metaweb scale, geographic range size was positively associated with species centrality of all node types across both insect and avian networks (Figure 3.2). However, once accounting for potential co-occurrence through z -transformation, geographic range size showed no association with metaweb centrality for avian or insect floral visitors, or insect-visited plant species, and instead showed a negative association with the centrality of avian-visited plant species ($POP_2 = 0.020$). Niche breadth had a negative impact on centrality for plants and insects within the insect-plant metaweb after controlling for geographic range size ($POP_2 < 0.001$, $POP_2 = 0.028$). After z -transformation, niche breadth instead had a positive impact only on plant centrality in both insect and avian metawebs ($POP_2 < 0.001$ across both networks) while showing no effect for floral visitors. There was no significant interaction effect between geographic range size and niche breadth on unstandardized metaweb centrality across node types. However, after z -transformation, plants in both network types showed positive interaction effects ($POP_2 < 0.001$ and $POP_2 = 0.009$ for insect and bird-visited plants, respectively).

For species that occurred across multiple sites, interaction turnover varied from 0 (complete conservation or nestedness of interactions across sites) to 1 (no overlap in interactions across sites), with a mean value of 0.644. The tensor product of interaction turnover and number of sites was statistically significant in predicting both raw and z -transformed metaweb centrality ($p < 0.001$), but varied widely in the amount of variance explained. For the GAM model predicting measured metaweb centrality $R^2 = 0.231$, with interaction turnover exhibiting a positive partial effect across much of the realized parameter space, particularly for species with the highest centrality values (Figure 3.3). After z -transformation, the tensor product of turnover and number of sites still exhibited a statistically significant effect ($p < 0.001$) but

explained little of the overall variation ($R^2 = 0.054$). Taken together these results further suggest that positive relationships between species' geographic range size and measured metaweb centrality are largely driven by interaction turnover across sites.

Local Webs

Average species centrality in local floral visitation networks was positively correlated with their centrality in their respective global metawebs ($r = 0.480$ and $p < 0.001$ for insect webs, while $r = 0.079$ and $p = 0.035$ for avian webs). Species that occurred at multiple sites showed variable centrality across their respective local webs (Figure B.3). Geographic range size was negatively related to the local centrality of insect-visited plant species (Figure 3.4, $POP_2 = 0.018$). Conversely, niche breadth was positively related to the average local centrality of nodes within insect networks ($POP_2 < 0.001$ for plants, $POP_2 = 0.004$ for insects). Insect-visited plants also exhibited a positive interaction term between geographic range size and niche breadth on average local centrality ($POP_2 = 0.022$). We failed to find any significant effect of range size, niche breadth, or their interaction on the local centrality of species within avian networks.

Discussion

Generally we find species' geographic range size and niche breadth are more clearly linked to their centrality in aggregate plant-pollinator metawebs than within local interaction networks. The positive impact of geographic range size on metaweb centrality we observed across all node classes may be caused by a number of potential drivers. Species with broad geographic ranges are likely to co-occur with more potential interaction partners across their range (Figure B.5). In this case, even if species are relatively peripheral across local networks, they may be relatively central at the metaweb level if interaction turnover is high across their range (Martins et al., 2025).

Our results support this interpretation, as while species with large geographic ranges tended to be more central in the global metaweb, they did not tend to be more central than null expectations based on their connections across all local webs (Figure 3.2). Additionally, generalized additive models indicated a large partial effect of interaction turnover on measured metaweb centrality, but this effect largely disappeared once these centrality values were z -transformed (Figure 3.3). Together, these patterns suggest that, at the metaweb scale, species centrality is shaped more by neutral processes—namely the accumulation of interactions across space—than by the specific identity of those interactions.

Contrary to our expectations, standardized centrality was negatively related to geographic range size of plants in avian-plant networks after accounting for niche breadth. This pattern may indicate slower accumulation of interactions across space (lower interaction turnover) than otherwise expected. This pattern may be driven by wide-ranging, disturbance-adapted species that occur in many networks but are less likely to form interactions with narrowly distributed specialists. Nectarivory has evolved numerous times within birds, but contemporary biogeographical distributions of avian nectarivore diversity are clustered in the neotropics and parts of Australasia (Kissling et al., 2012). This spatial clustering of potential partner diversity may drive lower rates of interaction turnover of avian-pollinated plants across large spatial scales.

After z -transformation, metaweb centrality was positively related to niche breadth in plants of both metawebs, but was unrelated in floral visitors. If plant species are limited in their dispersal by the availability of mutualists (Duffy & Johnson, 2017; Warren & Bradford, 2014), species with broad ranges of potential partners may be able to exploit a more climatically diverse set of environments. The availability or efficacy of mutualists may increase the abiotic tolerances of some species (Fowler

et al., 2023; Warren & Bradford, 2014), effectively expanding their climatic niche breadth. Batstone et al. (2018) outline how environmental variability may favor interaction generalism by altering the availability or efficacy of mutualists (Batstone et al., 2018). Similar mechanisms may occur across an organism’s range; plants with climatically diverse ranges may experience parallel changes in partner availability or efficacy, promoting interaction generalism (Hurtado et al., 2024). Alternatively, in systems characterized by widespread generalism, plant species with broad niche breadths may participate in more interactions as they co-occur with a broader complement of potential pollinators. Understanding the mechanisms of this interplay between a species’ climatic niche breadth and its position within local and global networks warrants further study.

In contrast to global metawebs, both geographic range and niche breadth show a reduced effect on species centrality at the local scale. In insect networks, average local centrality was positively related to niche breadth across both node types, but negatively related to geographic range size in insect-visited plant species. We also detected a positive interaction between range size and niche breadth on the centrality of plant species in insect-floral visitation networks, indicating that their combined influence shapes local structure of these networks (Figure 3.4). Species with broad geographic ranges tended to be more central when they also had broader climatic niches, whereas species with narrow ranges were more central when they exhibited broader niche breadths. This suggests multiple ecological pathways or strategies that may result in high centrality in local networks. Widely distributed species with broad abiotic tolerances may be relatively generalized in their interaction patterns at local scales, increasing centrality (Hurtado et al., 2024). Alternatively, narrowly distributed climatic specialists may also be highly central, potentially through achieving high abundance in restricted environments (Suárez et al., 2023). The functional

consequences of a species' centrality within a network may vary greatly across local, regional, and global scales, and further investigation of how centrality scales across levels of organization is of great interest. In our study, networks are either analyzed at the site level, or aggregated into global metawebs. Fully understanding the processes of scaling from local to global requires hierarchical sampling designs replicated across large geographic extents. While understanding the scaling of network properties across space, environments, and time has received a great amount of conceptual attention in recent years (Burkle & Alarcón, 2011; Morales & Vázquez, 2008; Moulattlet et al., 2023; Tylianakis et al., 2008), the logistical challenges of research across these scales have limited our ability to fully explore these questions in some cases.

The emergent consequences of a species' centrality may ultimately differ across local to global scales. For example, at local scales the perturbation of highly central species may propagate farther or more strongly through a mutualistic community than the same perturbation applied to more peripheral species (Cagua et al., 2019b; Delmas et al., 2019; Martins et al., 2024). Across larger scales, highly central species may play key roles in allowing mutualistic metacommunities to persist (Filotas et al., 2010), especially in the face of habitat heterogeneity or disturbance (Prakash & de Roos, 2004). Patterns of interaction turnover across space are closely linked to both the causes and consequences of centrality scaling relationships. While the temporal turnover of mutualistic interactions is fairly well appreciated in phenological contexts (Chávez-González et al., 2020; Olesen et al., 2008), empirical patterns of interaction turnover across space, habitats, and environments warrant further study (Tylianakis & Morris, 2017).

Understanding the factors that determine species positions in mutualistic networks across scales is a fundamental question in network ecology. Our study identifies geographic range size as a key determinant of species centrality in floral visitation

metawebs, largely driven by the spatial accumulation of interactions across sites. At the local scale, we also find that climatic niche breadth interacts with geographic range size to predict plant centrality in insect-plant networks, highlighting the multiple pathways through which species may become highly central at the site level. Together, these results emphasize the need for a multiscale perspective to understand the drivers and consequences of changes in ecological network properties across environments.

Data accessibility: Data and *R* code necessary to recreate this analysis are available on on figshare at <https://figshare.com/s/ab0b46493fd69cf4a2c8>

Figures

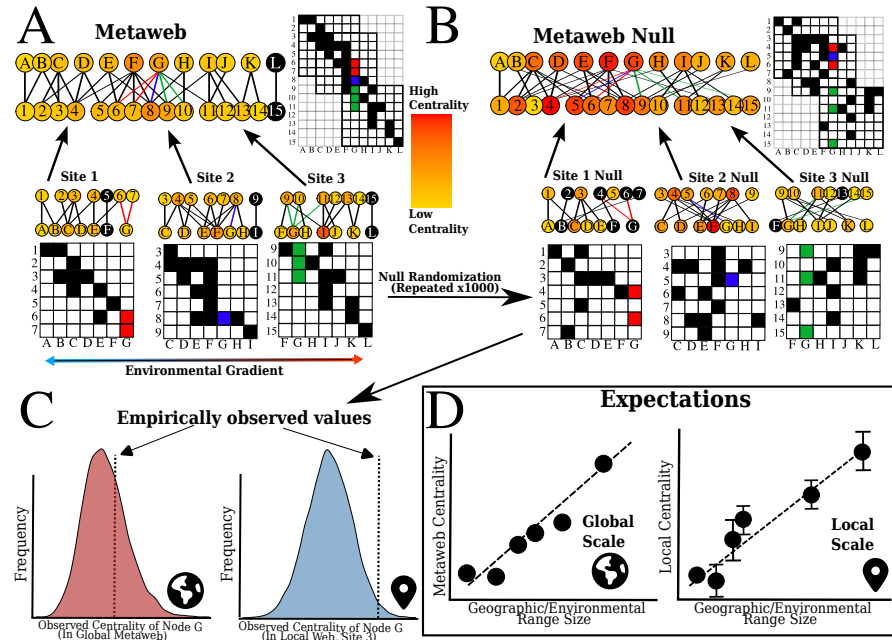


Figure 3.1 Local interaction networks are embedded along natural environmental gradients (A), with species occurring in one or more sites and contributing interactions to a global metaweb, defined as the union of all observed interactions across sites. At the metaweb scale, species with broad geographic ranges or wide abiotic tolerances may accumulate interactions across multiple sites. At the local scale, such species may also be more central if interaction generalism covaries with environmental or habitat generalism. Species centrality can be quantified within individual site-level networks as well as in the global metaweb (node color). However, metaweb centrality is jointly influenced by the identity of interactions and by interaction turnover across sites. To isolate these effects, we implement a site-level null randomization (B) that shuffles interaction identities within each site while preserving the total number of interactions per species. Repeating this procedure generates null distributions of expected centrality values for each species at both local and metaweb scales (C), given the constraints of observed local network structure. Empirical centrality values can then be standardized (e.g., z -transformed) relative to these expectations, allowing scale-dependent effects of range size on network centrality to be evaluated (D).

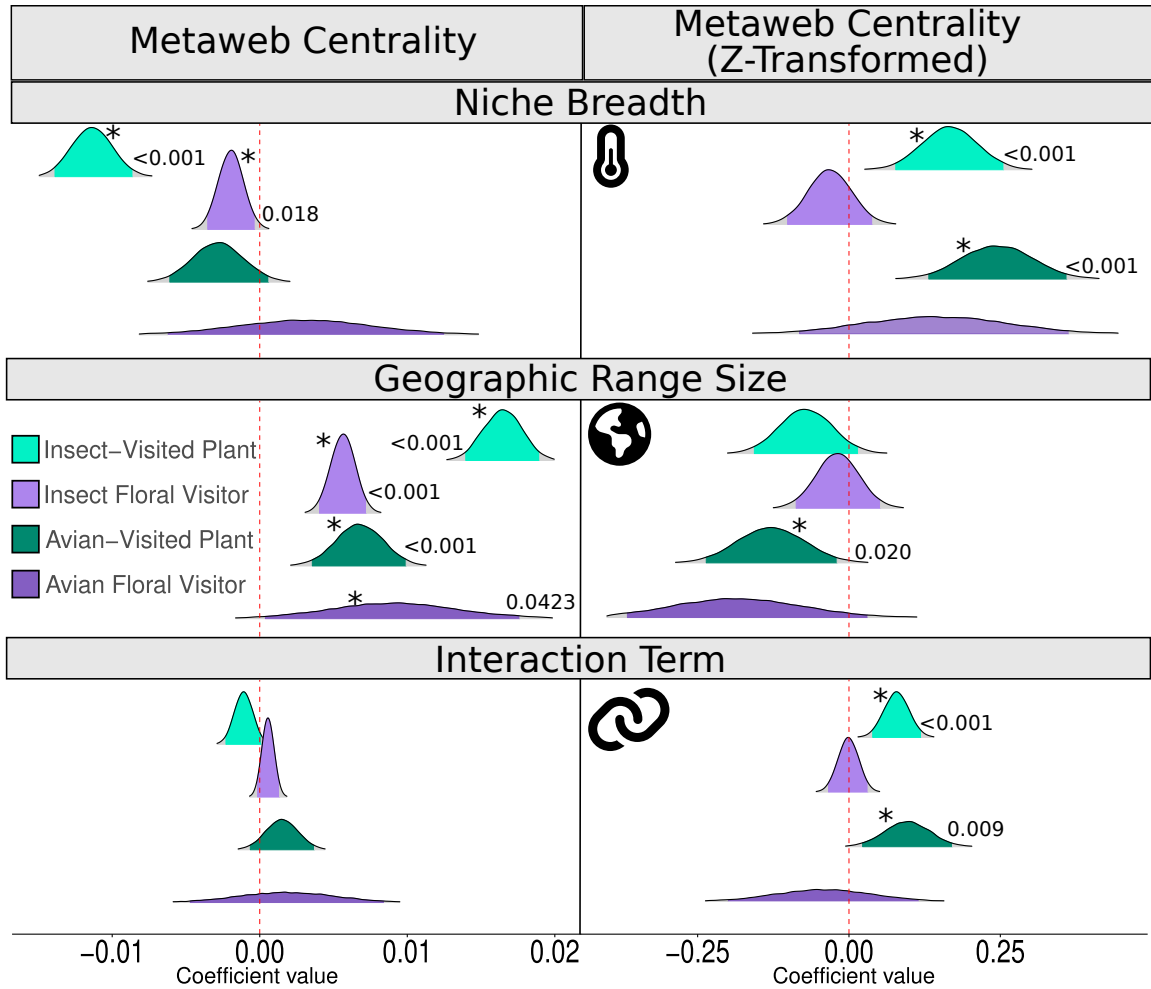


Figure 3.2 Sampled posterior slope densities for multivariate Bayesian generalized linear models of species centrality in the global metaweb. Colored areas represent 95% credible intervals of slope terms of the effect of niche breadth (top row, thermometer icon), geographic range size (center row, globe icon), and their interaction term (bottom row, chain icon). Numbers and asterisks indicate approximate two-sided posterior probability < 0.05. Left columns represent models of measured closeness centrality as the response variable; right column represents models predicting centrality values z -transformed based on null randomization. After accounting for niche breadth, geographic range size has a strong positive effect on species centrality, but this effect disappears when z -standardized. In plant species, both niche breadth and the interaction between niche breadth and geographic range size positively impact z -transformed metaweb centrality.

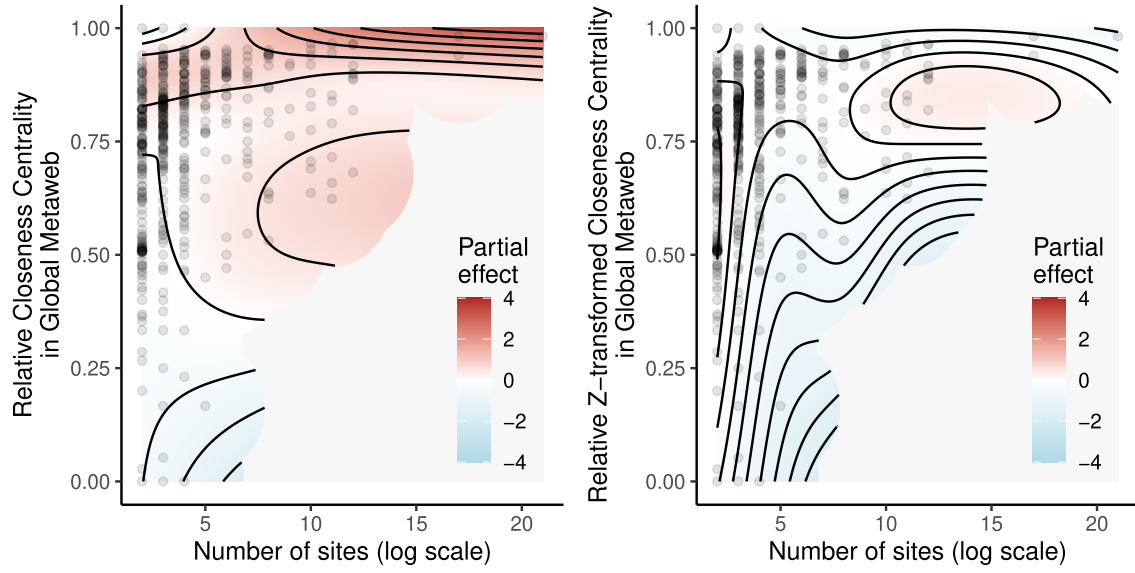


Figure 3.3 Partial effects of interaction turnover on measured metaweb centrality (left) or z -transformed metaweb centrality (right) response values and number of unique local sites, according to fit tensor product GAMs. For comparison, both responses are rescaled to a mean of 0 and standard deviation of one. Across the observed range of centrality values, the tensor product of interaction turnover and number of local sites occupied by a species had a positive effect on global metaweb centrality ($p < 0.001$, $R^2 = 0.231$), especially for the most highly central species. In contrast, after z -transformation while the tensor product of turnover and number of local sites had a statistically significant negative effect, this relationship explained less than 6% of variation in metaweb centrality ($p < 0.001$, $R^2 = 0.054$).

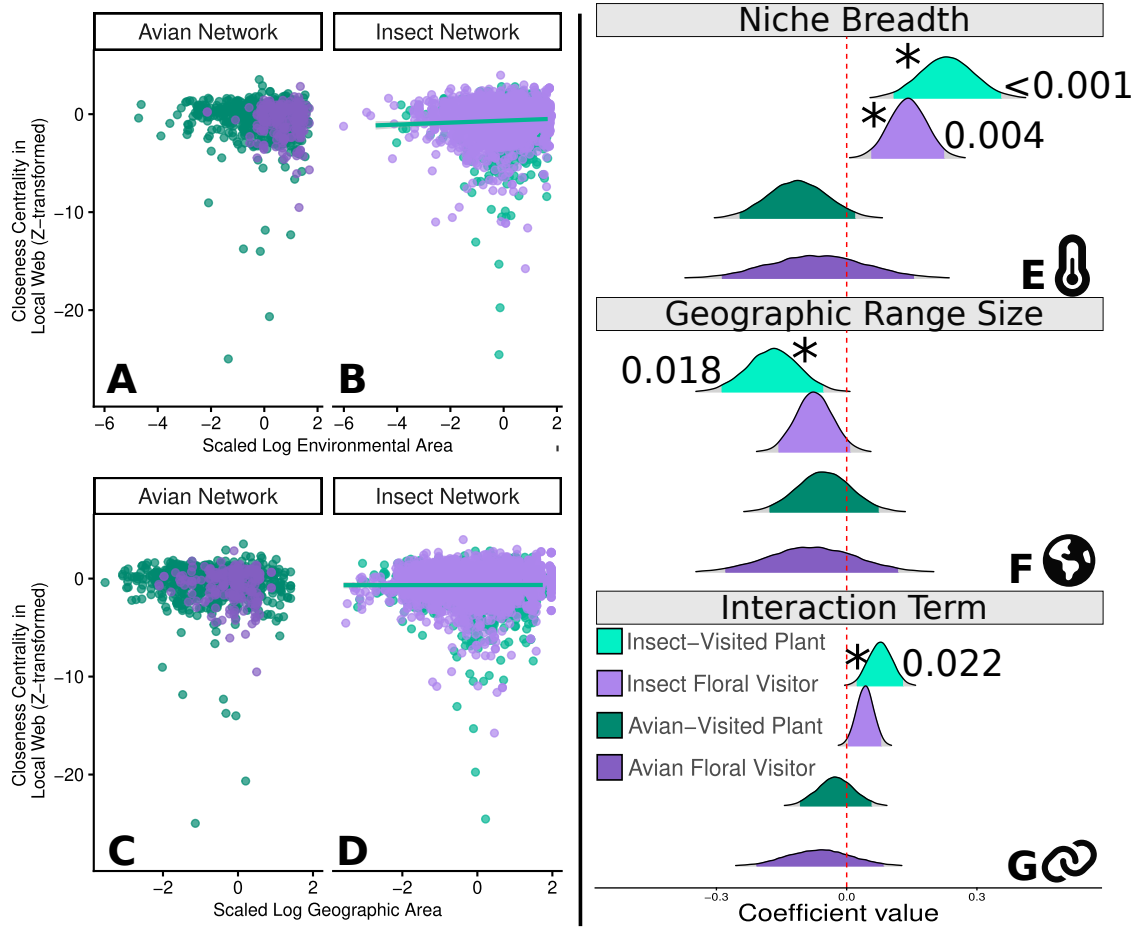


Figure 3.4 Relationships between average z-standardized centrality in local networks, geographic range size, and niche breadth. Species-level closeness centrality (z -transformed) in local interaction networks as a function of geographic range size (A,B) or environmental niche breadth (C,D). Panels (E-F) show sampled posterior slope densities for multivariate Bayesian generalized linear models of average species z -transformed closeness centrality in local metawebs. Colored areas represent 95% credible intervals; numbers and asterisks indicate approximate two-sided posterior probability < 0.05 . Niche breadth had a positive effect on the local centrality only of both node types within insect networks (E). Additionally, we also observe a positive interaction between niche breadth and geographic range size on standardized centrality for these node types as well (G).

CHAPTER 4

ISLAND BIOGEOGRAPHY OF HOST-HELMINTH INTERACTION NETWORKS

Abstract

Island biogeography theory has been widely applied to predict properties of species assemblages across islands, but its relevance for species interaction networks remains less clear. Here, we take a macroecological approach to investigate how bipartite interaction networks between free-living hosts and helminth parasites vary across gradients of island area and isolation. Using the current largest published global collection of host-helminth interaction records, we tested the relationship between island characteristics and host richness, helminth richness, and interaction nestedness. We found that host and helminth richness were positively associated with island area and mainland connectivity as measured by surrounding landmass proportion, though unrelated to distance to the nearest mainland. In contrast, nestedness showed only a weak positive relationship with island area and was unrelated to mainland connectivity. These results suggest that island characteristics shape the size of host-parasite assemblages more predictably than the organization of interactions within them. Our findings extend island biogeographic perspectives to antagonistic interaction networks and motivate future work examining how host-mediated colonization and parasite specialization contribute to the assembly of antagonistic network structure across space.

Introduction

Island biogeography theory posits that island area and distance to the mainland combine to constrain species richness on islands through effects on extinction and colonization, respectively (MacArthur & Wilson, 2001). Despite being developed well over fifty years ago, it remains an active research area, with numerous open questions (Patiño et al., 2017). This is perhaps a result of the ways that island biogeography theory can be used to inform other ecological relationships. For instance, island biogeography theory is quite closely linked to the examination of species-area relationships (Losos & Schluter, 2000; Peay et al., 2007). Further, island biogeography theory is a simple base on which to build, as seen by extensions to species functional trait variation (Jacquet et al., 2017), predator-prey interactions (Gravel et al., 2011), and compositional differences between communities (Lu et al., 2018). It is clear that island biogeography theory has the potential to inform trophic and community-scale phenomena.

The increasing availability of ecological data has encouraged the study of large-scale processes such as island biogeography theory. This data availability has also promoted the study of species interaction networks (Delmas et al., 2019; Dormann et al., 2017), which characterize the interactions between two (or more) trophic groups (e.g., plant species and their pollinators). This has permitted examinations into how spatial and environmental gradients influence the structure of these species interaction networks (Tylianakis & Morris, 2017), with structure defined using measures such as nestedness and modularity (Dalsgaard et al., 2013; Dormann et al., 2017), which capture the tendency of species interactions to form proper subsets of species with more interactions (nestedness) or cliques of interacting species (modularity). Regardless of the measure used to capture aspects of species interactions, there is mounting evidence that species interactions follow the same patterns as species richness with

regard to island area and isolation (Dormann et al., 2017; Sugiura, 2010; Tylianakis & Morris, 2017). For instance, Traveset et al. (2016) compared mainland and island plant-pollinator networks, finding that isolated oceanic islands had the lowest species richness – in agreement with predictions from classic island biogeography theory – and the lowest interaction diversity and highest plant niche overlap. That is, as the richness of plants and pollinators decreased with isolation, so did the structure of the interactions between plant and pollinator species. Some also exists to indicate that pollination networks may be more modular on islands (Dalsgaard et al., 2013). But how do we interpret the effects of island isolation and area on species interactions?

Classical biogeography studies posit that the species richness of oceanic islands is largely determined by the equilibrium between species colonization and extinction rates. Islands closer to mainland sources are colonized by new species at a higher rate, while islands with larger areas exhibit lower extinction rates. However, the interpretation of the influence of island isolation and area on the structure of species interaction networks must be different. That is, the richness of interacting species, even if driven by a balance between colonization and extinction rates, should not necessarily influence the overall structure of species interactions. For instance, if species interactions are conserved between mainland and island populations (Carstensen et al., 2018), then differences between mainland and island species interaction networks should be minimal. However, there are other reasons why island area could influence network structure. Larger islands tend to have more diverse habitat types than smaller islands, and species habitat preferences can contribute to modularity of interaction networks by reducing niche overlap between groups of habitat specialists (Santos et al., 2016). Meanwhile, island isolation, as noted above, is likely unrelated to network structure, provided that species interactions are not overly plastic (i.e., species cannot readily change the set of species they interact with), or that species

dispersal ability is directly related to specialization (i.e., good dispersers interact with many species).

It may be infeasible to provide tests of the underlying mechanisms which drive relationships between island isolation or area and properties of species interaction networks. However, demonstrations of the existence of the scaling relationships themselves are useful in identifying groups of species or types of interactions which modify the strength or directionality of effect of island characteristics on species interactions. For instance, while studies examining mutualistic network structure across islands are becoming increasingly common (Dalsgaard et al., 2013; Sugiura, 2010; Traveset et al., 2016; Trøjelsgaard et al., 2013), comparable studies of antagonistic networks have lagged behind (but see Gravel et al. (2011)). However, antagonistic and mutualistic networks often have quite different structural properties, potentially owing to the intrinsic difference in the effect of the species interaction (i.e., the cost of parasitism and the benefits of mutualism). Here, we examine the relationship between host-parasite network structure and island isolation and size, using host-helminth interaction data from 61 islands. First, we examined the relationship between species richness and island isolation and size, in accordance with traditional island biogeography theory. Second, we examined how a key measure of ecological network structure, interaction nestedness, scales with island characteristics. Overall, our findings suggest that while the diversity of host-parasite assemblages varies predictably with island characteristics, the structure of their interaction networks shows a more limited association. Reconciling the differences between mutualistic and antagonistic network structures among islands may provide insight into how island isolation and area are mechanistically related to species interactions and network structure.

Methods

Host - helminth interaction data

Records of helminth parasite occurrence on host species were obtained from the parasite database of the London Natural History Museum (LNHM) (Gibson et al., 2022), and accessed programmatically using the *helminthR* package (Dallas, 2016). These data currently represent one of the largest sources of host-parasite interaction data (Dallas et al., 2018), despite being restricted to helminth parasites, including Platyhelminthes (trematodes and cestodes), Acanthocephalans, and Nematodes (Gibson et al., 2022). We started by identifying a total of 62 islands for which host-helminth interaction data were recorded (Figure 4.1). After filtering out host and helminth species not identified to the species level, this comprised a total of 7294 host-helminth species pairs across 2651 helminth species and 1980 host species.

Island isolation and size measurement

Data characterizing the geography of islands were retrieved from (Weigelt et al., 2013), including island area (km²). We measured island isolation through distance to the nearest mainland (km), and surrounding landmass proportion (SLMP), calculated as the proportion of land area within 100, 1000, and 10,000 km buffers. This metric accounts for both spatial variation in mainland connection as well as nearby island connections, and has been shown to be more robust than distance per se in explaining insular plant diversity (Weigelt & Kreft, 2013). To roughly account for historical connectivity, we also extracted whether an island had terrestrial connections to the mainland during the last glacial maximum (GMMC). Records within the LNHM dataset that were georeferenced to an island chain or series (e.g. Trinidad & Tobago) were assigned to the largest island within the group (e.g. Trinidad). Data from (Weigelt et al., 2013) were available for all but one island in our data (South Shetlands), an Antarctic island with only seabird and pinniped infection records,

which we excluded from our analysis.

Quantifying host-helminth network structure

We related island characteristics to both species richness estimates and to measures of interaction network structure. Measures of species richness for both hosts and helminths were obtained from the London Natural History Museum data, as described above. To quantify network structure, we estimated interaction nestedness, a commonly measured aspect of network structure found to be related to network structural stability (Dallas & Cornelius, 2015; Thébault & Fontaine, 2010; Valdovinos, 2019) and resilience (Suweis et al., 2013; Thébault & Fontaine, 2010). However, some measures of nestedness may be related to network size (number of interacting species) (Bascompte et al., 2003; Nielsen & Bascompte, 2007), which creates an odd circularity where a significant effect of island area or isolation on species richness would automatically manifest in an effect on network structure.

One method for addressing the difficulty in comparing network structure measures between networks is to measure the difference between the empirical statistic and a distribution of the statistic obtained from null model randomization. This procedure has recently been criticized as not removing the effects of network size (Chagnon, 2015), specifically for nestedness (Song et al., 2017; Strona & Fattorini, 2014). Instead, we opted to use $NODF_c$, a normalized version of the classic nestedness metric based on overlap and decreasing fill proposed by Song et al. (2017). This value is calculated as $NODF_c = NODF_n / (C \cdot \log(S))$, where $NODF_n$ represents the relative observed connectance of a given graph given its maximum possible value ($NODF / \max(NODF)$), and C and S represent the connectance and the geometric mean of the number of nodes of each class, respectively. Song et al. (2017) show that this normalized metric is insensitive to network size, and thus appropriate for comparing network structure across biogeographic scales. Computation of $NODF_c$ was

performed using a hill-climbing algorithm as implemented in the *maxNODF* package (Hoeppke & Simmons, 2021). For some networks with a small number of interacting species, the set of possible network configurations is too small to make meaningful inference; as such we excluded islands with five or less species within their networks ($n = 12$). Across the remaining 47 islands, most contained networks with multiple disconnected components (modules of species with no direct linkages that connected them, $n = 35$). As NODF is not directly measurable in disconnected networks, for networks containing multiple connected components we added the minimum number of links necessary to create a single component when calculating the maximum potential NODF value (Hoeppke & Simmons, 2021). We demonstrate in the Supplemental Materials how the use of network structure measures that contain the influence of network size would lead to incorrect conclusions regarding the influence of island area and isolation on network structure.

Statistical Analysis

Relationships between species richness and island characteristics were modeled as multivariate negative binomial regressions as implemented in the *MASS* package (Venables & Ripley, 2002). We utilized negative binomial regressions as both Poisson and Quasi-poisson regression models exhibited significant overdispersion (see Supplemental Materials). As our alternative metrics of isolation (log distance to mainland and SLMP) were significantly negatively correlated ($\rho = -0.768$, $p = 1.94\text{e-}12$), we did not include both in the same model and instead used AIC to compare across metrics. The impact of island characteristics on $NODF_c$ was modeled as a multivariate linear model using a Gaussian error distribution and logit link. Model fits were inspected through analysis of deviance and visual inspection of standard model diagnostic visualizations. Models of the effect of historical mainland connectivity (GMCC) on diversity and network characteristics were fit separately under the same framework

as described above, as well as through a Wilcoxon signed-rank test.

Results

Species richness and island characteristics

Island biogeography theory predicts that species richness will be negatively related to distance from a mainland (as a result of low dispersal success to distant islands), and positively related to island area (larger islands support more "stable" populations). We found evidence for an influence of area on species richness of both hosts ($\beta = 0.419$, $p = 2.43\text{e-}12$) and helminths ($\beta = 0.391$, $p < 1.42\text{e-}10$). Interestingly, host and helminth richness increased at nearly the same rate with island area. In contrast to the influence of island area, we failed to detect any influence of distance to the mainland on host ($\rho = -0.189$, $p = 0.182$) or helminth ($\beta = -0.221$, $p = 0.116$) species richness (Figure 4.2). However, when accounting for island connectivity through SLMP, we detected a positive relationship between surrounding land area and richness of both hosts ($\beta = 1.54$, $p = 0.00161$) and helminths ($\beta = 1.74$, $p = 0.000262$).

Network structure and island characteristics

Many measures of nestedness are sensitive to the effects of network size. In this case, network size is the species richness of host and parasite species in the network, which we found to be related to island area and SLMP (but not distance to the mainland; Figure 4.2). Using a nestedness measure that is related to network size would therefore lead to the conclusion that network structure follows a similar gradient as species richness (see Supplemental Material). Here, we use a measure of nestedness that is insensitive to network size, $NODF_c$. Using this measure, we detect a positive relationship between island area and interaction nestedness; larger islands had more nested host-helminth networks ($\beta = 0.325$, $p = 0.0473$, Figure 4.3). We failed to detect any relationship between network nestedness and maximum elevation, distance to

mainland, or SLMP.

Prehistoric Connectivity

Independent of contemporary island area or distance to mainland, historical land-bridge connections may alter species assemblages or network characteristics by influencing dispersal processes. We found that islands that had terrestrial mainland connections during the last glacial maximum tended to have higher host ($\beta = 1.61$, $p = 9.58e-05$) and helminth diversity ($\beta = 1.59$, $p = 8.11e-05$, Figure 4.4). Despite differences in richness, interaction networks did not differ in nestedness across GMMC status ($p = 0.71$).

Discussion

Overall, the relationships we observed between island characteristics and the species richness of both hosts and helminth parasites are broadly consistent with predictions from classical island biogeography theory for free-living species, but nestedness showed a more limited pattern, responding weakly to island area and not to either measure of island connectivity. This suggests that island characteristics shape the size of host-parasite assemblages more predictably than they shape the organization of interactions within those assemblages. The framework of island biogeography predicts higher species richness on larger islands, in part because larger areas can support larger populations and thus lower extinction rates (MacArthur & Wilson, 2001), as well as greater habitat heterogeneity that accommodates a wider diversity of species (Hortal et al., 2009; MacDonald et al., 2018; Triantis et al., 2008). Consistent with these expectations, we find a significant positive correlation between island area and the species richness of both hosts and helminths (Figure 4.2).

Classical theory also predicts reduced diversity on islands that are more isolated from mainland source pools due to lower colonization rates (MacArthur & Wilson,

2001). While we do not detect a direct effect of distance to the mainland on species richness, we do observe a significant positive relationship between species richness and island connectivity, as measured by surrounding landmass proportion (SLMP), for both hosts and helminth parasites. Although distance and connectivity are related metrics, SLMP incorporates both proximity and the size and configuration of surrounding landmasses (including large neighboring islands), and may therefore provide a more accurate proxy for colonization potential (Barreto et al., 2021; Weigelt et al., 2013; Weigelt & Kreft, 2013).

Species interaction networks on larger islands tend to be more nested than those on smaller islands (Figure 4.3), but this effect was generally weak ($R^2 = 0.086$). In diverse communities, patterns of asymmetric specialization (i.e., specialist parasites tend to interact with hosts with high parasite richness, while the parasite communities of hosts with few associated parasites are dominated by generalist species) result in interaction nestedness (Lima Jr et al., 2012; Vázquez, Poulin, et al., 2005). Significant interaction nestedness has been observed in a number of parasitic systems (Guerra et al., 2016; Krasnov et al., 2011; Lima Jr et al., 2012), and island host-helminth networks may be smaller subsets of more regionally nested source networks. If so, larger islands may more closely reflect the structural properties of these regional pools, while the comparatively depauperate networks of smaller islands may not fully preserve this nested architecture. Alternatively, the relationship between island area and nestedness may reflect non-random assembly if colonization or extinction probabilities are associated with host or parasite specialization. For example, parasites with broad host ranges, high prevalence, or those associated with abundant or highly mobile hosts may be more likely to colonize and persist on islands, while specialist parasites may be disproportionately absent from smaller islands.

While some attention has been paid to the island biogeography of mutualistic networks (Dalsgaard et al., 2013; Nogales et al., 2024; Sugiura, 2010; Traveset et al., 2016), the processes underpinning them are fundamentally different. Among these is the role of co-colonization. In many mutualistic networks, potential interaction partners may disperse and establish on islands relatively independently. In contrast, helminths generally have much lower dispersal capacity than their free-living hosts (Poulin, 2004). As such, the dispersal of parasites to new islands is largely dependent on patterns of host dispersal (Bataille et al., 2017; Font, 2003). Parasites that have high prevalence of infection or broad host ranges may be more likely to colonize new islands. For example, parasites with sufficiently broad ranges may be able to disperse to new areas via transient hosts, even if those species do not establish persistent populations on a given island (Ellis et al. (2023) give an example with avian malaria). Alternatively, specialist species with close host associations may be more likely to be retained during colonizations (Schatz & Park, 2023). Thus, generalist parasites may have more opportunities to colonize new islands, while some specialists may be more likely to persist when their hosts successfully establish, making the relationship between specialization and colonization potential non-linear. These linkages between host range and colonization potential may induce a biotic filter that ultimately affects network structural properties. Understanding how parasite colonization dynamics scale up to affect resulting species interaction networks across gradients is an outstanding question.

The mechanisms underlying island-area effects on host-helminth network structure remain difficult to resolve in part because parasite occurrence and interaction data may be shaped by both biological features and sampling limitations (Llopis-Belenguer et al., 2023; Nielsen & Bascompte, 2007). The sampling of parasite communities is inherently difficult, and is limited by the availability of survey effort and taxonom-

ic expertise. Additionally, for many helminth taxa, we have incomplete information about which hosts they are capable of infecting in nature, making it difficult to determine whether unobserved interactions represent true absences or merely incomplete sampling (Farrell et al., 2022; Poulin et al., 2016; Walker et al., 2017). As a result, the networks analyzed here are unlikely to represent complete characterizations of island host-helminth communities, especially on very small islands with limited sampling effort or very large islands with highly diverse host and parasite faunas. More complete and effort-standardized surveys of host and parasite communities, paired with better characterization of helminth host ranges through surveillance and experimental infection studies, may help clarify the mechanisms determining parasite colonization, persistence, and network structure across biogeographical gradients.

However, in their current form, large-data repositories like the London Natural History Museum (Gibson et al., 2022) still provide the opportunity to investigate the structure of parasitic communities at macroecological scales, and can motivate these further data collection efforts. To our knowledge, ours is the first study to examine the structural properties of host-parasite networks in an island biogeography context. The increasing availability of open species interaction data spanning wide geographic, taxonomic, and temporal scales provides new opportunities to better understand large-scale determinants of network structural properties and their consequences for community assembly. By looking across geographic and environmental gradients, we show that island characteristics may shape the size of host-parasite assemblages more predictably than the organization of interactions within them, while also highlighting the need to understand how host-mediated colonization shapes antagonistic network structure across environments.

Figures

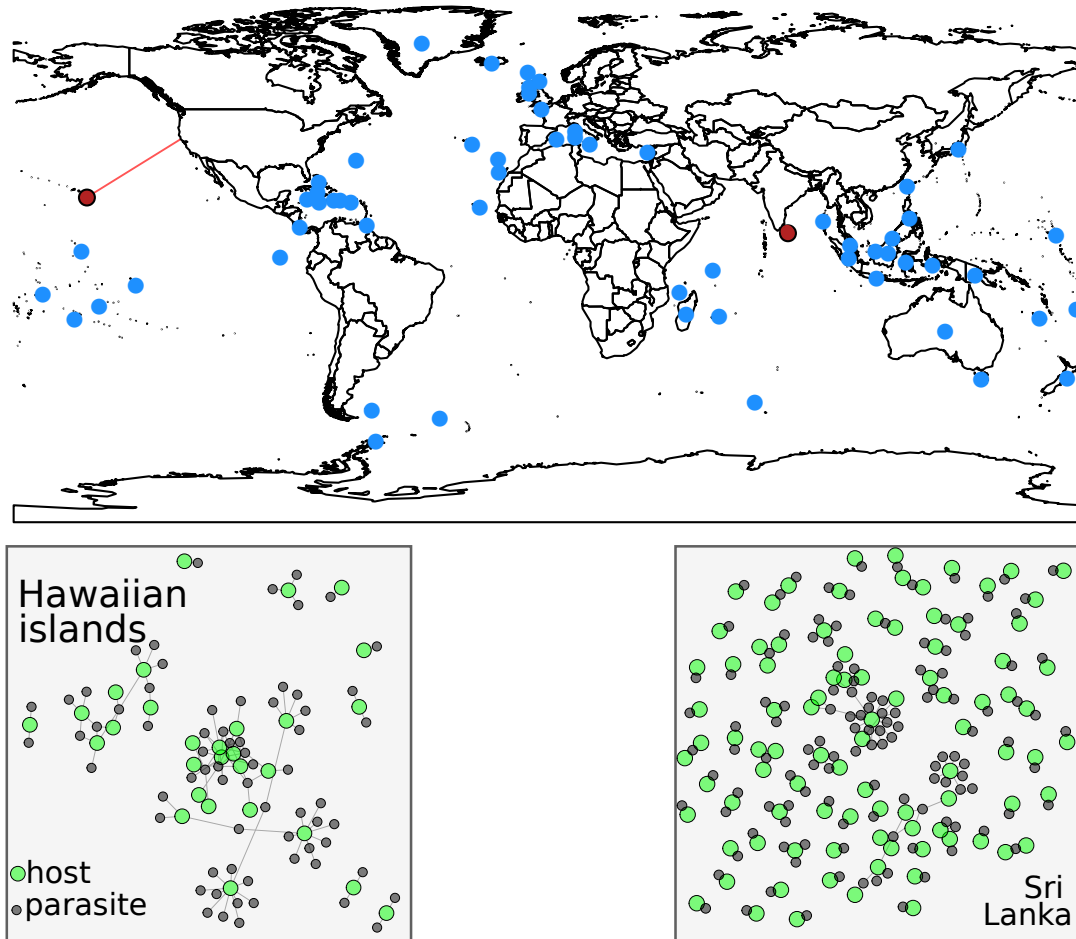


Figure 4.1 Both distance to mainland and island area may influence species richness and host-helminth network structure. Islands considered in the analysis are in blue, with two example networks from the Hawaiian islands and Sri Lanka highlighted in red. Hawaii is much smaller and more distant from the mainland, and also has lower species richness relative to Sri Lanka.

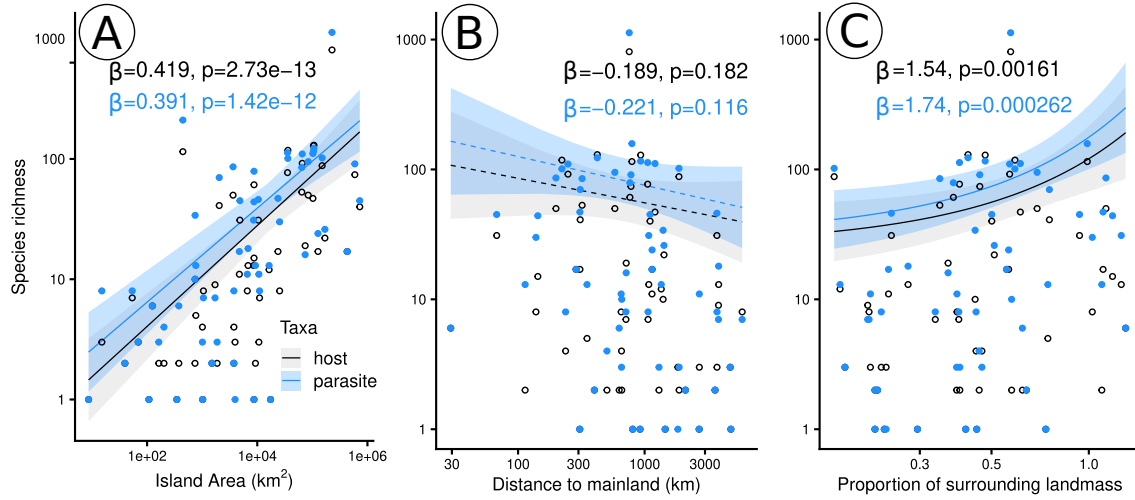


Figure 4.2 Traditional island biogeography theory relates species richness to the distance to the mainland (i.e., isolation) and island area. We detect a significant positive effect of island area on both helminth parasite and host richness (A), but detect no significant effect of isolation as measured by distance to mainland (B). When measuring island isolation through surrounding landmass proportion, we did detect a weak but significant positive effect of island connectivity on both helminth and host richness (C). Plotted lines represent fitted relationships from a negative binomial multivariate regression model, assuming mean values of non-focal predictor variables. Significant relationships are represented as solid lines; shaded regions represent 95% confidence intervals.

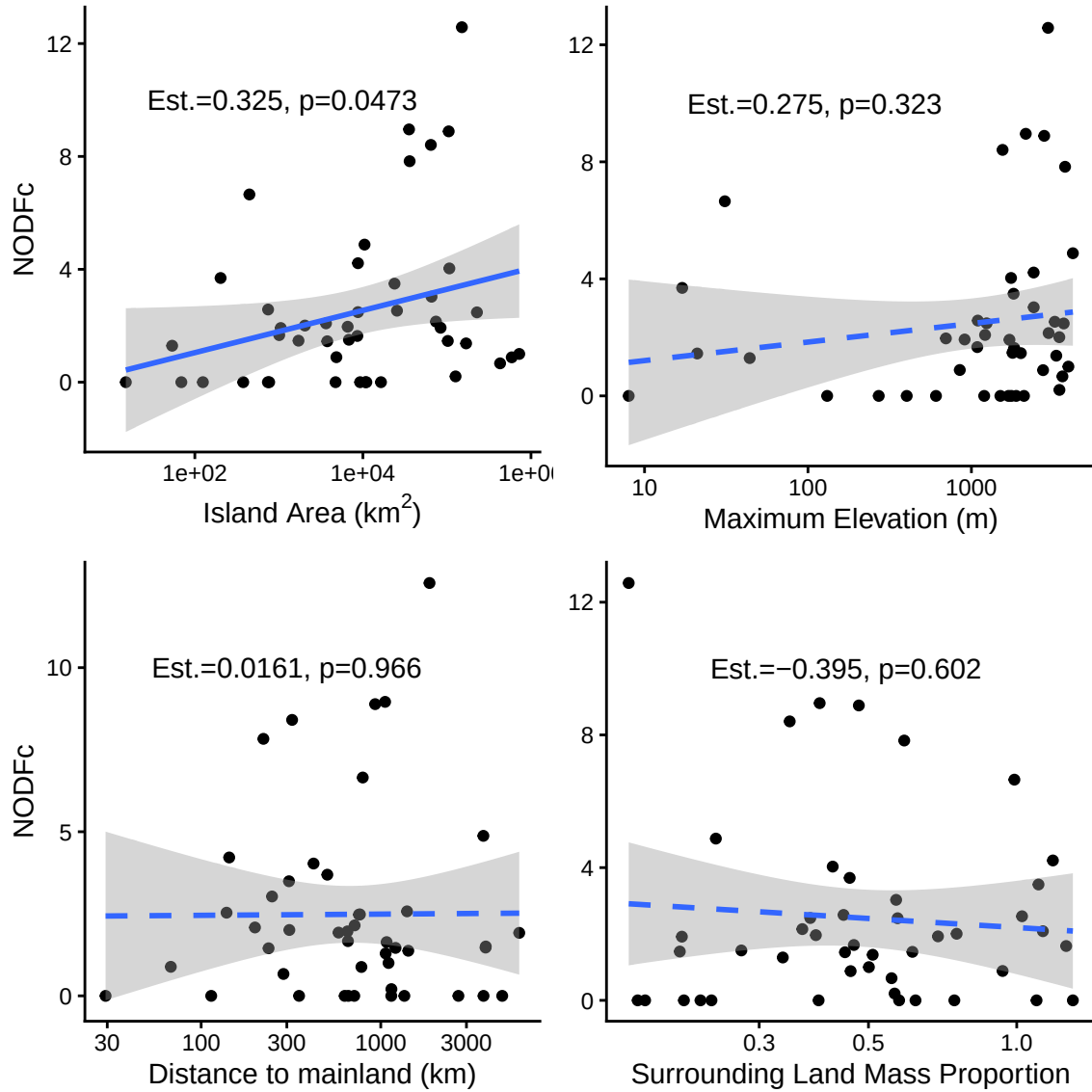


Figure 4.3 In addition to impacting network size, island area was positively related to $NODF_c$ such that larger islands tended to be more nested on average (top left). In contrast, neither island elevation (top right) nor measures of island isolation including distance from the mainland (bottom left) or SLMP (bottom right) were significantly related to interaction nestedness within island networks.

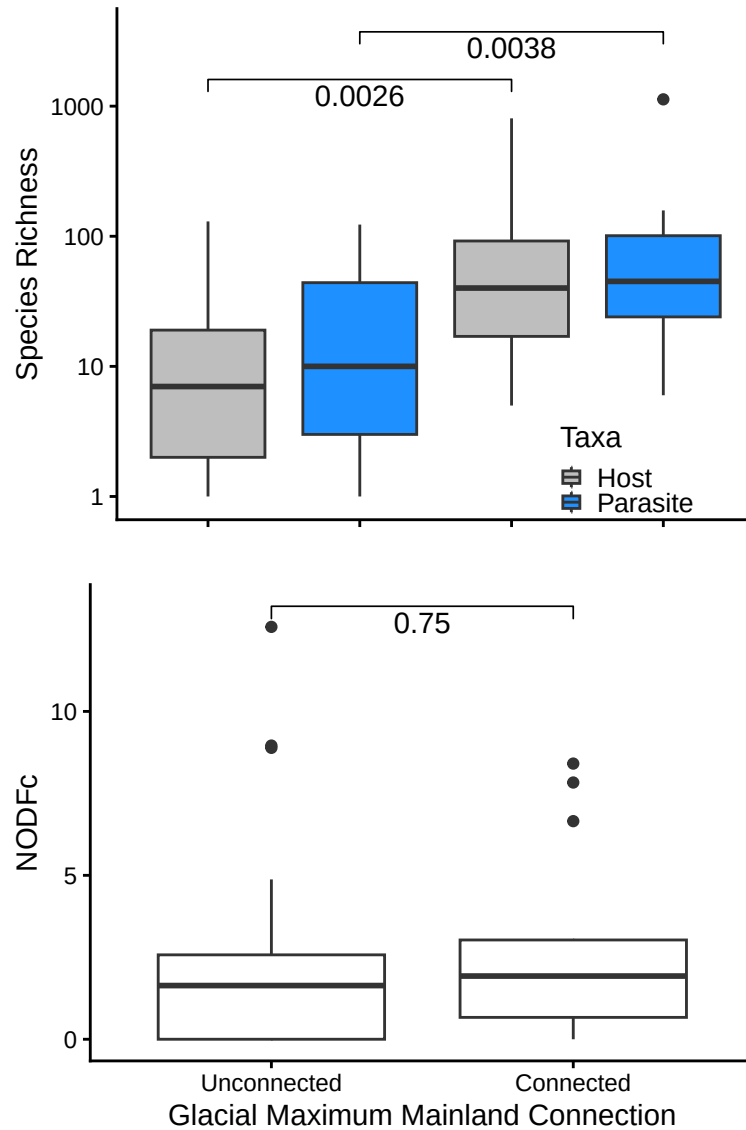


Figure 4.4 Islands with a terrestrial connection to the mainland during the most recent glacial maximum have significantly higher host and helminth parasite diversity (top), but exhibit no significant differences in network nestedness. Comparison labels indicate p-values from a pairwise Wilcoxon test.

CHAPTER 5

STRUCTURAL CHANGES IN MUTUALISTIC NETWORKS UNDER GLOBAL ANTHROPOGENIC PRESSURE

Abstract

Plant-animal transport mutualisms, such as seed dispersal and pollination, represent a diverse suite of interactions that impact the function and dynamics of many ecological communities. Network representations of these interactions exhibit a wide range of structural properties (e.g. nestedness, modularity) that ultimately mediate emergent properties of ecological relevance (e.g. stability, resilience, robustness to secondary extinction). Understanding how these structural properties are affected by anthropogenic forcing and how they scale up to shape emergent properties is vital for predicting and conserving mutualistic networks in the face of global change. Here, I review the current state of knowledge on how three major classes of anthropogenic impact: shifts in global climate, land use intensification, and human-mediated invasion, affect the structural properties of plant-animal transport mutualism networks. In particular, I synthesize emerging patterns and persistent knowledge gaps in how these drivers alter network structure across systems and scales. Finally, I highlight a series of open questions in the field, the answers to which may help elucidate how these structural shifts scale up to changes in community dynamics, and ultimately improve our ability to predict and conserve mutualistic systems.

Introduction

Mutualisms are important to the function of ecosystems

The term "mutualism" covers a broad suite of species interactions, many of which are collectively vital to the function of many ecosystems as we currently know them (Boucher et al., 1982), and iconic organismal features such as flowers or fruits are living homages to the mutualisms that they support (Weiblen & Treiber, 2015). The flow of resources facilitated by these and other mutualistic interactions can serve key ecological functions, and ultimately provide significant ecosystem services to human

communities across the planet. Globally, pollinators may provide as much as \$400 billion annually in agricultural pollination services alone as of 2020 (Porto et al., 2020), and are relied upon by 85% of extant plant species across the globe (Ollerton et al., 2011).

Due to their outsized impact on ecosystem function and services, understanding how mutualistic interactions change in response to perturbation is of great interest. These influential interactions do not happen in a vacuum, and taking a network approach to studying mutualisms can better allow us to understand how mutualisms as a whole affect the structure, function, and resultant services provided by an ecosystem. Network approaches allow us to more clearly understand how perturbations propagate through a system, and have been used to study consumer-resource interactions (Rooney & McCann, 2012), nutrient cycles, (Christian et al., 1996), host-parasite interactions, (Dallas & Cornelius, 2015), and mutualistic interactions (Rohr et al., 2014; Vanbergen et al., 2017). Together, these approaches have enhanced the ability to predict how diverse communities may respond to environmental change. In order to understand how communities of interacting species may change in the face of growing anthropogenic forcing, it is vital to understand how human-induced changes tend to affect the structure and function of mutualistic networks.

Global ecosystems are subject to anthropogenically driven changes at increasing rates, and as a result, so are the mutualistic interaction networks that occur within them. Structural changes in mutualistic networks can be driven by a diverse set of drivers, including shifting climates, urbanization and land use changes, and invasion of non-native species. Here, I aim to characterize the current state of understanding about the ways these diverse drivers affect the structural properties of two major classes of plant-animal transportation mutualisms (Bronstein, 2015): seed dispersal and pollination networks, as well as how we can leverage our knowledge of these

patterns to better understand, predict, and conserve these critical natural systems.

Emergent Properties of Mutualist Networks

The structure of relationships within mutualistic systems can generate dynamic system-level behavior, including several emergent properties useful for understanding how mutualistic systems may respond to perturbation (Figure 5). These include properties describing how mutualistic networks behave around equilibrial states, a number of which can be estimated through eigenanalysis of the Jacobian matrix representing mutualistic communities at equilibrium. Both local stability (the tendency of the system to dampen small perturbations from equilibrium) (Okuyama & Holland, 2008; Valdovinos, 2019), as well as resilience (here, the speed at which the system returns to equilibrium after perturbation) (Suweis et al., 2013; Thébault & Fontaine, 2010), are important for predicting how mutualistic networks respond to anthropogenic change. Other properties include the size of the feasibility domain, the range of parameter values that permits non-zero equilibrium abundances for all species in a community (Grilli et al., 2017; Saavedra et al., 2016; Wang et al., 2023). This property, sometimes referred to as structural stability, defines the safe operational space within which all species can persist (Bascompte & Scheffer, 2022). Given an existing network structure, the extent to which disturbances spread widely or remain confined to local subsets of species (Martins et al., 2024; Suweis et al., 2015) is yet another emergent feature of interest in the context of anthropogenic change, and is closely related to the structural property of modularity.

These metrics collectively provide important insights into how existing network structure may mediate the impact of perturbation, but we are also interested in how networks may be sensitive to changes to their existing structure. By definition, mutualistic partners improve one another’s fitness; if the magnitude of benefit scales in some way with population size (Valdovinos, 2019), positive feedbacks between

partnered populations or threshold responses (Hale & Valdovinos, 2021) may occur. Disrupting these positive feedbacks of networks can facilitate critical transitions, and structural properties of networks can be important for determining behavior at the critical boundary or determining how robust a network is to loss of certain nodes or links (van Der Heide et al., 2021). For example, simulated plant-pollinator networks with greater connectance are more likely to undergo large critical transitions as extinctions occur, undergoing larger community-level breakdowns instead of many small breakdowns (Lever et al., 2014). Positive feedbacks may also induce hysteretic effects crossing the critical transition boundary; that is community assembly and disassembly processes may often be non-symmetric in mutualistic systems, with the outcome depending on precursor system states (Cai et al., 2020). Understanding how these processes occur and are influenced by network structural properties is vital for both prediction and conservation of mutualistic systems.

The Importance of Structural Properties

Many of these aforementioned emergent properties of interest (stability, resilience, hysteresis, etc.) are heavily impacted by network structural properties (Landi et al., 2018). A significant body of existing literature has been devoted to interrogating relationships between the structural properties and emergent properties of species interaction networks, though at times with counterintuitive results. For example, network modularity (the property of having more connections within smaller subunits of a network than between them) tends to reduce the propagation of direct or indirect perturbations through a mutualistic system (Okuyama & Holland, 2008; Thébault & Fontaine, 2010). Invasion by non-native generalist species increases connectance and decreases modularity of some networks, potentially leading to broader or faster propagation of disturbances in the face of a changing environment (Fricke & Svenning, 2020; Martins et al., 2024; Vollstädt et al., 2022). Because of their important and

often complex effects on emergent properties, understanding how network structural properties tend to change in response to anthropogenic forcing is vital for prediction and management of mutualistic systems.

Structural properties often have nuanced effects on multiple emergent properties, making it important to identify which emergent properties are most relevant in a given context. For example, one widely observed property of mutualistic networks is nestedness, in which specialist species with few interaction partners tend to interact with generalist species that have many partners, effectively embedding specialist interactions within a broader, generalist core (Bascompte et al., 2003; Bascompte et al., 2006; Thébault & Fontaine, 2010). As nestedness increases, the overall robustness of the system to species loss may increase due to functional redundancy, while simultaneously making the final breakdown of the system more dramatic (increasing magnitude of critical transitions) (Bascompte & Scheffer, 2022). The nestedness of strong interactions tends to increase the size of a mutualistic system’s feasibility domain (Saavedra et al., 2016), yet decrease its stability at equilibrium (Allesina & Tang, 2012). However, even if more nested mutualistic networks are less stable, they may lead to higher overall community abundance, an effect stronger in systems characterized by abundant but weak mutualisms (Suweis et al., 2013). Robert May first pointed out in 1973 the “paradox of complex interactions”, where, under a Lotka-Volterra style modeling framework, larger species assemblages with many strong and randomly assigned species interactions lead to correspondingly smaller feasible domains of community stability (May, 1973), a pattern that tends to be even more pronounced in mutualistic systems (Allesina & Tang, 2012; Saavedra et al., 2016). In reality, however, interactions are not randomly assigned within a community, and empirically observed suites of mutualistic interactions often tend to favor stability in practice, though to varying degrees (Grilli et al., 2017; Thébault & Fontaine, 2010).

How structural properties translate into emergent ones may be further complicated by plasticity in species interaction preferences (Corro et al., 2021) or the degree of nodes' dependence on mutualisms (van Der Heide et al., 2021). Valdovinos et al. (2016) found that accounting for potential adaptive foraging behaviors reversed the effects of nestedness and connectance on species persistence times (Valdovinos et al., 2016). In general, the relationship between structural and emergent properties is complicated and can lead to counterintuitive results (Landi et al., 2018).

The mechanisms under which emergent properties arise in networks across a landscape still warrant further investigation, especially in the context of mutualistic systems. In food webs, Borrelli et al. (2015) found that stabilizing motifs were overrepresented relative to null expectations across a diverse set of 50 networks, potentially indicating the presence of stabilizing processes present across landscape scales (Borrelli, 2015). This is supported by prior work looking at the constancy of network structural properties across scales in both trophic (Fahimipour & Hein, 2014) and host-parasite (Dallas & Poisot, 2018) networks, even as individual species turn over across space. The conservation of network properties despite species turnover may indicate that separate assembly mechanisms operate at the level of interaction networks and species assemblages per se (Burkle & Alarcón, 2011; Dehling et al., 2020; Dupont et al., 2009; Petanidou et al., 2008; Ten Caten & Dallas, 2024). One potential set of mechanisms acting on network structure may be non-Darwinian selection processes (Saravia et al., 2022), where more stable networks endure longer through time as less stable networks break down, a potential phenomenon supported by theoretical modeling work (Cai et al., 2020). However, one must be cautious in interpreting the prevalence of stabilizing structural properties as evidence for these processes; Saravia et al. (2022) found little difference between realized local networks and random draws from the regional metaweb in trophic networks (Saravia et al., 2022), emphasizing

the importance of proper null models in the context of network structure (Maynard et al., 2018; Pellissier et al., 2018). Additionally, while different structural measures capture different aspects of network topology, they are seldom independent, a feature well-appreciated in food webs (Vermaat et al., 2009). Understanding how structural properties covary is another reason null model approaches are important and looking at joint properties may be useful (Dormann et al., 2017; Fortuna et al., 2010; Pellissier et al., 2018; Poisot & Gravel, 2014).

Widely Observed Structural Properties of Mutualistic Networks

There are a number of properties pervasive in empirically observed frugivory and floral visitation networks. Empirical networks often exhibit high degrees of interaction nestedness (Bascompte et al., 2006; Fortuna & Bascompte, 2006), a product of the asymmetric specialization widely observed in both seed-dispersal (Schleuning et al., 2014; Yohe et al., 2025) and floral visitor networks (Bascompte et al., 2003). This feature tends to increase the strength of highly connected nodes, often resulting in long-tailed degree distributions. Especially in pollination systems there may be trade-offs in specialization stemming from this pervasive interaction asymmetry; (Arceo-Gómez et al., 2020) found that a plant’s degree specialism was positively related to heterospecific pollen delivery, likely due to associations with generalist pollinators (Arceo-Gómez et al., 2020). Nestedness may increase positive indirect interactions within guilds, ultimately increasing the potential diversity of species a community can support (Bascompte et al., 2003; Guimaraes Jr et al., 2017). It also can reduce effective interspecific competition, increasing equilibrial species richness (Bastolla et al., 2009), a pattern potentially observed in empirical networks (Martín González et al., 2015).

Modularity is another prevalent feature of mutualistic networks, many of which may be separated by functional guilds (Ramos-Robles, Andresen, et al., 2018) or phylo-

genetically conserved groups of interaction partners (Martín González et al., 2015). Species with certain traits or within certain taxonomic groups may be more likely to make cross-module connections. Sebastián-González (2017) found that a small number of highly frugivorous birds and high energy content fruits had higher connectivity both between and within modules (Sebastián-González, 2017). In their study of Peruvian highland pollination networks, Watts et al. (2016) found connector species were likely to be comparatively generalist and taxonomically clustered (*Diptera: Syrphidae* and *Asterales: Asteraceae*). The introduction of a relatively small number of species with higher or lower propensity to establish cross-module connections can fundamentally alter the resultant modularity of a given network (Russo et al., 2014), an idea further explored in the section **Invasion**.

Agnostic to changing environments, some evidence shows that network structural properties vary across preexisting environmental or spatial gradients. In a study of both insect and avian pollination networks across broad latitudinal coverage Trøjelsgaard and Olsen (2013) found that modularity was highest at low latitudes whereas nestedness peaked at mid-latitudes (Trøjelsgaard & Olesen, 2013). Across altitudinal gradients, multiple studies have observed an increase in phylogenetic generalism and an associated increase in nestedness and connectance with increasing elevation in plant-pollinator networks (Adedoja et al., 2018; Classen et al., 2020; Lara-Romero et al., 2019), though similar evaluations to my knowledge have not been made in frugivory systems.

Both the mean and variability of temperature may also play a role in shaping network structural properties; Welty et al. (2015) found that both seed dispersal and floral visitation networks were generally more nested in areas with higher inter-annual temperature variability (Welty & Joern, 2015). González et al. (2015) found that in mainland plant-hummingbird networks warmer and more stable climates were

associated with higher modularity, though these variables had no effect on island networks (Martín González et al., 2015). Both current and past climatic features may have significant effects on network structure (Dalsgaard et al., 2013), further complicating the question of how and at what rate networks may respond to future climatic change.

Anthropogenic Change and Mutualistic Network Structure

While existing environmental conditions have already played a role in shaping mutualistic networks as we know them, human impacts have also drastically altered biotic and abiotic conditions across most of the globe. As natural ecosystems are subject to increasingly severe anthropogenic impacts, so too are the ecological networks that occur within them. Understanding how human-mediated changes affect structural properties and resultant dynamics of mutualistic networks is vital for predicting and conserving species interactions. These changes can occur through several mechanisms, including (1) alterations in community composition, (2) shifts in the probability of interactions occurring, and (3) disruptions to the coevolutionary processes underpinning these networks (Tylianakis & Morris, 2017). Here, I focus on how three major types of anthropogenic disturbance—climatic shifts, changes in land use, and human-facilitated invasion by non-native species—act through these mechanisms to reshape the structural properties of natural mutualistic networks.

Climate

Changes in climate can affect mutualistic network structure through a number of mechanisms, including inducing changes in species distributions across time and space, altering species abundances, or altering the physiology of interacting organisms. Many plant-animal mutualisms may only be able to occur during relatively short, yet ecologically important developmental or life history windows. For example, animal

pollination can only occur when both the flowering time of the plant and the seasonal activity period of the animal intersect. As climate shifts the timing of these developmental periods, phenological shifts and subsequent mismatches have the potential to occur, altering the strength of mutualistic interactions or even removing them altogether (Burkle et al., 2013; Teixido et al., 2022; Zettlemoyer et al., 2019) . For example, in a forest-prairie floral visitation network Memmott et al. (2007) showed that up to 50% of floral visitors have reduced overlap between their seasonal activity period and known floral resources over the past century (Memmott et al., 2007). Link rewiring, in which species form new linkages in response to environmental change, may help alleviate some of the effects of phenological mismatch, but fitness reductions due to reduced phenological overlap with mutualists have been observed empirically (de Manincor et al., 2023; Kudo & Cooper, 2019). Corresponding reduction in pollinator fitness have yet to be observed in field systems, though these reductions are likely already occurring in some contexts (Gérard et al., 2020). These altered interaction probabilities can scale up into altered network structural properties; Encinas-Viso et al. (2012) found in simulated communities that increasing phenological overlap between competitors led to less stable, less nested, and ultimately less speciose communities (Encinas-Viso et al., 2012). As climate shifts phenological periods, even if mutualistic interactors remain phenologically coupled, intraguild shifts in species' activity periods (such as overlapping flight or flowering periods) can change competitive interactions with cascading effects through the community (Wang et al., 2023).

Even if mutualistic partners do not become decoupled in time, climate-driven range shifts may also decouple partnerships by reducing spatial overlap. Warming conditions may shift species ranges northward latitudinally or higher in elevation. Differences in features such as generation time and dispersal capacity mean that two mutualistic partners are unlikely to be able to shift their distributions at the same

rate. Or, if biotic interactions limit the distributions of some mutualistic species, partners may be unable to shift ranges without commensurate shifts of the distributions of their partners (Warren & Bradford, 2014). This may be especially salient in the case of seed-dispersal mutualisms, where the ability of plant species to shift their ranges in response to climatic changes may be dependent on their interaction partner (Warren & Bradford, 2014).

While species may shift their distributions or phenology to track preferred environmental conditions, climate change can also alter species abundances, potentially leading to local extirpation. Complete extirpation is not necessary for ecological disruption, however; functional extinction can precede or occur independently of true extirpation when populations become too small to effectively fulfill their roles as mutualists (Costa-Pereira et al., 2018). Moreover, mutualistic partners often differ dramatically in their adaptive capacity due to contrasting life histories, such as generation time and population size (Toby Kiers et al., 2010). For example, the adaptive potential of semelparous insects with annual life cycles differs substantially from that of the long-lived plants they pollinate. Existing work has already shown plant-animal mutualisms are generally weakened by climate change across broad scales (Tylianakis et al., 2008). Climate-driven extinctions are not necessarily random, and taxonomic or functional biases in abundance changes may have large effects on network properties. Schleuning et al. (2016) found that higher elevation montane mutualism networks were less robust to climate-mediated losses than their lower-elevation counterparts (Schleuning et al., 2016). Limitations to upslope movement have already been shown to reduce population size or extirpate montane species (Freeman et al., 2018), potentially putting high elevation mutualistic networks at higher risk of network breakdown.

Changes in environmental conditions may also alter the strength of a given interaction through changes in physiology or interaction preference (Cruz et al., 2023), potentially leading to interaction breakdown. Hoover et al. (2012) observed shifts in nectar chemistry as a function of altered pCO₂, warming, and nitrogen deposition, which then had downstream effects on experimental pollinator preference and longevity (Hoover et al., 2012). Similar changes in pollen quality may also affect nutritional rewards of interactions, altering the fitness or preference of plants or pollinators (Gérard et al., 2020). These sorts of processes may alter the coevolutionary processes underpinning the development of mutualistic networks, or cause negative fitness consequences in taxa without the plasticity to alter their interaction patterns accordingly. Warming conditions also tend to reduce body size in insects while exerting less predictable effects on the size of plants; this could potentially alter matching of specialized insect feeding morphologies with their plants (Gérard et al., 2020). These sorts of mismatches can degrade interactions as a function of environment, and understanding how both plant and pollinator traits might shift with changing environments is important for predicting the outcomes even for contemporary interactions (Cruz et al., 2023).

Land use

Human-mediated changes in land cover result in large shifts in habitat availability, both in absolute area and spatial distribution. At local scales, the impact of land-use change on network structure depends on the speed, type, and extent of conversion. For example, Theodorou et al. (2017) found previously agricultural areas actually increased in both floral and bee diversity when urbanized, emphasizing the importance of establishing useful bases of comparison (Theodorou et al., 2017). As natural areas are fragmented, specialized and rare interactions are lost disproportionately (Aizen et al., 2012), with remaining generalist interactions potentially forming a widespread

consistent core of interactions at landscape scales. This loss can be driven by the actual extirpation of specialists from the community itself (Maurer et al., 2024), or by shifting biotic or abiotic contexts reducing the prevalence or efficacy of those interactions (Birkenbach et al., 2024). In the case of habitat fragmentation, even if overall habitat loss is small the proportion of edge area is increased as existing land-cover types are subdivided. As edge-habitats become more prevalent across the landscape, networks may grow increasingly homogenized as turnover decreases over larger spatial extents, even if local assemblages retain their diversity or structure.

While land-cover conversion can often have profound impacts on the presence of individual species or interactions at a given site (Birkenbach et al., 2024; Maurer et al., 2024; Schneiberg et al., 2020; Theodorou et al., 2017), whether these changes scale up into differences in overall network structure may be more context-dependent. For example, a recent paper by Rossi et al. (2025) quantified changes in frugivore networks in disturbed Amazon forest plots, finding that while logging and burning across timescales did reduce richness of both interacting species and links between them, neither had predictable impacts on network structural properties (Rossi et al., 2025). Salazar-Rivera et al. (2020) found that modularity was maintained in periurban frugivory networks compared to undisturbed habitats, only observing a decrease in nestedness (Salazar-Rivera et al., 2020). Network properties at local scales may be maintained if land use changes exert too small of an impact on structural or functional properties of mutualistic assemblages to cause appreciable changes in network structure. However, a lack of difference across disturbance regimes may also be an indication of comparatively strong constraints on potential structural properties that can be realized in a network comprised of a given regional pool of mutualists.

Observed network structures may help mutualistic communities deal with disturbance, or at least mediate their responses. Fortuna and Bascompte (2006) found

that empirically observed networks were more sensitive to simulated small reductions in habitat area, but persisted longer than randomly assembled graphs in the face of severe disturbance, largely due to the large amount of nestedness observed within empirical networks providing functional redundancy (Fortuna & Bascompte, 2006). Further empirical observations of how structural properties both mediate the response of mutualistic networks to change, as well as how these changes feed back to alter structural properties are warranted. Even if structural properties are not negatively impacted by altered land use, specific ecosystem services may still be disrupted (Menke et al., 2012). For example, Menke et al. (2012) found that frugivory networks along forest edges were more nested and had greater connectance, but saw an overall decrease in dispersal of large-fruited forest-specialist plants (Menke et al., 2012). While helpful for predicting and understanding changes in mutualistic networks, relying on structural properties alone as indicators of network change can mask functionally critical losses. In practice, isolating the effects of land use change from other anthropogenic drivers is often difficult. Areas with large amounts of human-mediated land cover conversion are often subject to increased propagule pressure of nonnative mutualists (Liu et al., 2023), which may have their own impacts on network structural properties.

Invasion

Human-mediated dispersal has vastly altered species distributions across the globe, and patterns of non-native species invasion can have persistent effects on the structural properties of mutualistic networks. Species with generalist interaction strategies can more easily invade existing ecological networks (Piechnik et al., 2008) and species roles are often conserved between native and invaded ranges (Emer et al., 2016). Invasive plants are often effectively dispersed by generalist pollinators and seed dispersers (Richardson et al., 2000), though in general many successful invaders can also suc-

cessfully interact with relative specialists in their new range (Stouffer et al., 2014). How the traits of a potential invader relate to the existing community can affect its propensity for invasion; in simulated communities Minoarivelo and Hui (2016) found potential invaders with trait means different from the existing community were more likely to successfully establish (Minoarivelo & Hui, 2016). The generalism or specialism of a potential invader may also be predictive of its effect on network properties given establishment (Russo et al., 2014).

As synanthropic generalist species are introduced into more human-affected habitats, mutualistic assemblages and potentially network structures tend to homogenize across regional scales (Fricke & Svenning, 2020). Even when homogenization across space is limited in the context of the entire community, persistent "invader complexes" of introduced species that interact with each other may still alter network properties or invasibility (Prior et al., 2015; Richardson et al., 2000; Vollstädt et al., 2022). Generalist invaders may also have hysteretic effects on networks once well integrated into existing assemblages. In networks where invasive plants have already established, removing them may in fact further alter network structural properties than allowing them to persist (Aslan, 2019; Valdovinos et al., 2009). Reliance on non-native pollination partners may further exacerbate the effects of other anthropogenic disturbances. Zettlemoyer et al. (2019) found higher phenotypic plasticity and greater phenological shifts in introduced plant species compared to native taxa, potentially further increasing the risk of phenological mismatch between native pollinators and introduced plants they now rely on (Zettlemoyer et al., 2019). Alternatively, while introduced mutualists may provide some direct benefit to native partners, their net effect may remain negative when indirect impacts are considered (Vidal et al., 2021). If invading species exert competitive pressure that reduces the abundance of their native intraguild counterparts while simultaneously being less

effective at fulfilling mutualistic functions (pollination, protection, etc.), mutualistic networks may collapse functionally even as interactions (e.g. floral visitation, etc.) appear to continue. While tight coevolutionary coupling is seen between some specialist partners (Stebbins, 1970), the widespread generalism exhibited in many animal-plant networks can cause indirect impacts of invaders to play a large role in shaping their eco-evolutionary trajectories (Guimaraes Jr et al., 2017; Medeiros et al., 2018). Native preferences to associate with introduced mutualists could even cause invasional meltdown of native interactions, a phenomenon exemplified in avian frugivory networks in Hawaii (Vizentin-Bugoni et al., 2021).

If invasion favors generalist species (Fricke & Svenning, 2020; Piechnik et al., 2008), one might expect directional changes in network properties associated with widespread invasion, especially if coupled declines in native mutualists or shifts in their interaction patterns (Aizen et al., 2008). Supergeneralist invaders may significantly increase nestedness by creating cross-module connections (Traveset & Richardson, 2014), allowing perturbations to propagate through the network farther and with greater strength. However, invasive species may also provide functional redundancy of potential partners, potentially buffering mutualistic networks against species extinction. Such buffering may not always translate to greater robustness or resilience in the face of broader perturbations.

In addition to structural properties, emergent properties such as robustness or resilience may be important for predicting the invasibility of a given network (Minoarivelo & Hui, 2016). The presence of invasive mutualists can modify how networks respond to perturbation, potentially altering the severity or likelihood of coextinction cascades. By simulating secondary extinction cascades in invaded or uninvaded empirical networks, Vanbergen et al. (2017) found that coextinctions were less likely in disturbed networks due to functional redundancy of partners (Vanbergen et al.,

2017). However, given that an initial coextinction event occurred, the risk of a cascading collapse was actually higher in these networks, highlighting how expectations about how invasion may impact short- and long-term responses to change differently in some scenarios.

Expectations for how structural properties translate into emergent properties may be very different depending on the strength and structure of competition within the community (Pascual-Garcia & Bastolla, 2017; Wang et al., 2023). In addition to global homogenization of interactions across scales, Fricke and Svenning (2020) found that more highly-invaded frugivory networks had higher connectance and lower modularity (Fricke & Svenning, 2020). Some evidence exists that the increasing prevalence of synanthropic generalists in some pollinator communities increases interaction evenness (Schneiberg et al., 2020) or nestedness (Bartomeus et al., 2008), though the exact pattern of this may depend on factors such as pollination syndrome or local pollinator communities. Vanbergen et al. (2017) also found that disturbed pollination networks were larger, but actually lower in connectance and nestedness (Vanbergen et al., 2017). Even further complicating expectations, numerous other studies have also observed little to no predictable impact of invasion on overall network properties such as nestedness or modularity (Parra-Tabla & Arceo-Gómez, 2021; Stouffer et al., 2014), even when considering invasion of hypergeneralist plant species (Vila et al., 2009). Further investigation into the processes maintaining network properties despite significant node-level changes is needed to fully synthesize these variable results across systems.

Additional Considerations

The Role of Rewiring

Ecological networks are not static through time or space. While links between species

may erode through time in response to changing environments, new links may form as well, a process known as link rewiring. This phenomenon is an important consideration when assessing how mutualistic networks may respond to anthropogenic pressures. While it remains essential to account for processes that erode existing interactions (e.g., phenological mismatches, species extinctions, etc.), failing to account for potential rewiring may lead us to overestimate their effects (Vizentin-Bugoni et al., 2020). However, despite recent conceptual advances in understanding the rules and constraints that govern rewiring (Corlett, 2011; Fuzessy & Pizo, 2024), in practice it is often unclear how networks will rewire until they are perturbed, and they may at times fail to rewire as strongly as expected (Leimberger et al., 2023). Leveraging predictive approaches can be potentially powerful in this context in order to better understand how networks may respond to change (Eklöf et al., 2013; Nunes Martinez & Mistretta Pires, 2024; Strydom et al., 2023; Strydom et al., 2021; Vizentin-Bugoni et al., 2020), but to date there are relatively few case studies looking at how networks change through time (though see (Burkle et al., 2013; Leimberger et al., 2023; Olesen, Stefanescu, et al., 2011b; Petanidou et al., 2008)). This gap has been long-acknowledged in the literature (Bascompte & Jordano, 2007), but due to the logistical challenges and significant resource requirements of resampling interactions at a given site regularly it is one that we have been slow to fill as a scientific community. From the evidence that does exist, species traits or position within a network may be predictive of their persistence (Burkle et al., 2013) or rewiring (Sandor et al., 2024).

In some cases, phylogenetic conservation of traits or interaction preference may be useful for determining which species may be interacting in present assemblages, as well as which interactions may occur between novel sets of species. This approach has been applied in modern systems (Vizentin-Bugoni et al., 2021) and used to gain insight into paleoecological systems of interacting mutualists (Braga et al., 2021). In other

systems, however, phylogenetic relatedness may explain less variation in interactions than directly measured traits (Rafferty & Ives, 2013). Machine learning approaches can help overcome the problem of predicting potential species interactions as well as potentially identify traits associated with rewiring potential (Foster & Dallas, 2026; Pichler et al., 2020b), though care needs to be taken in applying many of these approaches to notoriously sparse ecological data (Poisot, 2023).

Species' capacity to rewire may be an otherwise hidden or underappreciated potential of functional redundancy, and affinity-based rewiring has been shown to maintain network properties and stability in theoretical systems (Sheykhali et al., 2020). There are also new exciting avenues for measuring the rate and scale of rewiring as well, with potential for application in ecological systems (Han & Goetz, 2019). However, it is also important to emphasize that mutualistic interactions are not the only relationships being affected by global change. Competitive and trophic interactions are also likely to be affected by anthropogenic drivers, potentially complicating the picture of the emergent outcomes of rewiring. Furthermore, shifts in environmentally-mediated interaction strengths can fundamentally alter the dynamics of a network, even without changing its overall structure. Mechanisms to maintain mutualisms may be developed on coevolutionary timescales, and species may not be able to alter interaction patterns to reflect current fitness benefits (Toby Kiers et al., 2010). However, this is not always true; flexibility in interaction preferences has been observed across numerous pollinator (Morán-López et al., 2022) and frugivore taxa (Nowak & Lee, 2013), and adaptive foraging may help ameliorate the negative impacts of environmental change in some animal taxa (Valdovinos et al., 2016).

The inherent issue of sampling

Network representations can be useful tools for studying ecological systems, but sampling and model-construction choices can introduce artifacts into inference (Blüthgen

& Staab, 2024; Olesen, Bascompte, et al., 2011; Young et al., 2021). Sampling capacity and efficacy varies widely across different types of mutualistic systems, and biases may potentially lead to artificial structural properties. For example, at least some of the persistence of patterns of high nestedness and low connectivity of many ecological networks may be a byproduct of poorly sampling them (Blüthgen et al., 2008; Blüthgen & Staab, 2024) or neutral evolutionary processes (Valverde et al., 2018). One solution to circumvent this shortfall is by combining data from multiple sources (Cirtwill et al., 2019; Olesen, Bascompte, et al., 2011), such as independent abundance estimates or alternative sources of interaction data. By properly combining data from multiple sources it becomes possible to account for these structural artifacts as well as characterize higher-order interactions and structure, even without a fully sampled network (Valdovinos, 2019).

When describing or characterizing networks, the scale of aggregation chosen should be biologically motivated. Combining data across too large a spatial or temporal extent can artificially decrease network connectance and lead to misinterpretations regarding species generalism or network turnover (Burkle & Alarcón, 2011; Duchenne et al., 2022). As potential partners that never co-occur in time or space are aggregated into the same network, this creates an overabundance of "missing" links. As such, the same network may exhibit different structural properties depending on the level of aggregation (Guimaraes Jr, 2020). Species considered generalists across their range may function as relative specialists at individual sites, with local populations only interacting with a narrow handful of partners in any given biotic context. Eco-evolutionary modeling (Medeiros et al., 2018) found that gene flow among generalist populations may actually increase their trait-matching to local mutualists. Even in the same habitats, network properties may change across fine scales (e.g. forest floor to canopy) (Schleuning et al., 2011). While network approaches necessarily simplify

complex ecological systems, methodological choices can substantially influence the structural properties observed and the conclusions drawn from them.

Additionally, many theoretical expectations for how structural properties translate into emergent properties rely on simulating truly mutualistic networks, where interactions between mutualists by definition confer a positive fitness benefit to both interacting species. However, interactions may not always convey a fitness benefit in practice (Thomson, 2003). The fitness impacts of opportunistic interactions, "cheaters", palynivory, and granivory are often difficult to quantify, especially as the interaction is often contributing to fitness of each partner in very different ways (e.g. the benefit of resource for a pollinator is difficult to measure against the fitness benefit of pollination for a plant). "True" mutualisms – where both organisms receive a positive fitness benefit – may actually be more conserved than interaction networks show (Burkle & Alarcón, 2011; Montesinos-Navarro et al., 2017). Additionally, even the fitness effects of existing interactions may shift along the mutualism-antagonism continuum as climate alters the cost-benefit relationship of an interaction (Toby Kiers et al., 2010). Empirically measuring the fitness effect conveyed by an interaction is logistically challenging but vital work, especially in the context of global change. There is some evidence, at least in pollination systems, that easier to measure properties such as interaction frequency can be a useful surrogate for fitness benefit (Vázquez, Morris, et al., 2005). However, it remains unclear whether this relationship holds across broader spatial scales or applies to other forms of transport mutualism.

Network approaches for conservation planning and assessment

The idea of conserving species interactions has been around for nearly a half-century (Janzen, 1974). Conserving mutualist assemblages and reducing the rate of anthropogenic impacts can help mitigate potential future degradation of existing networks, but a large amount of change has already occurred in many mutualistic systems.

Accounting for both existing and potential future changes during conservation planning and assessment is a vital reality for effective management of ecosystems. Clear links between management goals and measurable outcomes are also essential in network contexts. For example, practices allowing or promoting temporal variance in population size across a community generally have positive effects on network stability (Bascompte & Scheffer, 2022), but may be counter to management practice for individual species (Tylianakis et al., 2010).

While conceptually linked, not all transport mutualisms are equal. Plant-pollinator relationships require an individual pollinator to engage in multiple interactions with the same species for successful conspecific pollen transfer, whereas effective seed dispersal can occur as the result of a single frugivory interaction. The different fitness benefits and mechanisms of these interactions may translate into very different impacts of global change on their structure and coevolutionary underpinnings (Wheelwright & Orians, 1982). Regardless, taking network-level approaches to ecological questions can provide insight into numerous foundational ecological properties, and understanding how networks change may have relevance in the community beyond the confines of the network itself (Windsor et al., 2022).

Conclusions

Mutualistic networks represent complex webs of interactions that underlie essential ecosystems. Here, we have reviewed evidence that ever-increasing human impacts inflict non-random changes in the structural properties of these networks, which have the potential to scale up into emergent properties of ecological and evolutionary consequence. Anthropogenic shifts in global climate, land use, and species assemblages all have the potential to alter mutualist assemblages and the interactions between them. Understanding how resultant changes in the structural properties of mutualistic net-

works influence community dynamics can allow us to better predict how networks may respond to future perturbations, but additional research is needed to highlight the mechanistic causes and consequences of these changes. Across all three types of anthropogenic impacts discussed, there are examples of relatively drastic changes in node or interaction-level properties failing to scale up into expected changes in structural or emergent properties of networks. Further investigation of the rules that govern how these interactions scale up to network-level properties, especially in the context of empirically disturbed networks, is warranted.

In many ways, theoretical advancements in our ability to predict how networks may respond to environmental change have outpaced empirical evaluation of these predictions. Sampling a given network repeatedly through time is labor intensive, but a vital effort necessary to begin closing this gap. Existing empirical work looking at network change through time has made vital steps forward in our understanding of how mutualistic networks change throughout time. Non-Darwinian selection processes (Borrelli, 2015; Saravia et al., 2022), in which more stable network structures persist longer through time as less stable networks break down, have been predicted theoretically (Cai et al., 2020), but empirical investigation of this phenomenon in mutualistic systems remains limited. Clarifying the extent to which stabilizing dynamics operate at the network level and identifying appropriate null models for comparison remains an important area for future investigation.

Structural properties can be powerful for understanding how species interaction networks may respond to global change, but not if we lack a coherent understanding of how structural properties scale up to ecologically meaningful outcomes. Many existing structural metrics applied to ecological networks are borrowed from analysis of non-ecological graphs. Moving beyond single-metric summaries towards approaches that characterize the distribution, covariance, and constraints among multiple node,

interaction, and network-level properties may provide a more complete picture of the operational space these networks may occupy. A deeper understanding of the relationship among structural properties, and between structure and function in mutualistic communities, will be essential for revealing the principles that govern network responses to global change.

These questions are especially urgent in the face of accelerating anthropogenic change. As climates shift, land-cover changes, and species introductions rewire ecological communities, the persistence of mutualistic networks will depend on their capacity to reorganize in response. Identifying which structural properties matter, under which contexts, and for which outcomes, remains an essential need in the context of a changing world. Repeated, interannual sampling of mutualistic networks as they respond to shifting environments is especially important for bridging the gap between theoretical predictions and empirical models of network change. With a proper understanding of these processes, we may be able to move from simply describing how networks change in response to perturbations to actively managing them in ways that help them endure.

Outstanding questions

1. How does the relationship between structural and emergent properties change depending on the temporal or spatial scale of aggregation? Does choosing too broad or narrow a spatial scale obscure the impacts of structural properties on network dynamics?
2. How can considering distributions of node or interaction level traits alongside network structural properties improve our understanding or prediction of emergent properties?

3. Network structural properties inherently covary (Pellissier et al., 2018; Poisot & Gravel, 2014). Can we leverage our understanding of how these properties constrain one another to more effectively predict emergent network properties from suites of structural metrics, rather than from individual properties alone?
4. How is the feasibility domain of a given network (Bascompte & Scheffer, 2022) constrained by the underlying species pool and suite of possible interactions? Can we use characterizations of this space to better understand how mutualistic systems may respond to future anthropogenic change?
5. Does "non-Darwinian selection" (Borrelli, 2015; Saravia et al., 2022) on stabilizing properties play a role in maintaining mutualistic network structures over eco-evolutionary timescales? If so, when might these processes maintain mutualistic function in the face of changing environments?
6. Most existing work on the stability of mutualistic network structures has focused on static network properties; how might selection towards stabilizing properties influence sequential assembly processes?
7. Why do substantial changes in species or interaction-level properties so often fail to translate into measurable changes in overall network structure, and what does this reveal about the processes that maintain structural stability in mutualistic systems?
8. How can insights gained from characterizing network structural properties be translated into actionable conservation and management strategies?

Figures

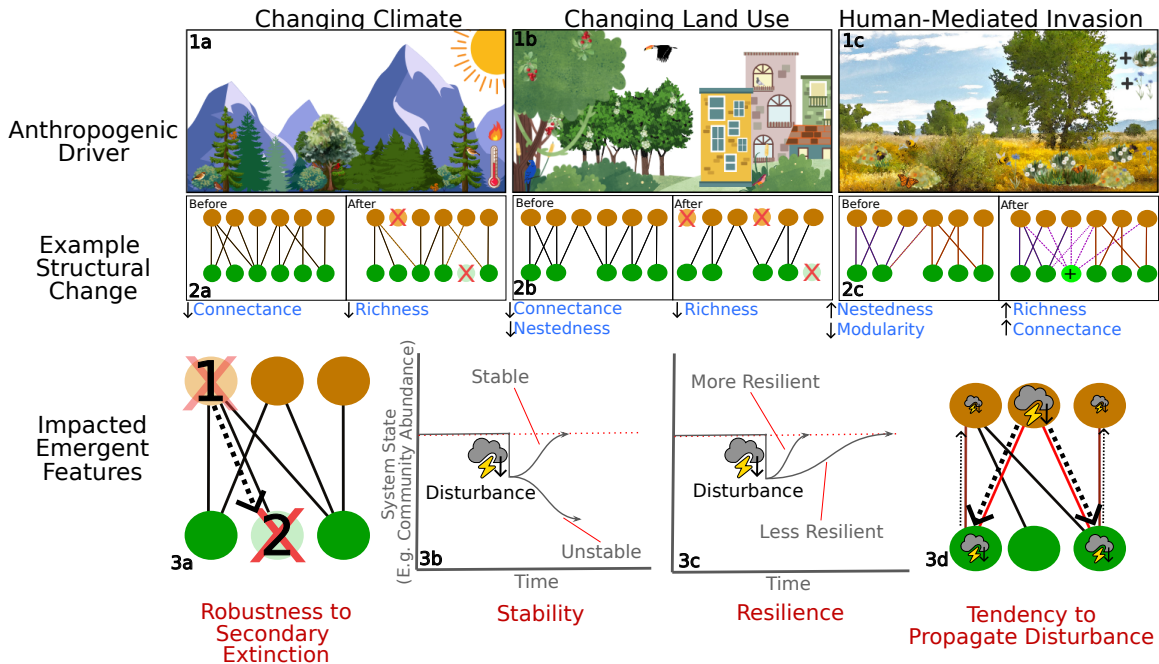


Figure 5.1 Across a diverse set of mutualistic systems, anthropogenic drivers including changes in climate (1a), land-use patterns (1b), and the human-mediated introduction of novel mutualists (1c) can alter the structural properties of the mutualistic networks they impact (2a–2c). These changes to component structural features may, in turn, impact emergent properties of ecological relevance (3a–3d). For example, robustness (3a) characterizes a network’s ability to avoid secondary extinctions through functional redundancy. Other emergent features describe the behavior of systems near equilibrium, including their stability—the tendency to return to equilibrium after disturbance (3b), and resilience—the speed at which they return (3c). Other features include the tendency of networks to propagate or buffer the indirect effects of disturbance (3d, where a large disturbance in the top central node’s population is propagated within the network through shared links). By improving our understanding of how anthropogenic forcing elicits structural change in mutualistic networks, and how those changes translate into altered emergent properties, we can better predict and conserve mutualistic systems.

CHAPTER 6

ENVIRONMENTAL CONTEXT ALTERS THE PREDICTABILITY OF YEAST COMMUNITY ASSEMBLY FROM MONOCULTURE DEMOGRAPHIC PARAMETERS

Abstract

Predicting the assembly processes of competitive communities is nontrivial and often requires estimating interspecific pairwise effects. However, interaction-explicit approaches become increasingly difficult to implement as community size increases. Demographic rates measured in single-species cultures may provide a simpler basis for prediction than interaction-explicit approaches, but the conditions under which monoculture-derived parameters scale to multispecies communities remain unclear. Here, we use experimental communities of phenotypically distinguishable strains of brewer's yeast (*Saccharomyces cerevisiae*) to test whether demographic parameters derived from monoculture growth experiments can predict multi-strain community outcomes across nutrient environments. We find that simulations parameterized by strain-specific estimates of finite rate of increase (R) and carrying capacity (K) closely match empirical communities under low-nutrient conditions, but match empirical communities less closely under high-nutrient conditions. Competitive differences between strains were also more pronounced in high-nutrient conditions. Together, our results show that single-species growth parameters can provide useful predictions of community assembly, but that their predictive value depends on environmental context.

Introduction

Understanding and predicting how biological communities assemble is necessary to understand how biodiversity is structured across space and time. The predictability of assembly outcomes has been a major research focus in classical community ecology (Chase, 2003; Diamond, 1975; Drake, 1990; Keddy, 1992; Tilman, 2004; Weiher & Keddy, 2001). Traditional community assembly paradigms emphasize the importance of abiotic and biotic filters on determining how regional species pools translate into local communities (HilleRisLambers et al., 2012; Weiher et al., 2011). However, at the site level assembly outcomes are influenced by both demographic stochasticity and species-level niche and competitive differences (MacDougall et al., 2008; Martorell & Freckleton, 2014; Tilman, 2004; Zhou & Ning, 2017), and untangling the relative importance of stochastic and deterministic processes is difficult, though increasingly possible through advanced statistical approaches (Maynard et al., 2020).

Modern coexistence theory emphasizes that members of coexistent communities should be able to invade equilibrational communities when rare (Chesson, 2000; HilleRisLambers et al., 2012). Demographic studies that characterize how the growth rates of one species depend on the frequency of competitors can be used to predict both coexistence and assembly outcomes (Harpole & Suding, 2007; Hille Ris Lambers et al., 2002; Wills et al., 1997). However, pairwise competition experiments are difficult, and become increasingly intractable as community size increases (Maurer, 1999). Additionally, predictions based on pairwise interactions may fail when multispecies outcomes depend on emergent higher-order effects or environmental context dependence (Chesson, 2018; Lee et al., 2023; Miller & Max, 2025). Alternative approaches look to predict community outcomes from lower dimensional data, including community abundance timeseries (Carrara et al., 2015) or single-species experiments (Nestor et al., 2023). At one extreme, neutral theory, which ignores species-level differences and

simulates community composition through stochastic processes, while inherently simplifying can recapitulate patterns observed in some empirical communities (Hubbell, 2001; Ofițeru et al., 2010; Woodcock et al., 2007). Between fully neutral models and interaction-explicit approaches are models that retain measurable differences among species, such as growth rate or carrying capacity, while simplifying the structure of interspecific interactions between species.

Single-species vital rates are one possible source of simplified demographic information, and may help predict community outcomes (Nestor et al., 2023; Ram et al., 2019). However, the scaling from population to community outcomes is not always straightforward (Khristich et al., 2025). Demographic rates derived from monoculture describe how the vital processes of a given species scale with its own density. In communities strongly structured by antagonistic interactions, differences between intra- and interspecific competitive effects may strongly influence assembly outcomes (Barabás et al., 2016; Chesson, 2018; HilleRisLambers et al., 2012). Failing to account for these differences may therefore limit the ability of single-species vital rates to predict community outcomes. In some systems, functional traits or phylogenetic distances can help approximate these pairwise effects (Cadotte et al., 2013; E-Vojtkó et al., 2023). However, in communities where species are weakly structured by direct interactions, or where species contribute similarly to shared density limitation, single-species vital rates may be sufficient to predict community-level outcomes.

The utility of single-strain vital rates may also depend on environmental context. The strength and mechanisms governing species interactions can vary across environments (Van Dyke et al., 2022), potentially changing whether monoculture dynamics capture the processes driving final community composition. Nutrient availability may be especially important because it can alter both the densities that populations reach and the relative importance of different competitive mechanisms, including shared

resource limitation, metabolite accumulation, and interference competition (Albergaria & Arneborg, 2016; Salvadó et al., 2011). As a result, both community assembly outcomes and the predictability of those outcomes may be dependent on resource availability.

Microbial communities are especially useful for testing questions about community assembly because they are highly replicable and can be assembled under controlled environmental conditions (Nemergut et al., 2013). These features make it possible to manipulate both community composition and environmental context while directly measuring the outcomes of assembly. By experimentally standardizing the species pool, initial abundances, and abiotic environment, microbial systems allow local assembly processes to be isolated from broader biotic and abiotic filtering effects observed in natural systems (Weiher et al., 2011). Phenotypically distinguishable strains of *Saccharomyces cerevisiae* with a shared genetic background provide an especially controlled system for testing community assembly dynamics (Mitchell et al., 2015). By reducing many of the phylogenetic and physiological differences that complicate prediction in natural communities, this system provides a conservative test of whether monoculture growth parameters can predict community assembly under different environmental conditions.

Here, we use experimental communities of phenotypically distinguishable strains of brewer’s yeast (*Saccharomyces cerevisiae*) to test whether single-species vital rates scale to community-level outcomes. By comparing experimental communities to those simulated from single-species vital rates, we test whether nutrient availability alters the predictability and assembly of multi-species communities. We show that nutrient environment has no detectable effect on the finite rate of increase, but large effects on the carrying capacity for a given strain in monoculture. We further find that simulation models parameterized by single-species vital rates accurately predict multistrain

communities in low-nutrient environments, but not in high-nutrient environments. Together our results highlight the environmental-dependence of community assembly outcomes, even in highly simplified experimental communities.

Methods

Yeast System

We used a series of yeast strains with biosynthetic pigment pathways assembled on episomal VEGAS (Versatile genetic assembly system) assembly vectors created using a yeast Golden Gate approach, detailed in Mitchell et al. (2015). The result is a series of yeast strains with identical genetic backgrounds, but containing cytosolic shuttle plasmids that encode biosynthetic pigment pathways, along with selectable markers (KanMX and URA3), and as such are visually distinct due to pigment production. Three strains include combinatorial 3 or 4-gene beta-carotene pathways (hereafter the red, orange, and yellow strains), while a fourth contains the pathway to synthesize violacein (hereafter the black strain).

Monoculture Growth Curves

We used liquid cultures to estimate strain-specific vital rates. Liquid cultures of each strain were diluted 1,000-fold in full-strength media (25 g/L Difco™ YPD broth powder, Becton, Dickinson, and Company), and 20 μ L was then used to inoculate 180 μ L sterile cultures within wells of a 96-well plate. In high-nutrient treatments, sterile cultures were made of YPD solution. In low-nutrient treatments sterile cultures were prepared with DI water, making the final working concentration of 2.5 g YPD per liter (10% of laboratory standard concentrations). Culture plates were incubated for 20 hours at 30 °C inside a temperature-controlled spectrophotometer. Every ten minutes cultures were shaken for 5 seconds prior to recording optical density at 600 nm. Split across 3 trials, we measured 12 replicate growth curves for each strain at

low-nutrient concentrations, and 8 replicate curves for each strain at high-nutrient conditions. One growth curve trial (4 replicates) of orange in low-nutrient conditions showed anomalously high absorbance values, and was discarded from the analysis.

We used nutrient-specific linear standard curves generated from wild-type liquid culture, and applied them across pigment strains under the assumption that pigment effects on OD600 in liquid culture were small relative to differences in cell abundance. Visible pigment differences developed slowly relative to the timescale of our growth curve assay, becoming most distinct after several days of colony growth on agar plates. We fit the resulting abundances to a discrete logistic growth model under a Bayesian hierarchical framework in STAN (Carpenter et al., 2017) as implemented through the R package *cmdstanr* (Gabry et al., 2025), such that

$$N_{n,t+1} = N_{n,t} + R_n N_{n,t} \left(1 - \frac{N_{n,t}}{K_n}\right) \quad (6.1)$$

The finite rate of increase (R) and carrying capacity (K) were estimated separately for each strain-by-nutrient combination; each timestep of our discrete model represented a 10-minute interval between OD600 measurements. Replicate trajectories were allowed to vary in R around their corresponding strain-by-nutrient mean, with a single replicate-level variance parameter shared across all groups. Observation error was modeled with a shared lognormal error term, and weakly informative priors were placed on log-transformed vital-rate parameters and variances. Models were fit using four MCMC chains with 1000 timesteps following a 1000-timestep burn-in period. Models were checked for convergence by analyzing effective sample size (ESS) and $\hat{R}$ for each parameter estimate, as well as through posterior predictive visualizations.

Polyculture communities

We created multi-strain polyculture communities by inoculating sterile media with different combinations of single-species liquid cultures at the same relative abundance

across a community. To ensure initial abundances were the same across strains, cell density estimates were recorded via hemocytometer for each single-species culture. We adjusted initial inoculation amounts based on measured standard densities. For each monoculture, 1 mL sterile cultures were inoculated with 100 μ L of the least abundant culture, or the equivalent amount of more abundant standards. Polycultures were inoculated with an equivalent total amount of yeast, split evenly between present strains so that all strains began at the same relative and total abundances.

Communities were vortexed, and 200 μ L samples were added to wells of a 96-well plate, and passaged in an incubated spectrophotometer for 24 hours. To measure relative abundance of strains within each community, communities were subsampled and diluted to either 1:1000 (high-nutrient) or 1:100 (low-nutrient), and then used to inoculate two spread plates per community on agar media (25 g/L YPD, Kanamycin 50mg/L). We created 8 community types in all; four single-strain communities (yellow, red, orange, and black), two two-strain communities (yellow-black and yellow-red), one three-strain community (yellow-red-black), and one four-strain community (yellow-orange-red-black). We created eight replicates of each community per nutrient treatment; after incubation each community was sampled twice to improve estimates of relative abundance and plate-to-plate variation, for a total of 256 community samples. Spread plates were incubated at 30 °C for 48 hours, after which pigments were allowed to saturate by holding the colonies at approximately 2 °C for two weeks. After this period plates were photographed and the number of colonies of each strain were counted from photographs. 43 samples were discarded due either to contamination or low overall abundance (<5 colonies), leaving a total of 213 communities (Table D.1).

Strains were maintained on selective agar (Kanamycin 50mg/L) prior to experiments, but antibiotic selection was not maintained during the 20-hour liquid growth and polyculture assays. Although experimental communities were inoculated only

with colored strains, a minority of colonies did not express visible phenotypes and could instead appear wild-type appearing (mean = 15.2% across strain-nutrient combinations). Because loss or reduced expression of pigment pathways could produce wild-type-appearing colonies, and because these colonies in multispecies communities could originate from any strain present, we used a Bayesian hidden-state model informed by monoculture expression rates and observed phenotype counts to estimate underlying strain relative abundances in each final community. Details of the model structure and uncertainty around relative abundance estimates are presented in the supplement.

Community Simulations

To test whether monoculture vital rates were sufficient to predict community composition, we simulated each multistrain community using strain-by-nutrient posterior estimates of R and K . We began simulations at the mean initial starting abundance across all community replicates, and then simulated 144 timesteps (24 hours of development) of a multispecies discrete logistic growth model, using the equation

$$N_{n,t+1} = N_{n,t} + \max \left[R_n N_{n,t} \left(1 - \frac{\sum_i N_{i,t}}{K_n} \right), 0 \right]. \quad (6.2)$$

. For each focal strain (n), density dependence was determined by total community abundance, ($\sum_i N_{i,t}$), scaled by the focal strain’s monoculture carrying capacity, (K_n). This formulation assumes equivalent intra- and interspecific competitive effects among all strains, corresponding in a Lotka-Volterra framework (MacArthur & Levins, 1967; Volterra, 1928) to a competition matrix in which all interaction coefficients are equal to 1. Because our experiment was conducted over a short timescale during which cell division was expected to dominate changes in abundance over mortality, we constrained population size to increase monotonically. We therefore set population growth rate, ($R_n N_{n,t} (1 - \sum_i N_{i,t} / K_n)$), to zero whenever it would otherwise be negative. Therefore final relative abundances within a community are determined by

both the finite rate of increase (R) and carrying capacities (K) of strains within the community. We simulated 1000 iterations of each community using sampled R and K values for each strain from fit posterior distributions, and the distribution of simulated community outcomes was compared to those observed empirically. Wilson's score intervals were used to compare colony counts across plates within a given treatment. Relative abundances were compared using 95% credible intervals and pairwise contrasts across strains. We interpreted pairwise differences as supported when the 95% credible interval for a contrast did not overlap zero, and interpreted observed and simulated outcomes as different when their 95% credible intervals did not overlap.

Results

Monoculture growth parameters differed among strains and nutrient environments

Strain-specific carrying capacity differed strongly across nutrient environments (Figure 6.2). Across all strains, high-nutrient carrying capacities were approximately an order of magnitude larger than low-nutrient carrying capacities. Despite a large shift in absolute carrying capacity, the relative ranking of strains was broadly consistent across nutrient treatments. The black strain had the lowest estimated carrying capacity in both low- and high-nutrient environments, while red and orange had the highest estimated carrying capacities. Yellow and wild-type strains were intermediate, with overlapping 95% credible intervals in both nutrient environments.

In contrast, finite rate of increase varied more among strains than between nutrient environments (Figure 6.2). For each strain, 95% credible intervals for (R) overlapped between low- and high-nutrient treatments. Black, yellow, and wild-type strains generally had higher estimated (R) values than red and orange, although several strain-level intervals overlapped. Posterior estimates also suggested a negative relationship between R and K across strains (see supplement Figure D.4). Together,

these results indicate that nutrient availability strongly altered monoculture carrying capacity while having comparatively little effect on estimated finite rate of increase.

Empirical communities deviated from equal starting composition

Although all polyculture communities were inoculated at equal starting relative abundances, final strain composition was often nonequivalent (Figure 6.3). These deviations were most pronounced under high-nutrient conditions. In high-nutrient environments, the yellow strain increased in relative abundance across all community types in which it was present; in each case, the 95% credible interval for change in relative abundance was entirely above zero. Black, red, and orange strains were generally similar in final relative abundance, with 95% credible intervals for most pairwise contrasts overlapping zero, although black slightly exceeded red in the four-strain community (Figure 6.4). In low-nutrient environments, final community composition was closer to equal relative abundance across strains. The black strain tended to achieve slightly higher final abundance than the other strains across community types (95% credible intervals of relative abundance pairwise contrasts were slightly above, but not overlapping zero), while most other pairwise differences were small.

Monoculture-parameterized simulations predicted low-nutrient communities better than high-nutrient communities

Simulated community outcomes generally matched empirical community composition, particularly in the low-nutrient treatment (Figure 6.4). In low-nutrient communities, 95% credible intervals for observed and simulated pairwise contrasts overlapped in all cases except the red-orange contrast in the four-strain community, where red was more abundant and orange less abundant than predicted (Figure 6.4).

Observed and simulated communities diverged more strongly under high-nutrient conditions, especially in more diverse communities. 95% credible intervals of simulated and observed outcomes overlapped in both high-nutrient two-strain commu-

nities. However, in high-nutrient three- and four-strain communities empirical estimates of black relative abundance were lower than simulated estimates, with non-overlapping 95% credible intervals. Across comparisons, high-nutrient communities tended to show larger-magnitude deviations from both equal starting composition and monoculture-based predictions.

Discussion

Our results demonstrate that nutrient availability can alter both community assembly outcomes and the predictability of those outcomes, even in highly simplified experimental communities. The difference between low- and high-nutrient environments may reflect a shift in the density-dependent processes structuring these communities. Nutrient availability had no detectable effect on finite rate of increase, but strongly altered carrying capacity, suggesting that the two nutrient treatments differed primarily in the densities that populations could reach rather than in their early per-capita growth. In resource-poor, low-density environments, exploitation competition over shared nutrient resources, such as dextrose and peptides, may be more influential in shaping competitive outcomes (Hoek et al., 2016). In contrast, at high resource levels, interference competition (in the form of metabolite or toxin production (Greig & Travisano, 2008), cell-cell signaling (Honigberg, 2011), or other mechanisms (Pérez-Torrado et al., 2017)) may be more important for determining pairwise and ultimately community-level outcomes. Nutrient-dependent changes in density dependence may therefore help explain why single-strain growth parameters predicted community outcomes more accurately in low-nutrient than high-nutrient environments, emphasizing how the environmental contexts of competitive outcomes may be key to predicting community assembly.

Single-species vital rates scaled to community-level outcomes in some strain-by-nutrient combinations, but not others. Our simulation framework provides a simplified expectation for community assembly based on monoculture growth parameters (Nestor et al., 2023; Ram et al., 2019). In this sense, the model occupies an intermediate position between neutral models, which assume species equivalence (Hubbell, 2001), and interaction-explicit coexistence models (Chesson, 2000, 2018). The close agreement between simulated and observed communities in low-nutrient environments suggests that these parameters captured much of the relevant density-dependent structure in that treatment. In contrast, disagreement in high-nutrient environments indicates that monoculture growth parameters were insufficient to explain community composition under those conditions. This environment-dependent mismatch is consistent with nutrient conditions altering the interaction processes that shape community composition, reinforcing the view that niche-based and neutral processes may vary in relative importance across environmental contexts (Weiher et al., 2011).

Most of the mismatch between simulated and empirical communities was associated with the black strain reaching lower relative abundance than predicted in high-nutrient environments. Based on monoculture growth parameters, simulations predicted that the black strain would outperform red and orange strains and reach similar final abundance to yellow in high-nutrient communities. Instead, black had lower relative abundance than yellow across multistrain communities and showed little or no advantage over red or orange. One possible explanation is that the engineered pigment pathways differed in their physiological costs. In our system, red, orange, and yellow pigmentation is produced through beta-carotene-related pathways, whereas black pigmentation is produced through a violacein pathway (Mitchell et al., 2015). These pathways likely differ in biosynthetic demands (for example,

beta-carotenes are hydrocarbons (Maoka, 2020) while violacein is nitrogenous (Choi et al., 2015)), and those differences could alter competitive performance in ways not captured by monoculture growth parameters. The black strain also had the highest estimated single-strain finite rate of increase, suggesting that growth-competition tradeoffs could contribute to its lower-than-expected performance in high-nutrient communities (Kurihara et al., 1990). However, our experiment cannot distinguish among pigment-production costs, plasmid instability, metabolite sensitivity, or other mechanisms. Black also produced wild-type-appearing colonies more often than the other pigment strains, suggesting that pigment-pathway instability or loss of visible phenotype may contribute additional uncertainty to its inferred abundance (Figure D.2). However, as models of final strain abundances were informed by monoculture phenotype-expression rates, the observed differences between predicted and observed relative abundances in high-nutrient environments are unlikely to be caused by differential phenotype expression unless expression rates greatly differed between monoculture and mixed communities.

Together, our results show that single-species growth parameters can provide useful predictions of community assembly, but that their predictive value depends on environmental context. Our experiment used a deliberately simplified community composed of closely related yeast strains with a shared genetic background, yet prediction from vital rates alone remained imperfect. Monoculture-parameterized simulations predicted assembly in low-nutrient environments but not high-nutrient environments, suggesting that even minimal communities can exhibit context-dependent departures from models that simplify interaction structure. Community-level outcomes are difficult to predict, yet ecological processes such as stability, resilience, and ecosystem function depend on community structure (De Mazancourt et al., 2013; Tilman et al., 2014). Future work should identify how competitive mechanisms and density depen-

dence vary across environmental gradients, and when additional interaction-specific parameters are necessary to predict community responses to environmental change.

Figures

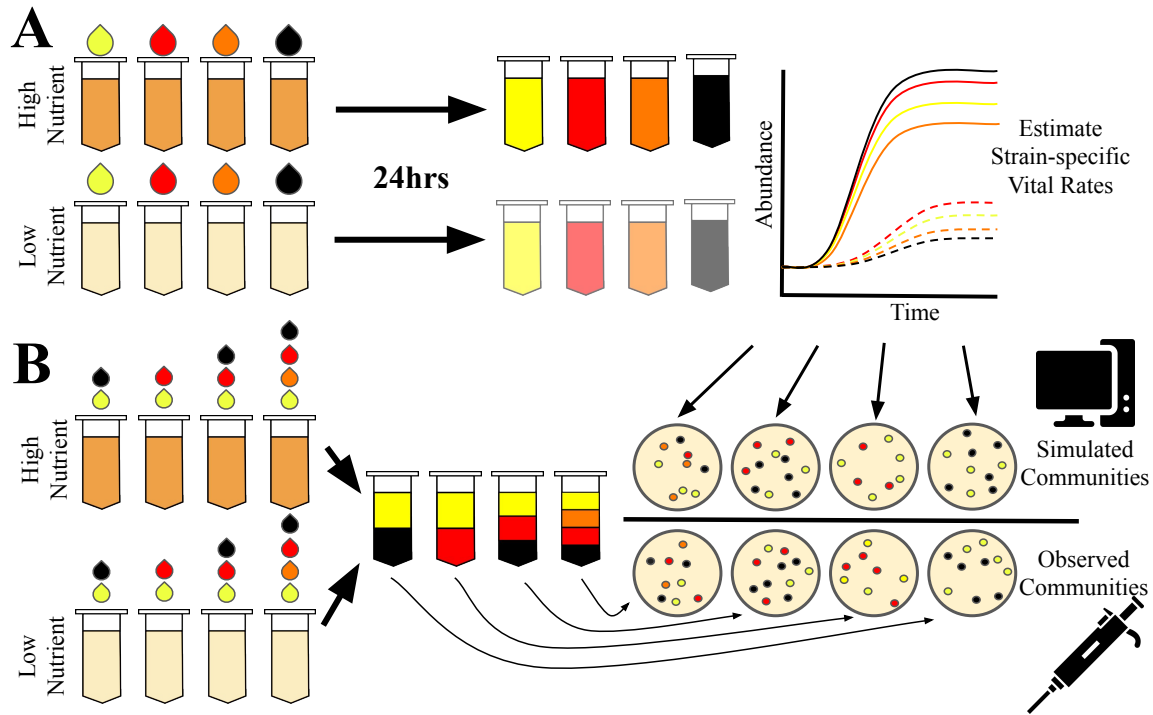


Figure 6.1 Conceptual overview of the experimental approach. A) Sterile liquid monocultures were inoculated with each yeast strain and passaged in an incubated spectrophotometer, allowing populations to approach carrying capacity. These monocultures were used to estimate strain-specific carrying capacity (K) and finite rate of increase (R) in both low- and high-nutrient environments. B) Yeast communities were then inoculated with equal starting abundances of multiple strains and incubated for 24 hours under the same nutrient treatments. Final communities were subsampled by spread plating, and strain relative abundances were estimated from colony phenotypes. Comparing observed community composition with communities simulated from strain-specific vital rates allowed us to test whether multispecies community assembly can be predicted from single-strain growth dynamics.

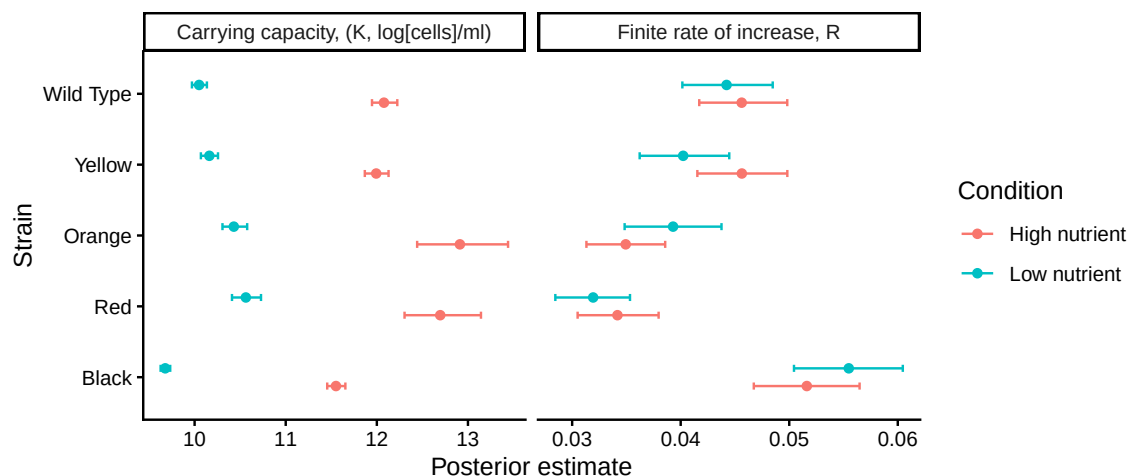


Figure 6.2 Strain-specific vital rates estimated from single-species liquid cultures. Points represent posterior mean estimates from the Bayesian logistic model, and horizontal lines represent 95% credible intervals. Carrying capacity (K , log scale) differed strongly between nutrient environments across all strains, with higher carrying capacities in the high-nutrient treatment. In contrast, finite rate of increase (R) showed comparatively little sensitivity to nutrient environment. Strains also differed in their estimated vital rates, with red and orange strains generally exhibiting higher carrying capacities and lower R across nutrient environments.

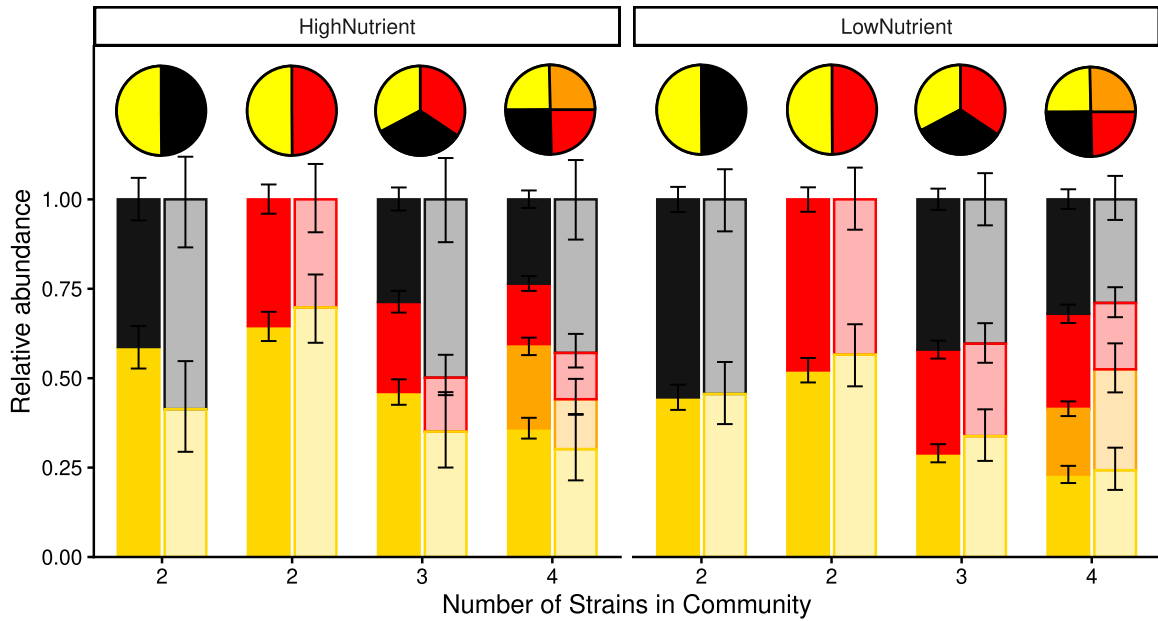


Figure 6.3 Modeled observed (opaque bars) and simulated (lighter bars) relative abundance of strains across community compositions; error bars represent 95% credible intervals. In high-nutrient environments (left), the black strain exhibits lower relative abundance in experimental multistrain communities than predicted by models parameterized by monoculture vital rates. In contrast, in low-nutrient environments (right) experimental communities generally match those predicted by monoculture vital rates.

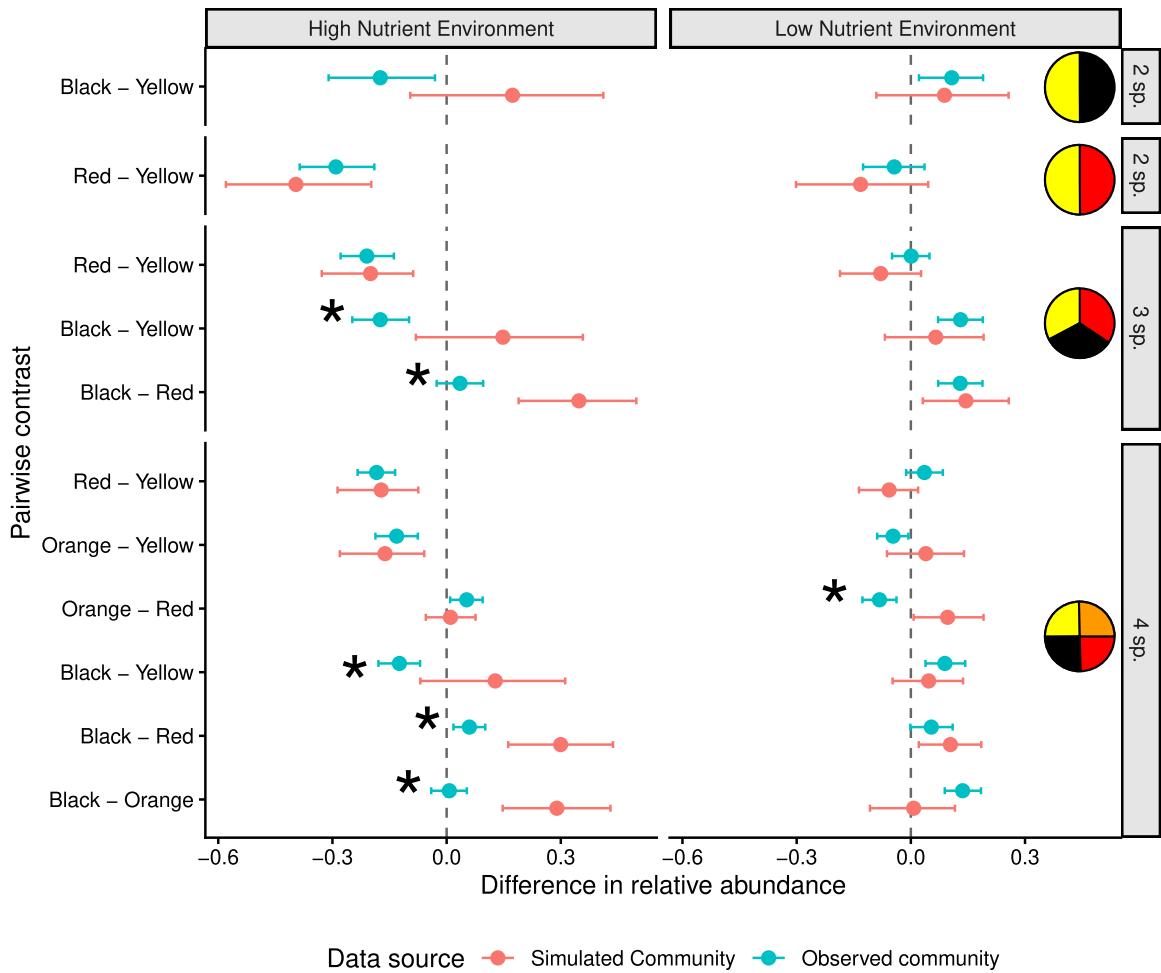


Figure 6.4 Pairwise contrasts of strain relative abundance estimates across simulated (red) and observed (blue) multi-strain communities. Points represent posterior mean estimates, while horizontal lines represent 95% credible intervals. Dashed vertical lines mark zero, indicating no difference in final relative abundance between the two strains in a given contrast. Stars indicate contrasts where the 95% credible intervals for simulated and observed estimates do not overlap. Overall, deviations from neutrality were larger in high-nutrient environments (left) than in low-nutrient environments (right). In three- and four-species communities, observed black-strain relative abundance was lower than expected from simulations parameterized using single-species vital rates.

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APPENDIX A

SUPPLEMENTAL INFORMATION FOR CHAPTER 2

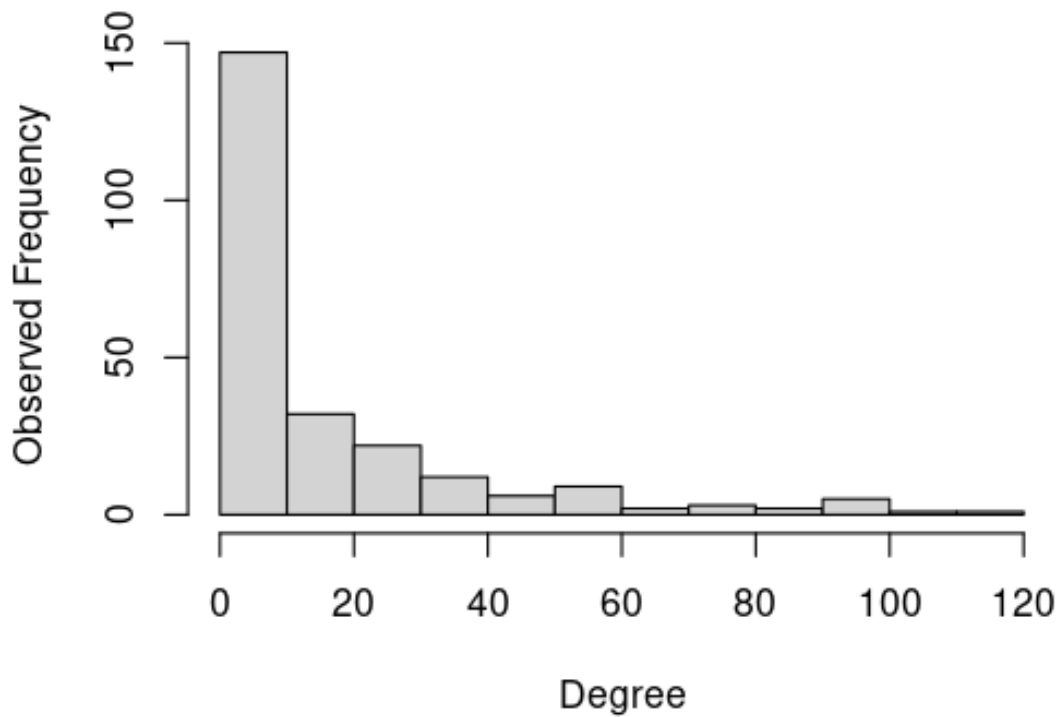


Figure A.1 Observed degree distribution of avian frugivore species included in our analyses. X-axis represents node degree, or number of unique partners. Degree for frugivores ranged from 1 (53 species) to 120 unique plant interactions recorded for *Turdus rufiventris*; median frugivore degree was 5.

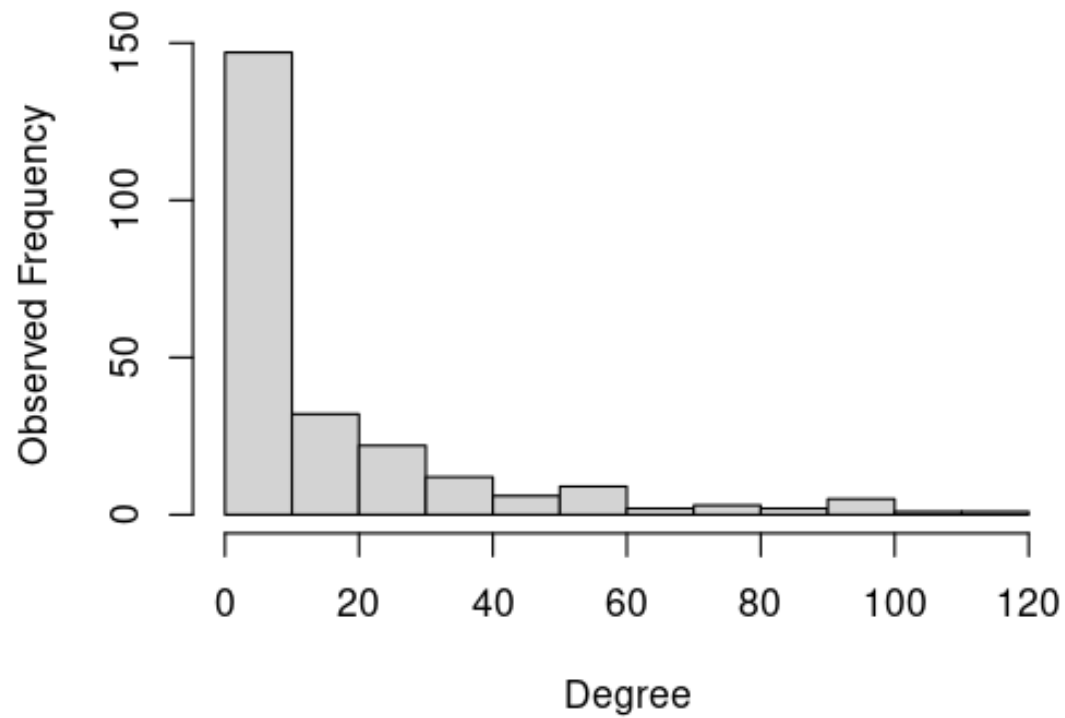


Figure A.2 Observed degree distribution of plant species included in our analyses. X-axis represents node degree, or number of unique partners. Degree on average tended to be lower for plants than frugivores; plant node degree ranged from 1 (128 species) to 80 unique frugivorous interactions recorded for *Myrsine coriacea*; median plant degree was 5.

Table A.1 Number of interactions, unique bird species, and unique plant species used in each model. Incomplete modeling cases were dropped from each model; the remaining number of complete cases was shown below.

Model	Number of Links	Number of Bird species	Number of Plant spp
Latent	3856	242	458
Phy	3276	225	377
Traits	2813	159	283
PhyLatent	3276	225	377
PhyTraits	2492	157	246
TraitsLatent	2813	159	283
Trio	2492	157	246

Impact of Phylogenetic Imputation

In our main text results, for the 46 plant species and 2 bird species which did not appear in the appropriate phylogenies but did have congeners, we randomly assigned those species to be a polytomy within their parent genera. When these species are excluded from the analysis, our results are still qualitatively indistinguishable. Below are versions of the three main-text figures with those 48 species removed from the network.

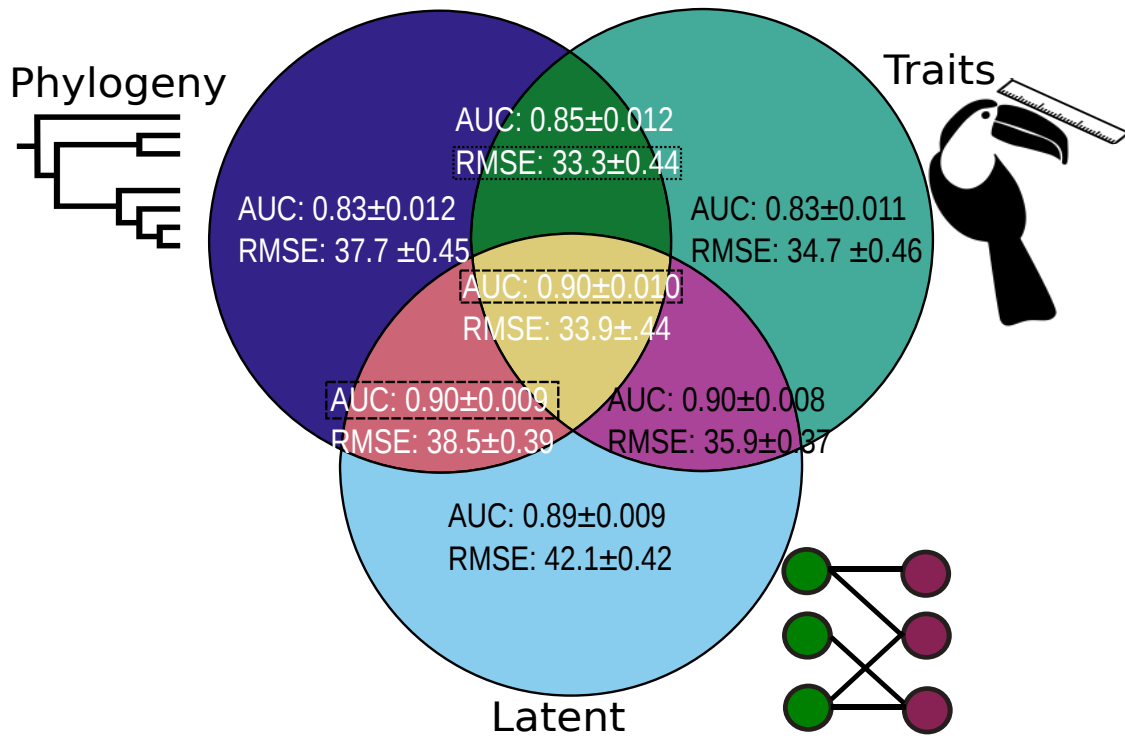


Figure A.3 Summary performance metrics of all 7 models *when species without full phylogenetic information are excluded*, as measured by area under the receiver operating characteristic curve (AUC) and root mean squared error (RMSE); highest performing models for each metric are outlined in dashed (AUC) or dotted lines (RMSE). Mean metric values are presented from 100 replicates of each model structure alongside standard deviation. As in the main text, model discriminatory power between links and non-links is maximized by including latent structural features, with the inclusion of trait, phylogenetic information, or both actually slightly decreasing discriminatory power. However, inclusion of trait and phylogenetic information, while not improving AUC, does increase overall model accuracy as measured by mean root squared error. Pairwise *t*-test adjusted for multiple comparisons showed no significant difference between the discriminatory power of the highest performing models (Trio, PhyLatent, $p = 0.24$)

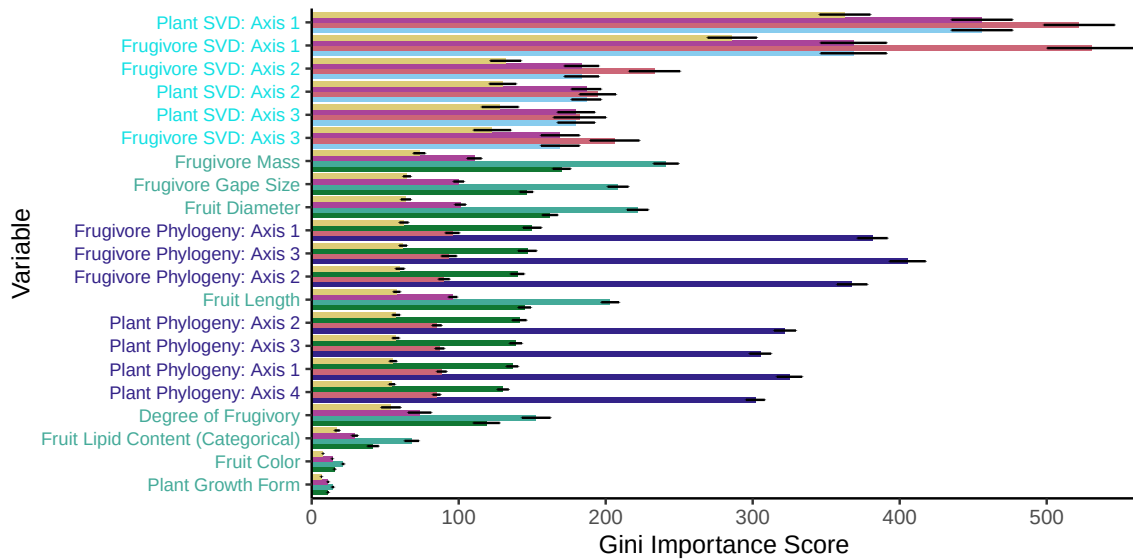


Figure A.4 Variable importance across all models as measure by Gini importance score *when species without full phylogenetic information are excluded*; color scheme is consistent with figure S3. As in the maintext, latent traits were consistently the most important variables for prediction. These were followed by continuous frugivore traits (body mass, gape size), and frugivore phylogenetic axes. Plant phylogenies and continuous trait information were generally less important for prediction than frugivore traits. Categorical plant traits (Lipid content, fruit color, growth form) were the least important variables for prediction.

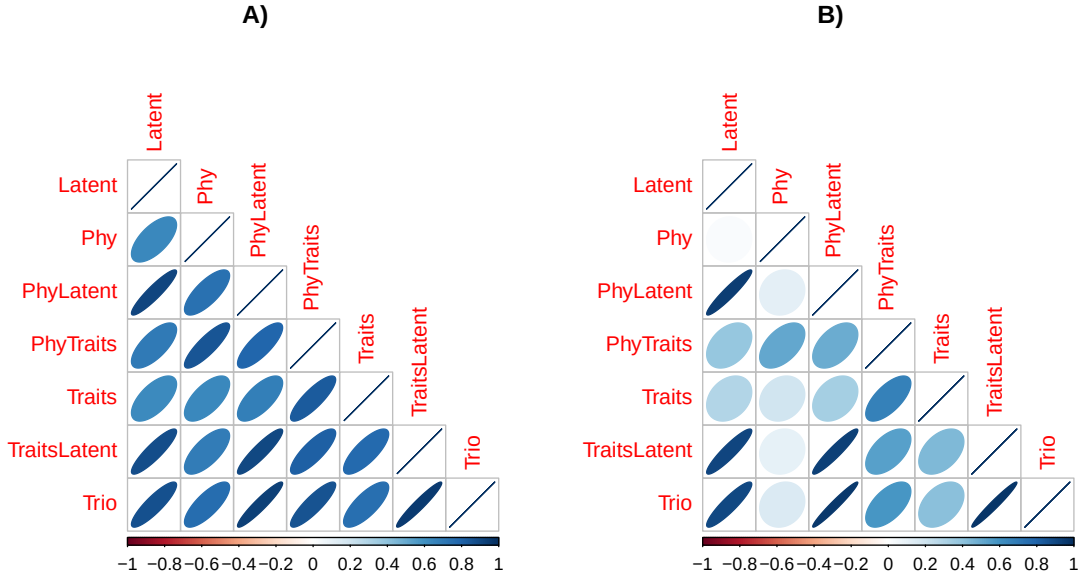


Figure A.5 Pairwise Spearman's rank correlations of link suitabilities across models for all potential interaction (A), as well as only unobserved interactions (B), *after species without full phylogenetic information are excluded*. The latter set of links represents both true forbidden links, as well as other potential interactions not observed in our data-set.

Sensitivity to Class Imbalances

In order to deal with the high degree of sparceness in our network, we trim the training set to enforce class balancing. In the maintext, we present the results of a 3:1 ratio unobserved:observed links. Here, show the methods is relatively insensitive to this exact proportion by repeating our analysis with alternative training ratios of 1:1 and 1:10.

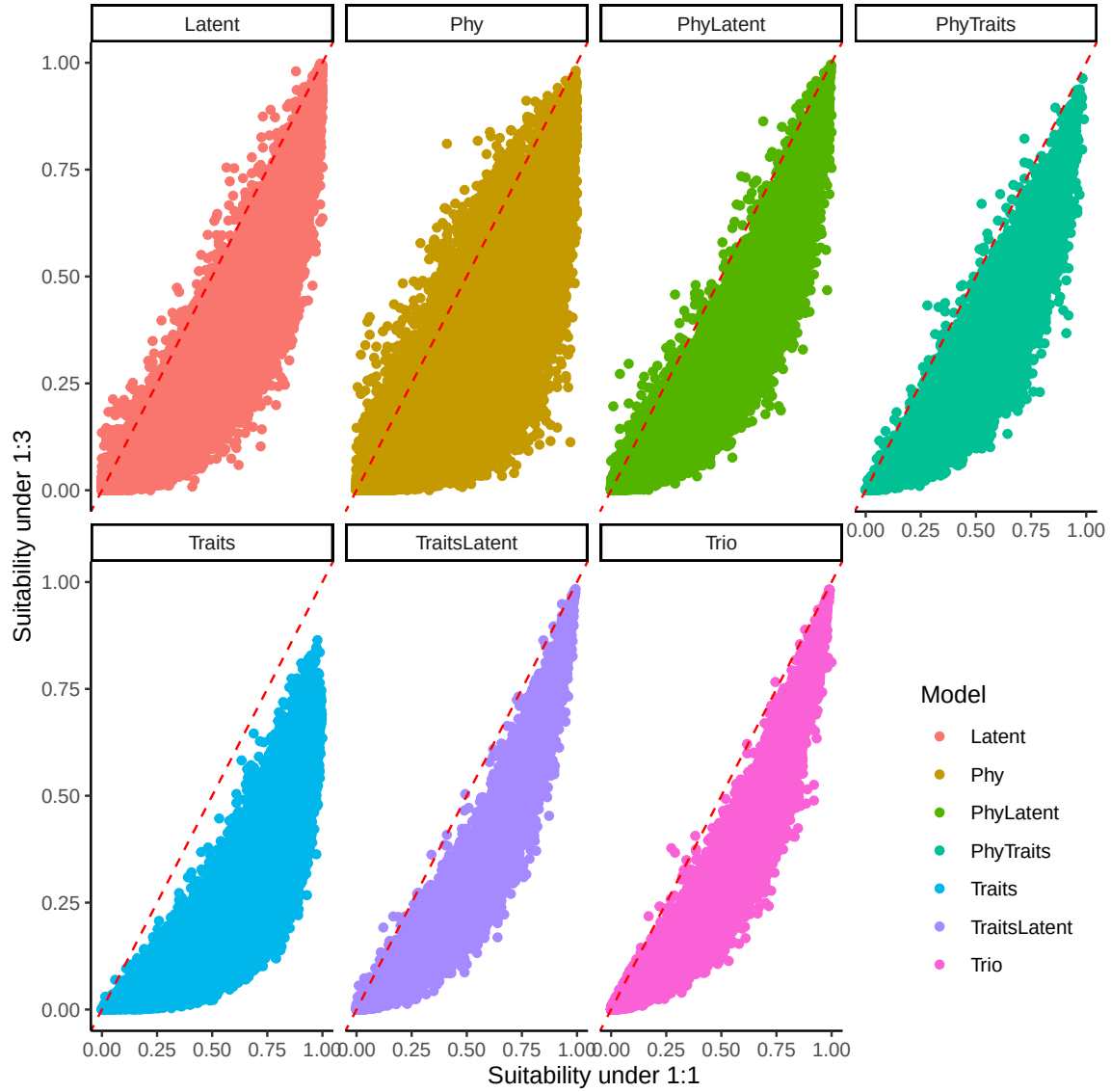


Figure A.6 Scatterplot of relative suitability values of the same models trained under either a 1:3 ratio of present to absent interactions (y-axis) or a 1:1 ratio (x-axis); red dashed lines represent a 1:1 line. While the absolute value of suitability values tend to be reduced when the prevalence of true positives is reduced (most points fall below the 1:1 line), the relative rank of suitability values tends to be very closely aligned (Spearman's $\rho = 0.964$ across all predictions).

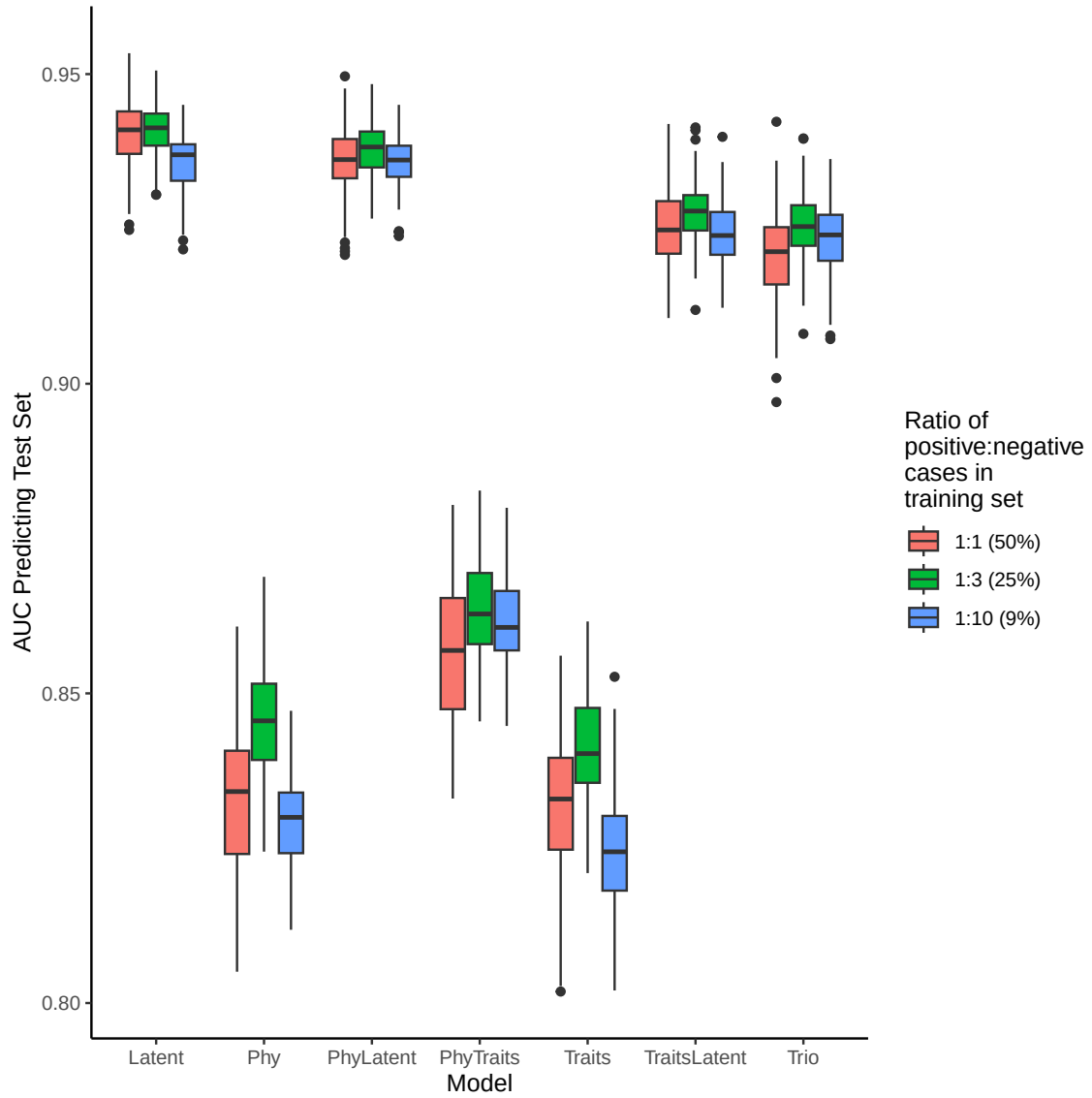


Figure A.7 Box plots of AUC values for model predictions on a 20% test set after being conditioned on training data with either 1:1, 1:3, or 1:10 ratios of presence to absence values. We see that while there is some variation in model performance according to training prevalence, the relative performance of each model is still qualitatively the same across the range of prevalence values.

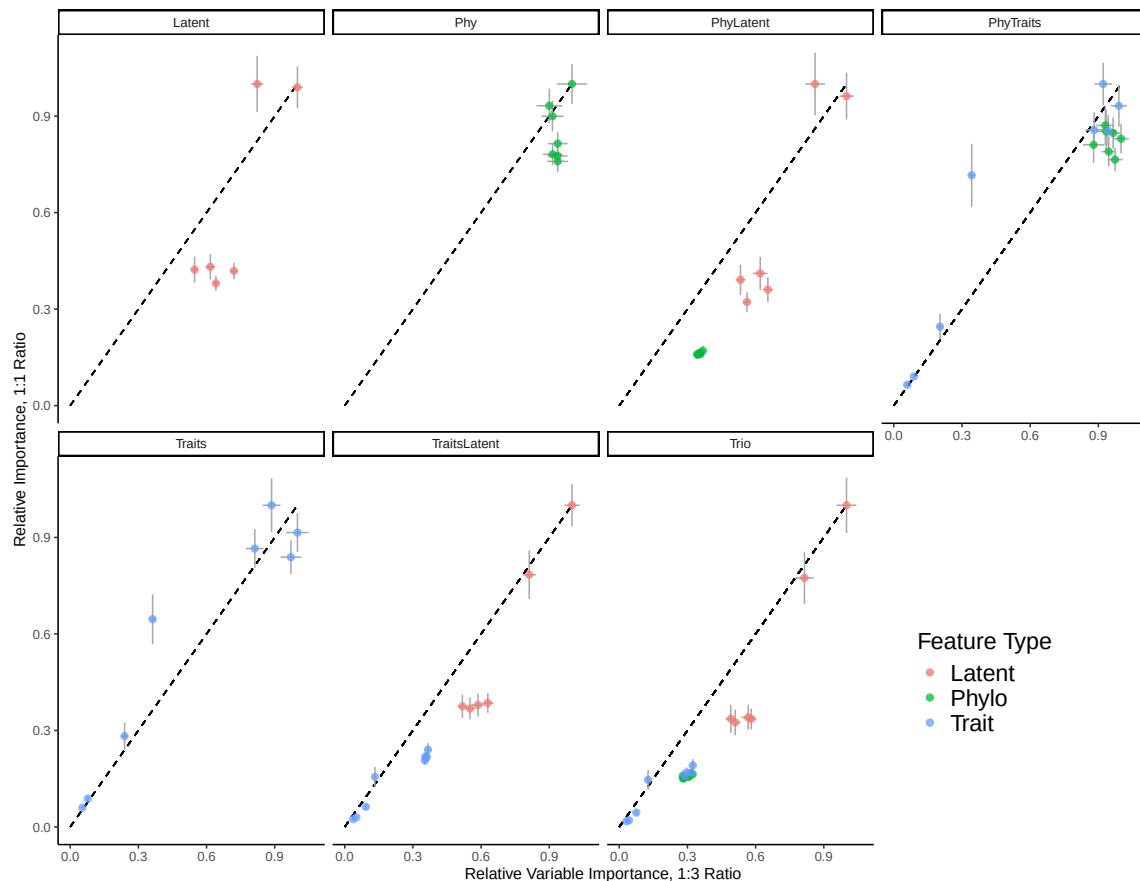


Figure A.8 Variable importance between the same models trained on either a 1:1 ratio of present to absent interactions (y-axis) or a 1:3 ratio (x-axis). Dotted black lines represent a 1:1 line; points that fall on the line indicate that relative importance of that variable was unaffected by training prevalence. We see that the majority of features fall very close to the 1:1 line, including the most important variables (top right of each inset plot). This indicates that models are relatively insensitive to training prevalence.

Sensitivity of Latent Traits to Incomplete Sampling

Latent traits are derived from the observed network, which is an incomplete sample of some underlying "true" network that we're trying to predict. We might expect the ability of latent features to effectively predict links to be different across networks with different sample completeness. Here, we, attempt to characterize that sensitivity

through a null model approach.

The function below takes an arbitrary number of plants, frugivores, and "true" links between them and simulates a network based on those characteristics. Each frugivore and plant species is assigned an abundance value randomly drawn from a lognormal distribution. Links are randomly assigned across all potential interactions according to the abundance of both potential partners. Links between abundant species are more likely than links between rare species. While a simplistic, neutral process, this graph generation feature recapitulates long-tailed degree distributions and asymmetric specialization patterns frequently observed in empirical networks.

Given this "true" observed network, we perform single-value decomposition to create latent features, extracting the first three axes of variation for all nodes as in the main text. From these features we create a euclidean distance matrix that represents the pairwise distances of each intraguild node combination in the three-dimensional space defined by these three latent trait axes. The result is $f \times f$ and $p \times p$ distance matrix, where 'f' and 'p' are the number of frugivores and plants, respectively.

We then degrade this network, removing 50 links from the full set, again sampling non-randomly so that less abundant combinations of species are more likely to go unsampled. This serves to approximate an incomplete sampling process in an empirical system, where links between abundant partners are likely to be observed, but links between rare partners are more likely to be missed. We then repeat the process of single value decomposition and creation of an intraguild pairwise distance matrix. We then use spearman's rank correlations to compare the "true" distance matrix with that produced from the degraded network. If latent features are able to separate the ranks of most nodes even despite degradation, spearman's rank correlation should be high. However, if degrading the network severely alters node clustering, spearman's correlation should be low between the true and observed distance matrices. By re-

peating this process across a range of degradation values (each time comparing back to the "true" network), we can start to characterize the sensitivity of latent features to incomplete sampling.

Below, we parameterize the "true" model with 787 plant species, 242 frugivores, and 3643 "true" links, the same size as our empirically observed network. We repeat the process of degrading the network 50 times, and present average decreases in spearman's rank correlation.

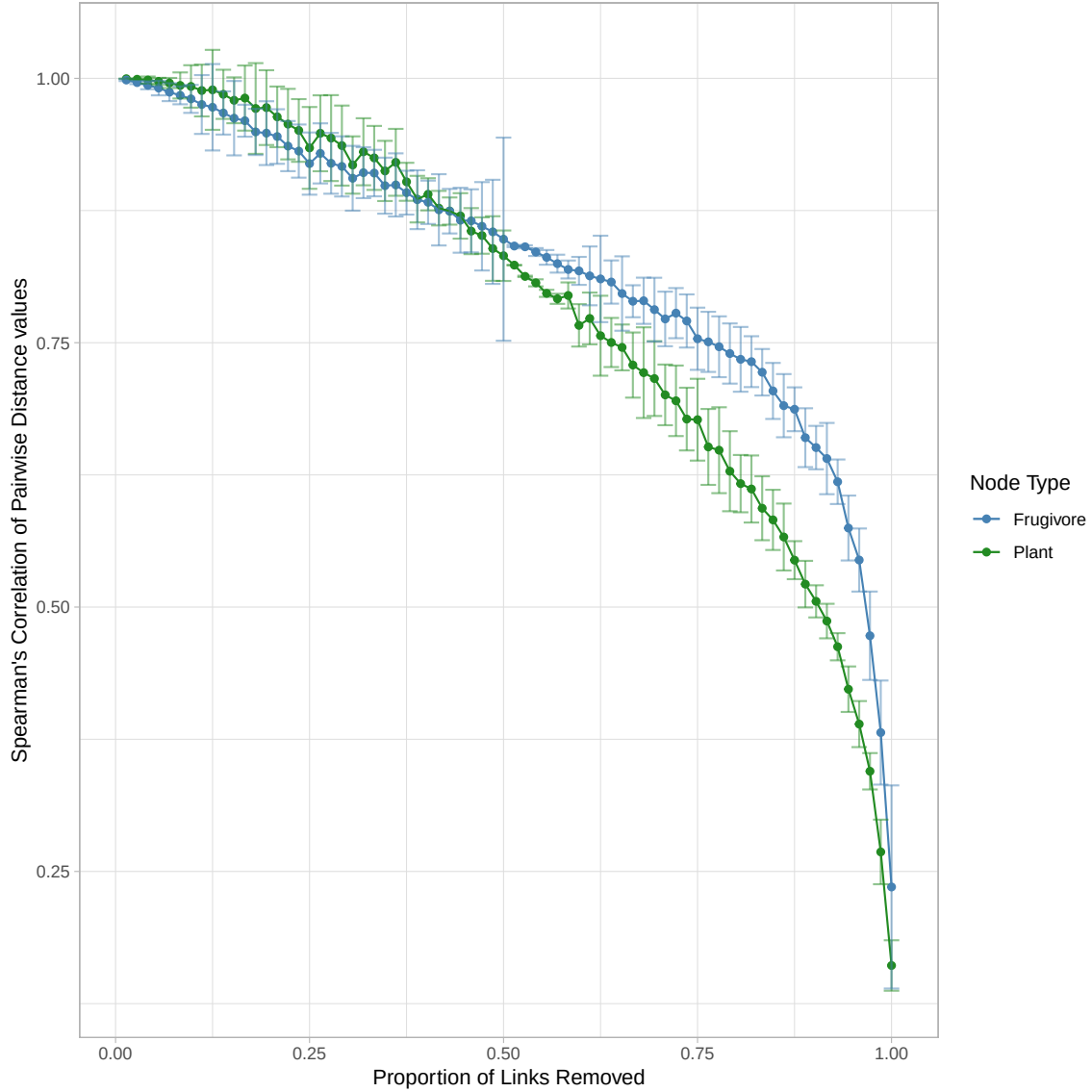


Figure A.9 Plot of intraguild latent pairwise distance matrices derived from compared to the true network (spearman's rank correlation); lines represent mean values across 50 iterations, while error bars represent standard deviation. Correlation decreases as links are removed for both frugivore (blue) and plant (green) nodes, but generally decreases slowly until a high proportion of links are removed. Distance matrices exhibit $\rho > 0.75$ for plant and frugivore species even after removing 64.5% and 76.9% of links, respectively.

We also are interested in the ability of latent features to capture neutral processes given the aforementioned feature of incomplete sampling. Below, we simulate

a number of "true" networks where attachment is again only a function of species abundances drawn from a lognormal distribution.

Next, we create a situation where empirical networks are generated using the generation process outlined above, and we again only observe a portion of the "true" links. We define a sampling rate, where we specify the number of observed links.

We use singular value decomposition on this observed network, and use those features to inform a random forest model trained on the observed positive interactions and enough background points to create a 25% prevalence. We evaluate this model through AUC on the held-out positive links as well as enough new background points to again achieve a 25% prevalence on the test set.

Below, we simulate this process across a variety of a) graph sizes, b) connectance values of the "true" graph, c) sampling completeness values, in order to characterize the potential of latent traits for prediction across this parameter space. This neutral case represents the *absolute simplest* case in which latent features may be useful. The only factors impacting species' probability of interacting are their respective abundances, while in real systems there may be many unmeasured factors impacting interaction structures which latent features may be able to detect. It is however a useful investigation into how latent features might change across this parameter space. The results of these simulations are found below.

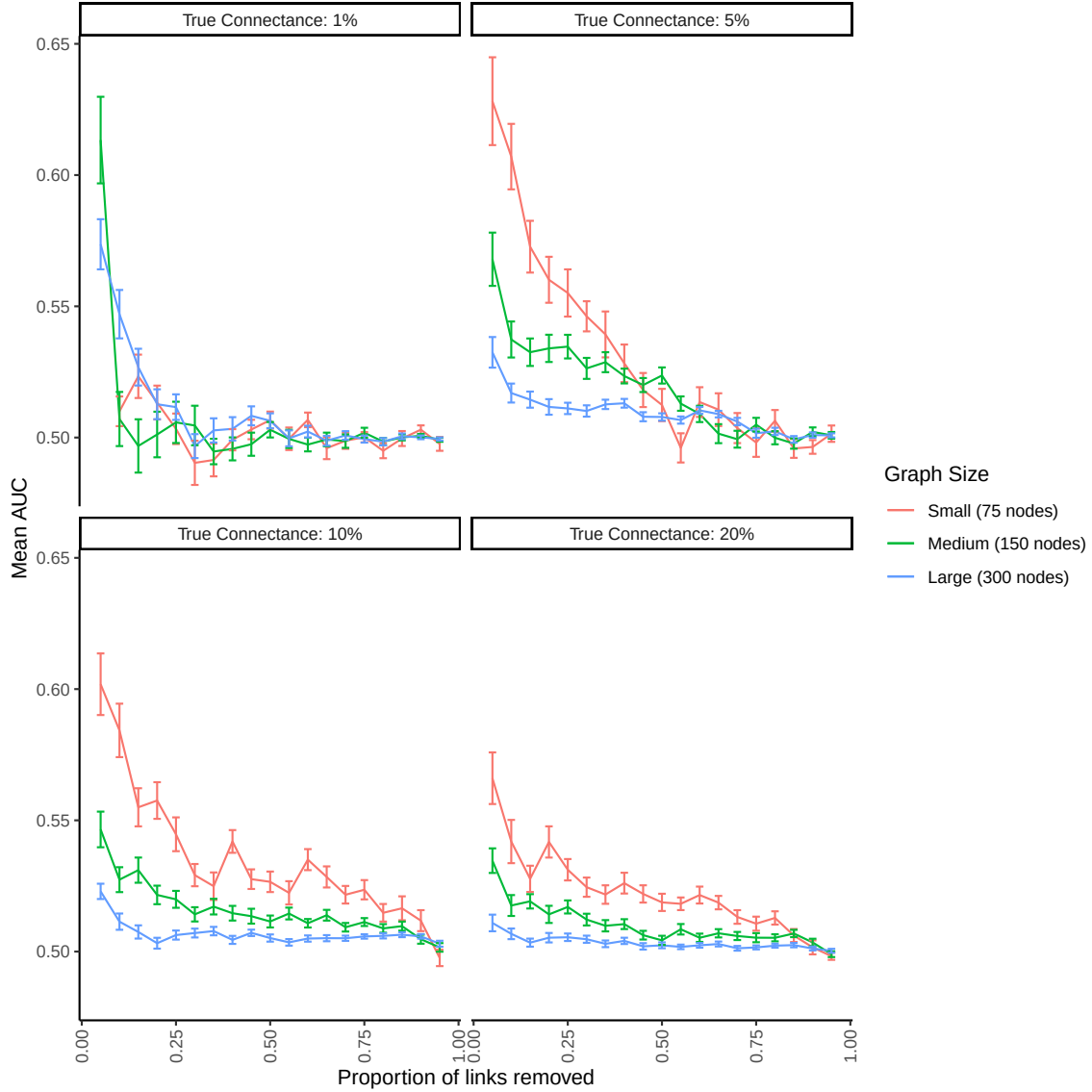


Figure A.10 Plot of mean decrease in AUC performance as a function of the proportion of links removed from the observed set. Plot facets represent alternative connectance values of the true graph; color represent alternative graph sizes. Aside from the extreme sparse case (connectance = 1%), latent features tended to perform best on the smallest graphs. Performance overall was highest on moderately sparse graphs (5% connectance); across all connectance values and graph sizes, model performance decreases with greater sample degradation.

In the main text version of this manuscript, while validating model performance (presented in Figure 1) we use a cross-validation approach suggested by Dr. Pedro

Perez-Neto. In this approach, latent features are created for each test set independently, minimizing the chance of information leakage between test and train sets and thus facilitating comparisons of the predictive capacity of latent and non-latent features. When trying to impute missing links from a network in practice however, using the entirety of the network to create latent features is likely to improve performance. Below, we compare latent features created from these training subsets and those created from the full network.

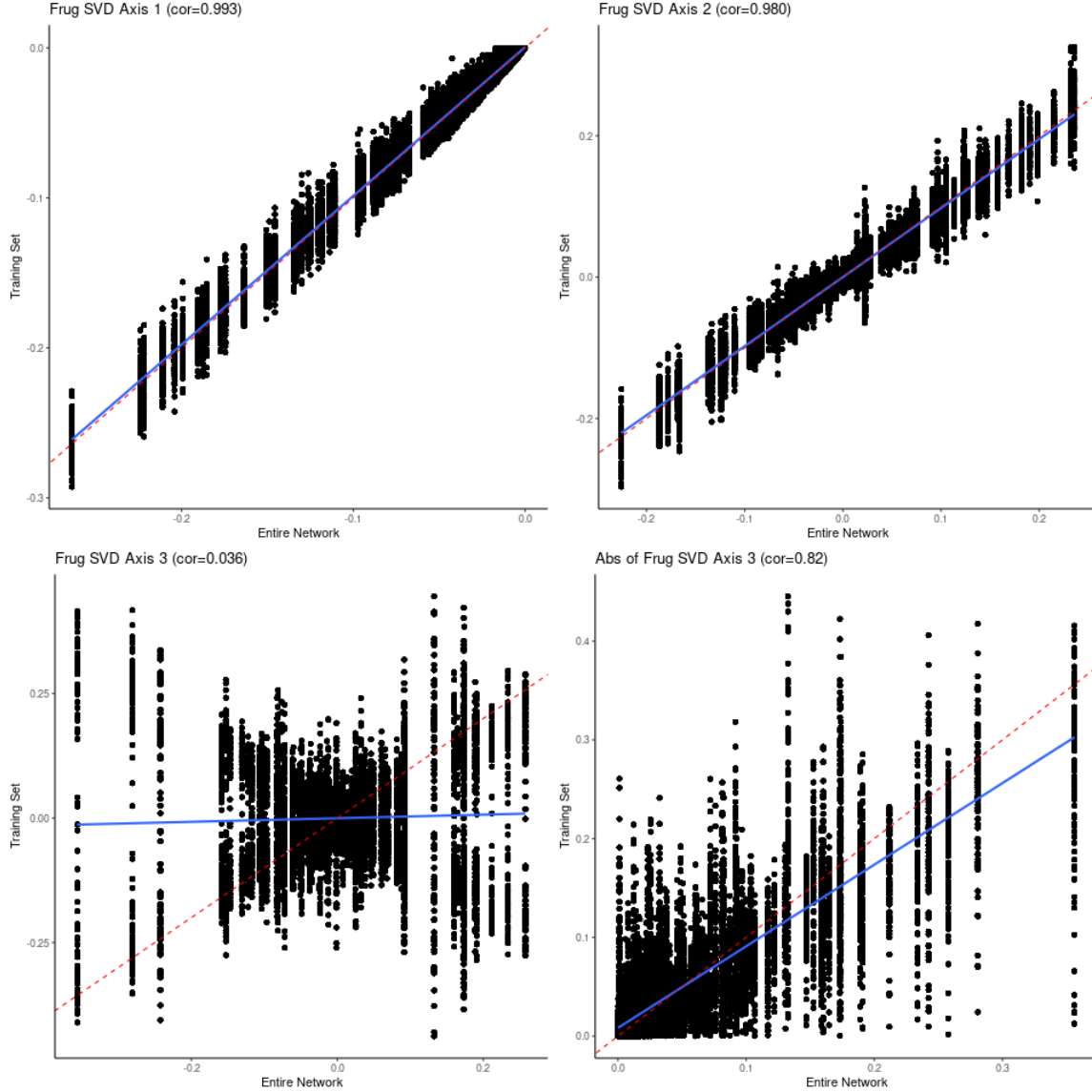


Figure A.11 The first three SVD values for each frugivore species when calculated from the entire network (x axis) and random 80% training sets (y-axis). Red dashed lines are 1:1; blue solid lines show a best fit linear model. Pearson's correlation coefficients are presented in plot titles. Axes 1 and 2 are strongly related, even when a 20% test set of interactions are withheld. While the exact relationship of axis 3 may be either positive or negative depending on the exact training values selected, the absolute values of the training subset are linearly related to the full networks.

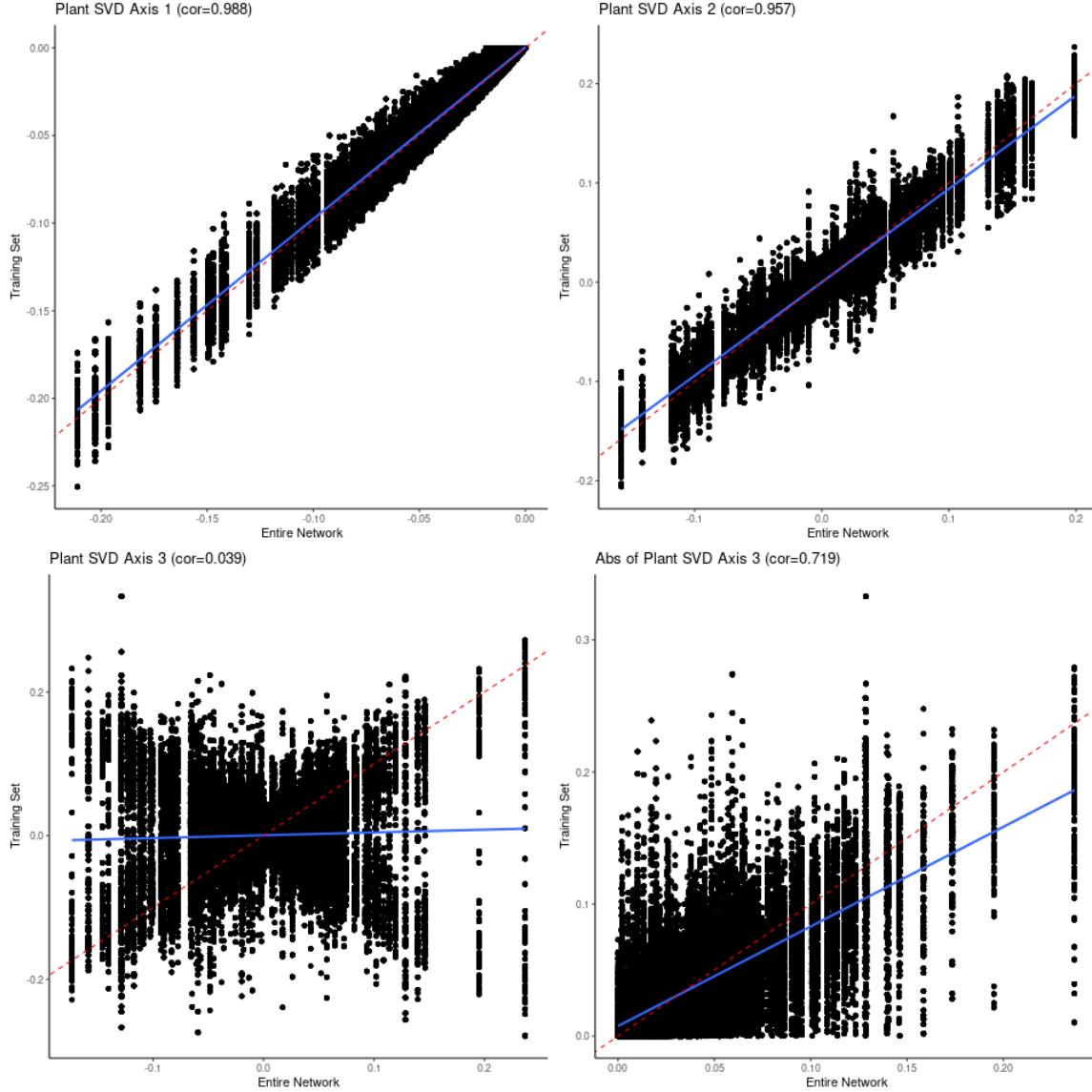


Figure A.12 The first three SVD values for each plant species when calculated from the entire network (x axis) and random 80% training sets (y-axis). Red dashed lines are 1:1; blue solid lines show a best fit linear model. Pearson's correlation coefficients are presented in plot titles. Similar to frugivores, axes 1 and 2 are strongly related, even when a 20% test set of interactions are withheld. Again, while the exact relationship of axis 3 may be either positive or negative depending on the exact training values selected, the absolute values of the training subset are linearly related to the full networks.

Below we repeat our main text analysis and illustrate predictive comparisons when

latent features are created from the entire network, and then afterwards subsampled into test and training sets. In this way each individual model is still only trained on 80% of observed interactions and corresponding background points, but the features associated with each interactions are created based on the entire network.

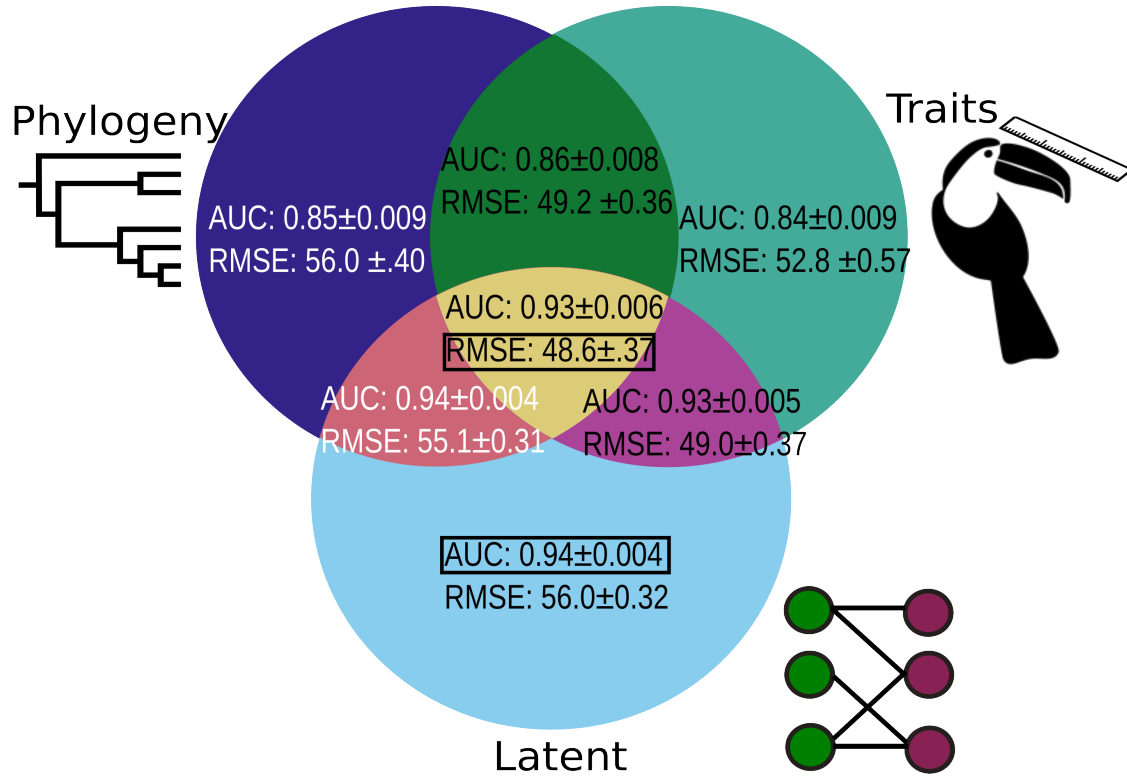


Figure A.13 Model performance metrics when latent features are created from the entire network prior to dividing data into test and train sets. While slightly increasing the performance of latent features (potentially through some amount of information leakage between test and train sets), qualitative patterns as presented in the main text are still the same.

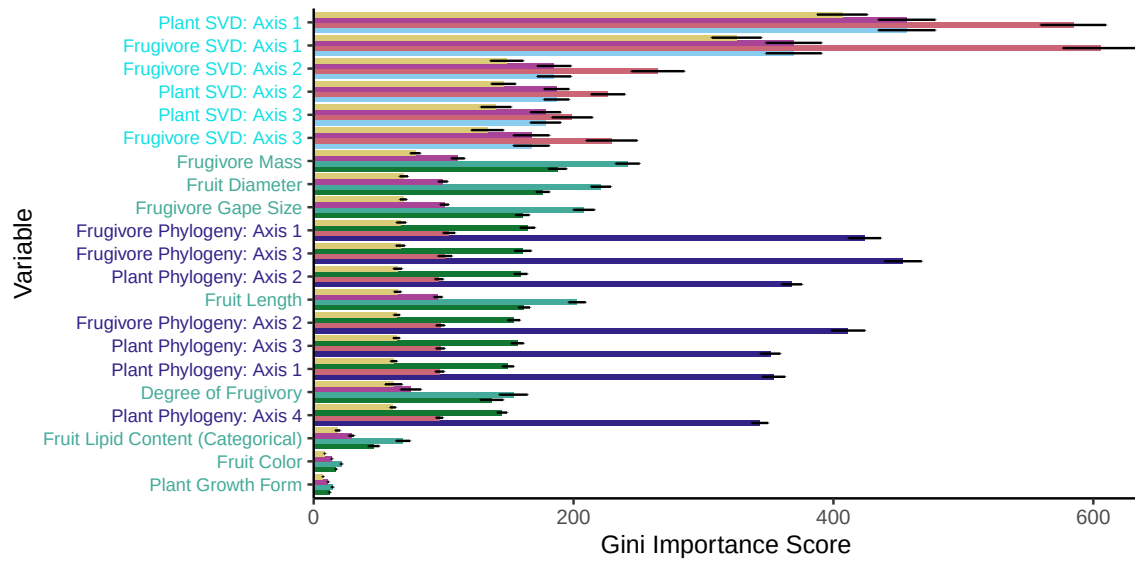


Figure A.14 Variable importance plots for models trained on 80% training subsets, with latent features recalculated within each subset. The qualitative patterns are consistent with those observed when models are trained on the full interaction network without train–test splits (shown in Figure 2 of the main text).

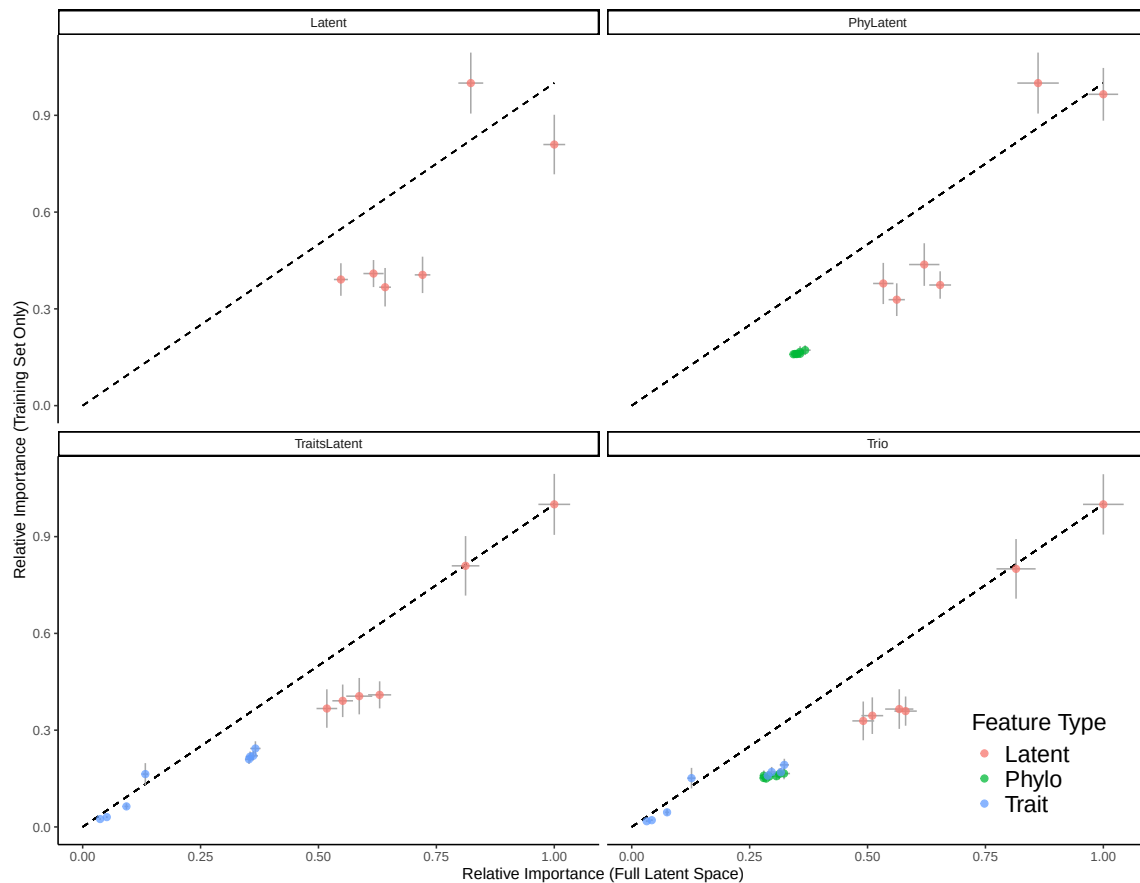


Figure A.15 Comparison of variable importance between the same models informed by latent features calculated from the entire network (x-axis) or only from 80% training subsets (y-axis). Dotted black lines represent a 1:1 line; points that fall on the line indicate that relative importance of that variable was unaffected by the cross-validation procedure. We see that the majority of features fall relatively close to the 1:1 line, including the most important variables (top right of each inset plot).

APPENDIX B

SUPPLEMENTAL INFORMATION FOR CHAPTER 3

Floral visitation networks used in the dataset, queried from the Web of Life dataset, spanned different geographic extents depending on taxonomic focus. Our Plant-insect interaction networks (N=66) spanned the globe, ranging in size from 3 to over 15,000 interactions Robertson, 1929. Avian-plant interaction networks (N=31) were exclusively sampled from the neotropics, ranging from solitary pairs of interactions to networks with as many as 248 unique interactions. Networks with less than four interactions were included in the global metaweb, but were excluded from analyses of local network properties, resulting in the exclusion of one insect and three avian networks from the analysis of local network properties.

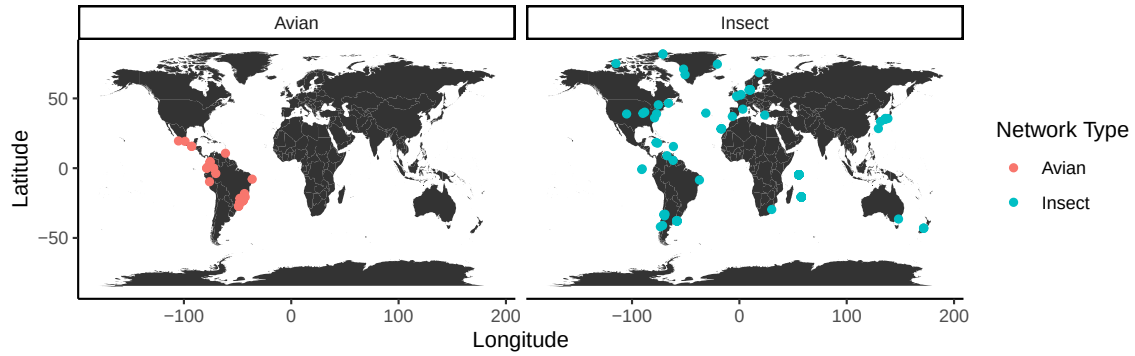


Figure B.1 Map of networks used in our study. Each point represents a unique local network, created by observing avian (Red, $N=31$), or insect (Blue, $N=66$) floral visitations.

Species that occur in multiple local webs may exhibit different centrality values, dependent both on the compositions and connections observed within any given local community. For species that occur in multiple webs of our dataset, we attempt to predict mean z -transformed centrality across all webs where a species is found. To show the variability species may exhibit across networks, we plot the distributions of closeness centrality for species that occur at the most sites across our dataset. Differences between raw and z -standardized metaweb centrality stem from the number of cooccurring extraguild partner species. Narrow-ranged plants that cooccur a small set of potential pollinator species are less likely to be highly central in the global metaweb, regardless of local connectivity. We show the relationship between geographic range size and the number of extraguild cooccurring species in Figure B.5.

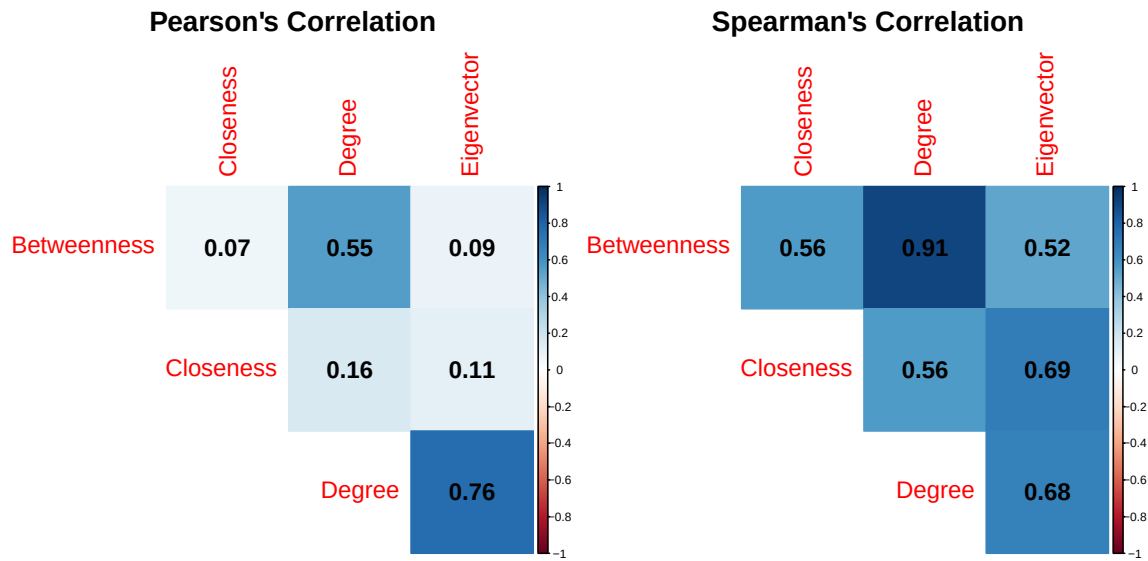


Figure B.2 Pairwise correlation of common centrality measure as applied to our global metawebs. While the pearson’s correlation of most measures are only slightly positive, spearman’s rank correlation shows the ordering of nodes according to alternative measure is relatively conserved. In maintext we present results of closeness centrality, a middle-ground between hyper-local and hyper-global measures.

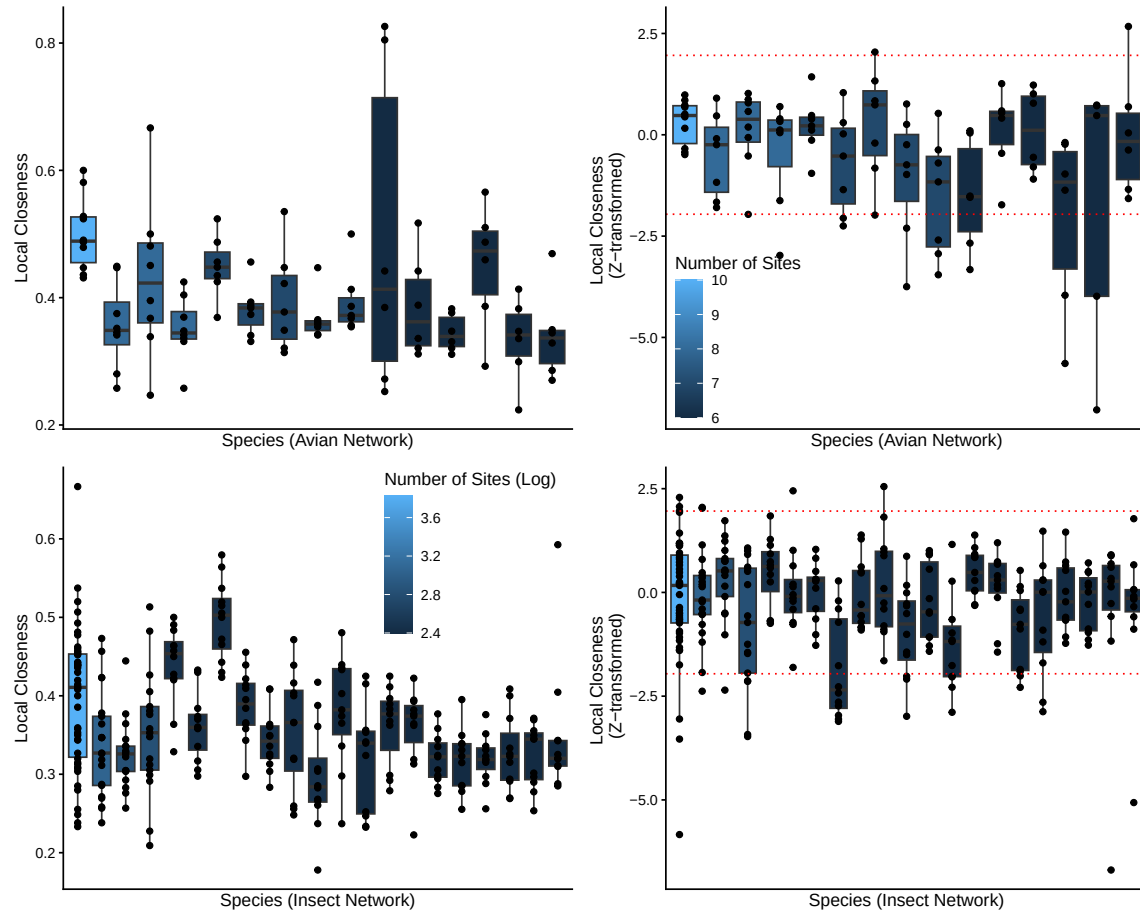


Figure B.3 Distribution of closeness centrality in local networks for species that occur at multiple sites. Top row of plots represent nodes in bird-plant networks with 7 or more occurrences. Bottom row represent nodes in insect-plant networks with 11 or more occurrences. Left columns represent unstandardized local closeness centrality values. Right column represents the same points after z-score standardization with respect to 1000 null model randomization (see text).

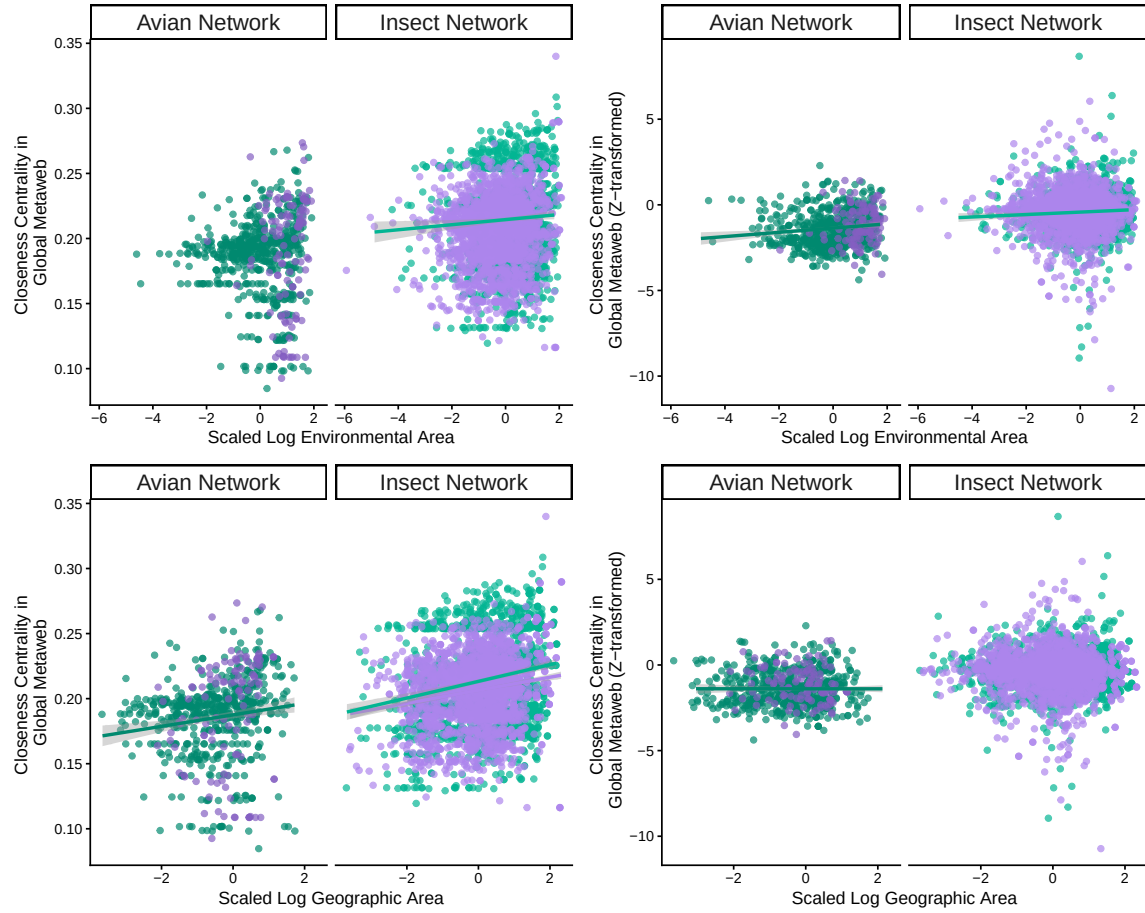


Figure B.4 Species-level closeness centrality in the global metaweb as a function of environmental niche breadth (A,C) or geographic range size (B,D). Plots in the left column show observed centrality values(A,B), while those in the right column show z -transformed centrality values based on permutations of local networks(C,D). Points and line colors respond to plant (green) or floral visitor (purple) nodes, respectively. Colored lines depict linear relationships between variables with POP_2 values < 0.05 in our Bayesian GLMs (included for illustrative purposes only).

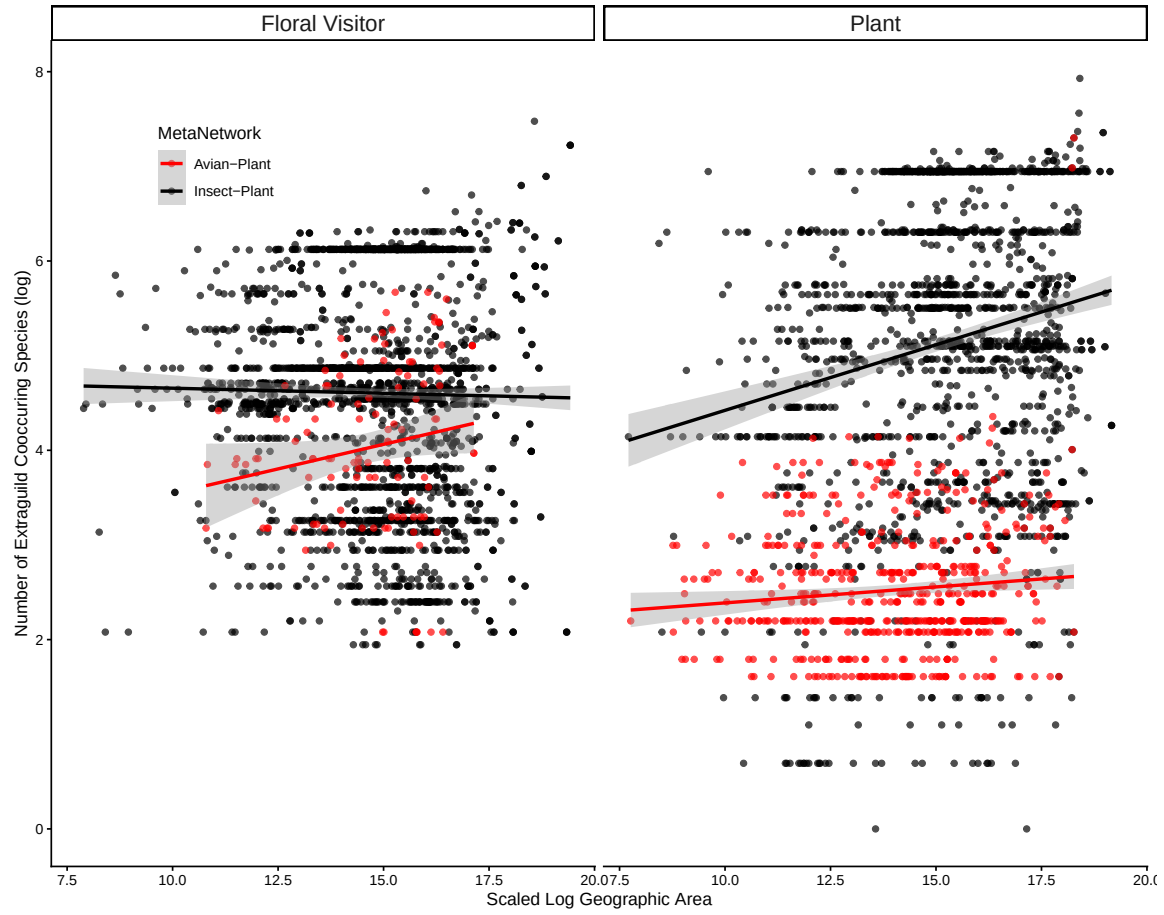


Figure B.5 Relationship between log geographic range size (x-axis), and number of potential partners (extraguild species) it cooccurs with in our dataset. Shaded lines represent 95% confidence intervals of a linear model. For insect-pollinated plants, we see a significant positive relationship between a species' geographic range size and the diversity of insect species it coocures with alongside its range (alpha, p). We see no significant effect of range size on potential interactor richness for avian-pollinated plants, or either pollinator group

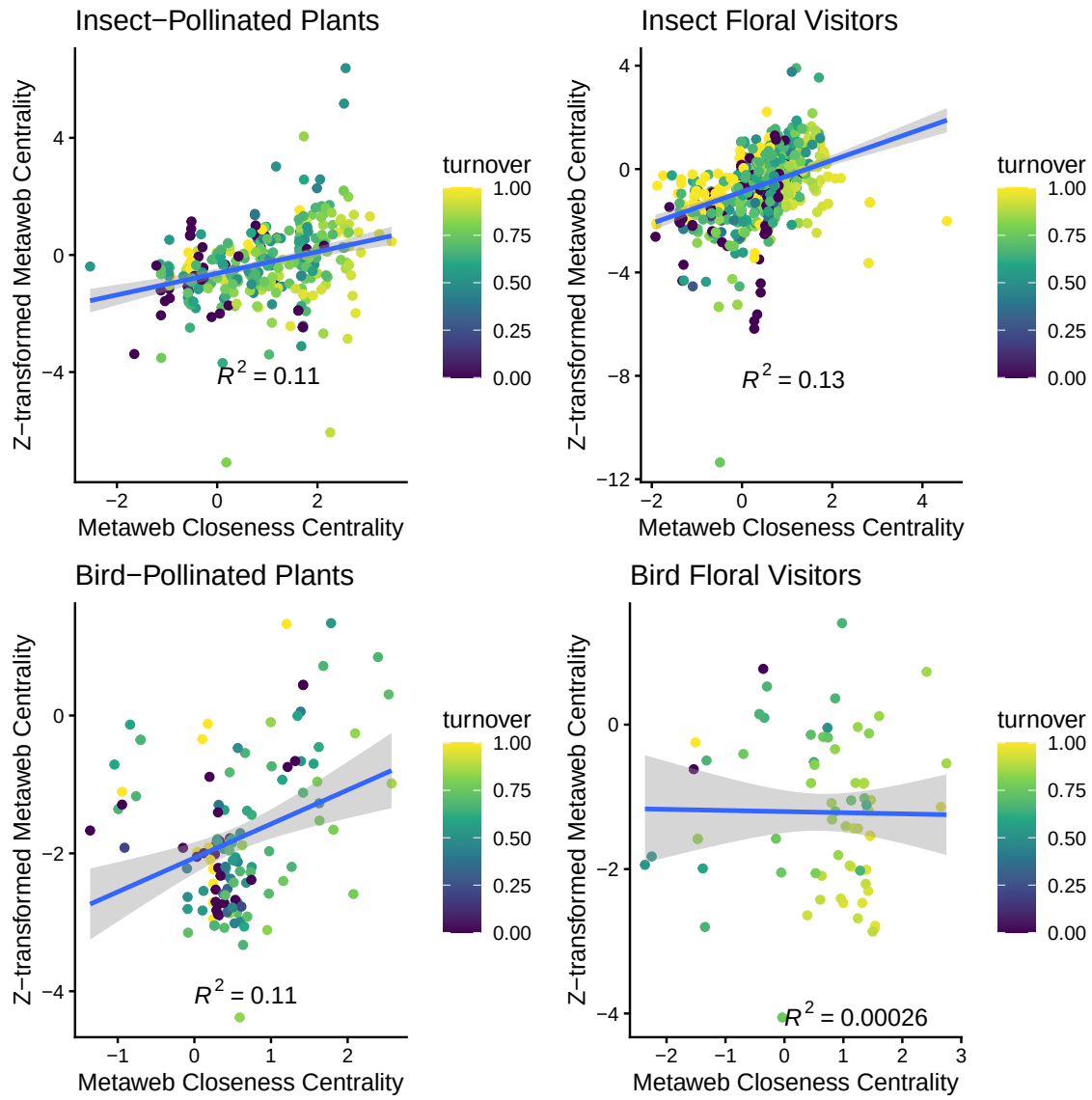


Figure B.6 Comparison of unstandardized (x-axis) and z-standardized (y-axis) closeness centrality in global metawebs across our four node types. Points are colored by interaction turnover across sites. While there is a weak positive relationship between our centrality measure across most node types, the lack of strong explanatory power indicated that our z-transformation procedure does indeed capture alternative aspects of node connectivity at the metaweb scale.

Below we present diagnostic pairs plots of all bayesian multivariate regression models presented in the main text. Despite a positive correlation between log geographic range size and log environmental range area (Pearson's $r = 0.56$) and negative correlation between fit slope values in some models, correlation structure did not prevent effective inference or MCMC chain convergence ($\hat{R} < 1.01$ across all presented models).

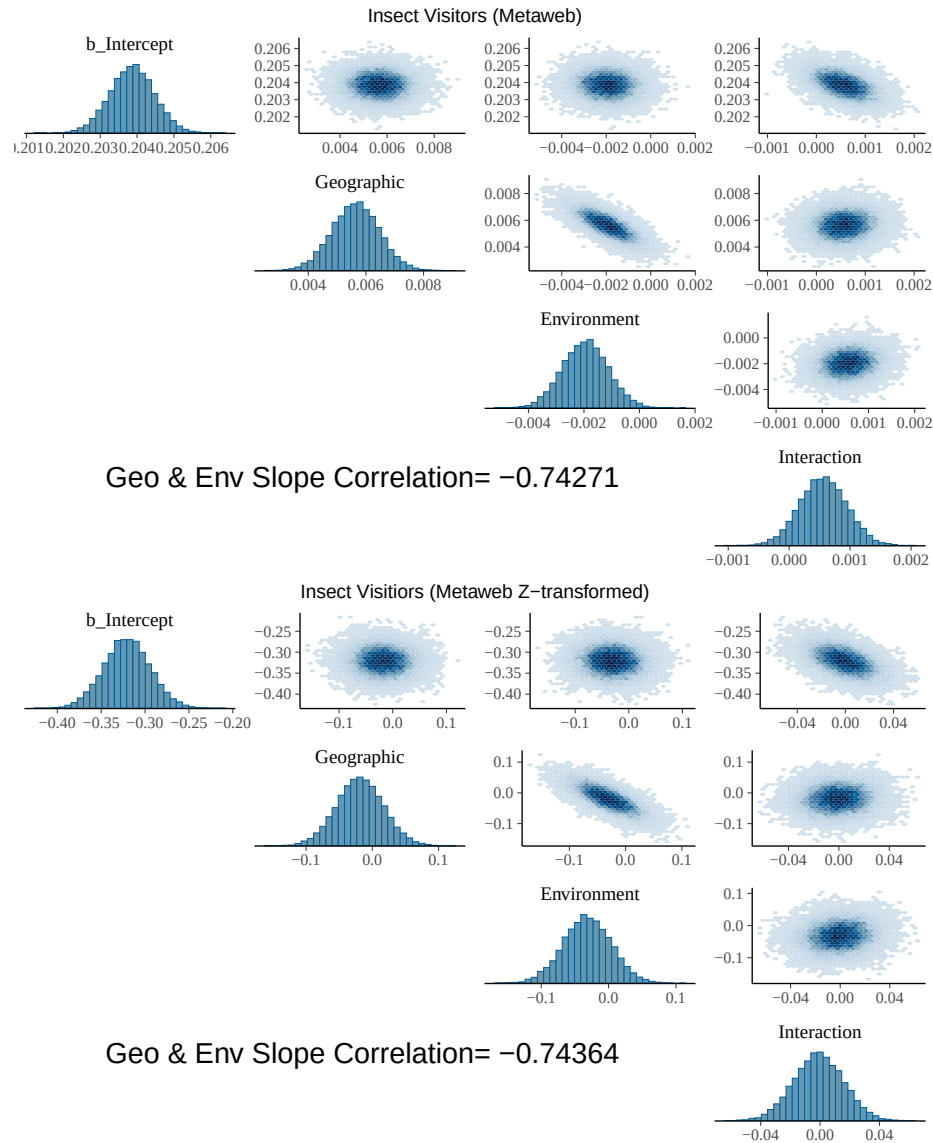


Figure B.7 Pairs plots of fit slope and intercept parameters for models of closeness centrality of **insect floral visitors** in the **global metaweb**. Hex plots represent the bivariate density of parameter estimates sampled from the posterior, while histograms represent univariate densities of each parameter. Top plots represent a model using raw closeness centrality values as a response variable; lower plots represent a model predicting z-standardized centrality. While there is some negative slope correlation between slope parameters for niche breadth and geographic range, observed $\hat{R}$ values <1.01 indicate adequate chain convergence for inference.

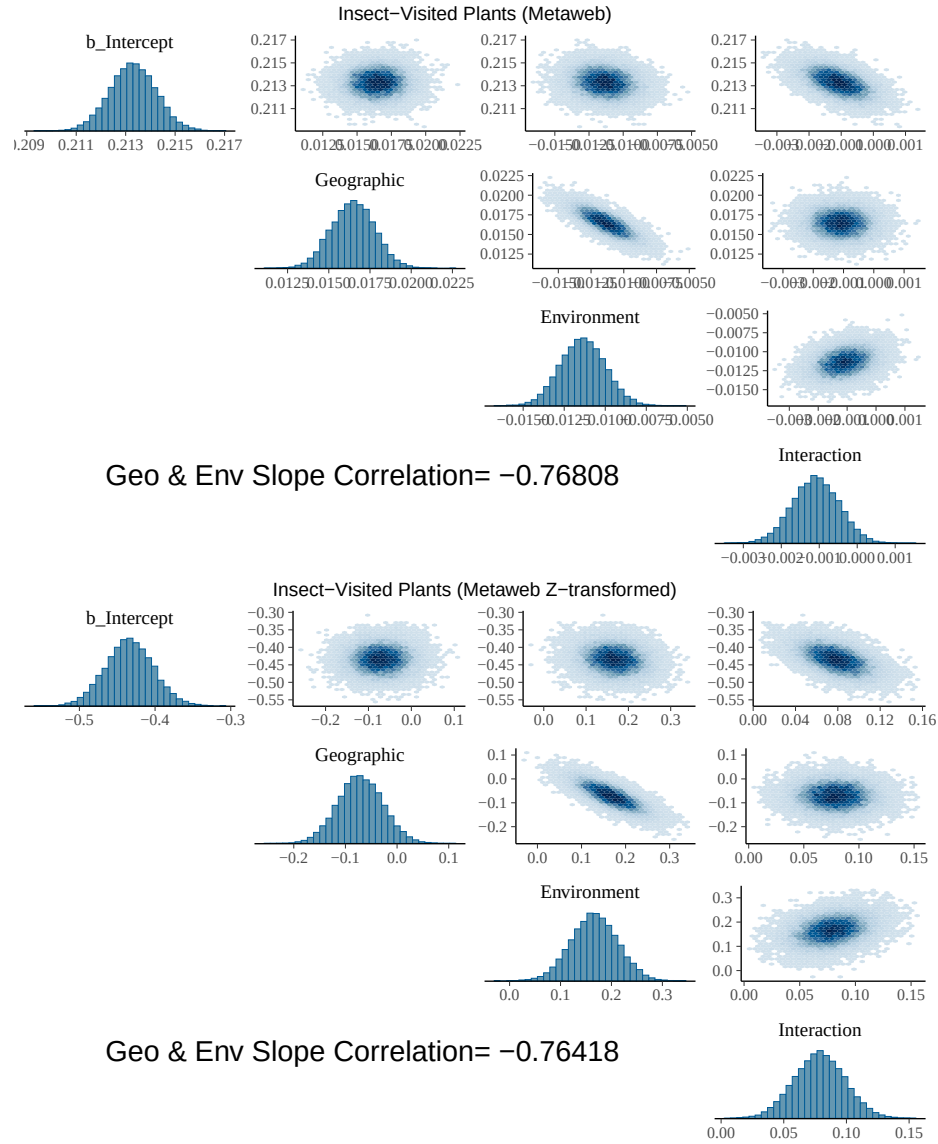


Figure B.8 Pairs plots of fit slope and intercept parameters for models of closeness centrality of **insect-visited plants** in the **global metaweb**. Hex plots represent the bivariate density of parameter estimates sampled from the posterior, while histograms represent univariate densities of each parameter. Top plots represent a model using raw closeness centrality values as a response variable; lower plots represent a model predicting z-standardized centrality. While there is some negative slope correlation between slope parameters for niche breadth and geographic range, observed $\hat{R}$ values <1.01 indicate adequate chain convergence for inference.

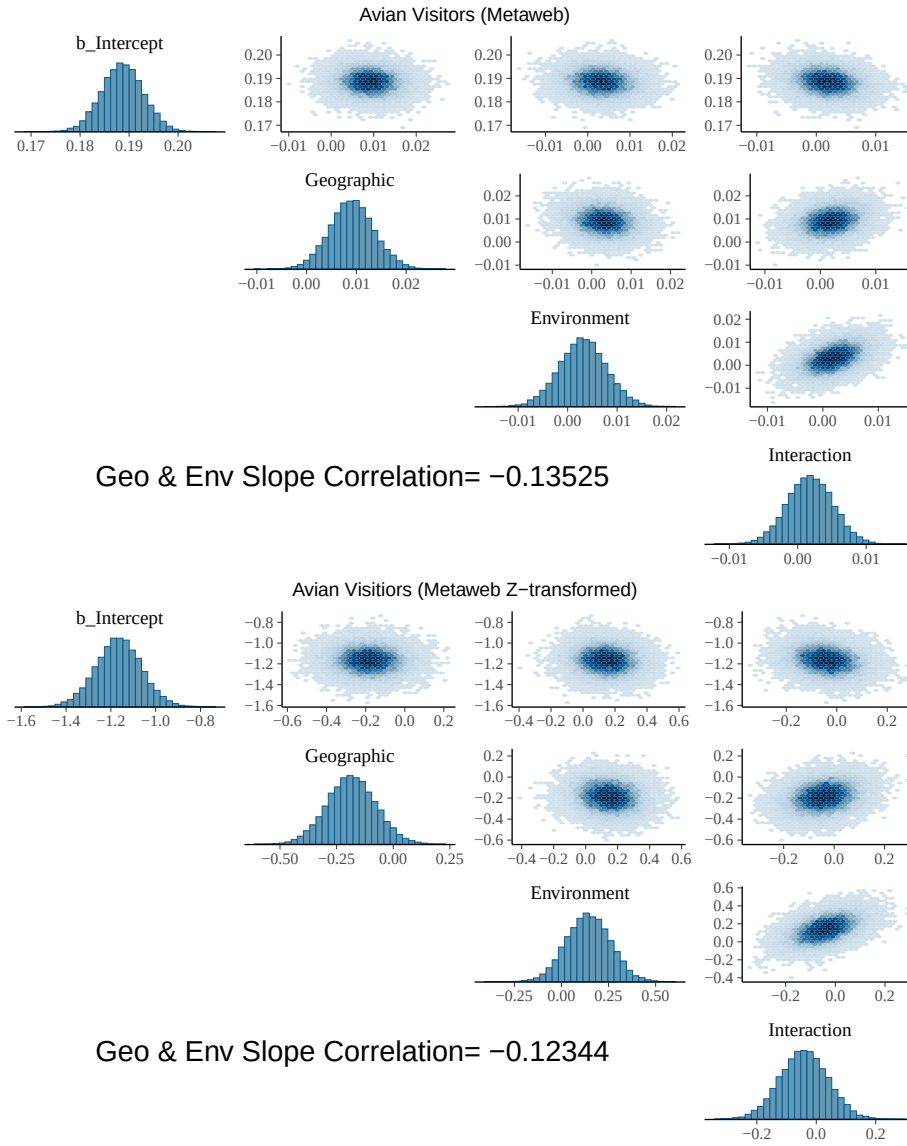


Figure B.9 Pairs plots of fit slope and intercept parameters for models of closeness centrality of **avian floral visitor** in the **global metaweb**. Hex plots represent the bivariate density of parameter estimates sampled from the posterior, while histograms represent univariate densities of each parameter. Top plots represent a model using raw closeness centrality values as a response variable; lower plots represent a model predicting z-standardized centrality.

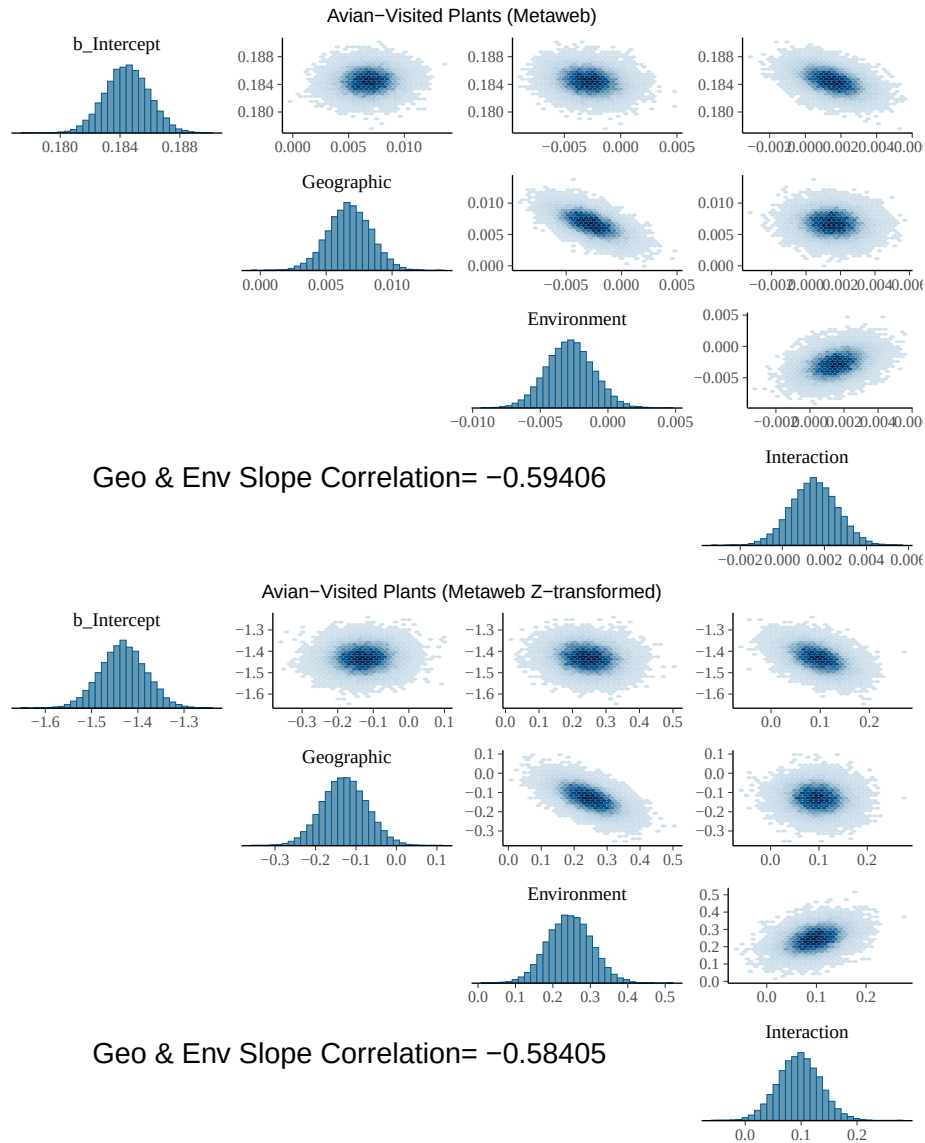


Figure B.10 Pairs plots of fit slope and intercept parameters for models of closeness centrality of **avian-visited plants** in the **global metaweb**. Hex plots represent the bivariate density of parameter estimates sampled from the posterior, while histograms represent univariate densities of each parameter. Top plots represent a model using raw closeness centrality values as a response variable; lower plots represent a model predicting z-standardized centrality. While there is some negative slope correlation between slope parameters for niche breadth and geographic range, observed $\hat{R}$ values <1.01 indicate adequate chain convergence for inference.

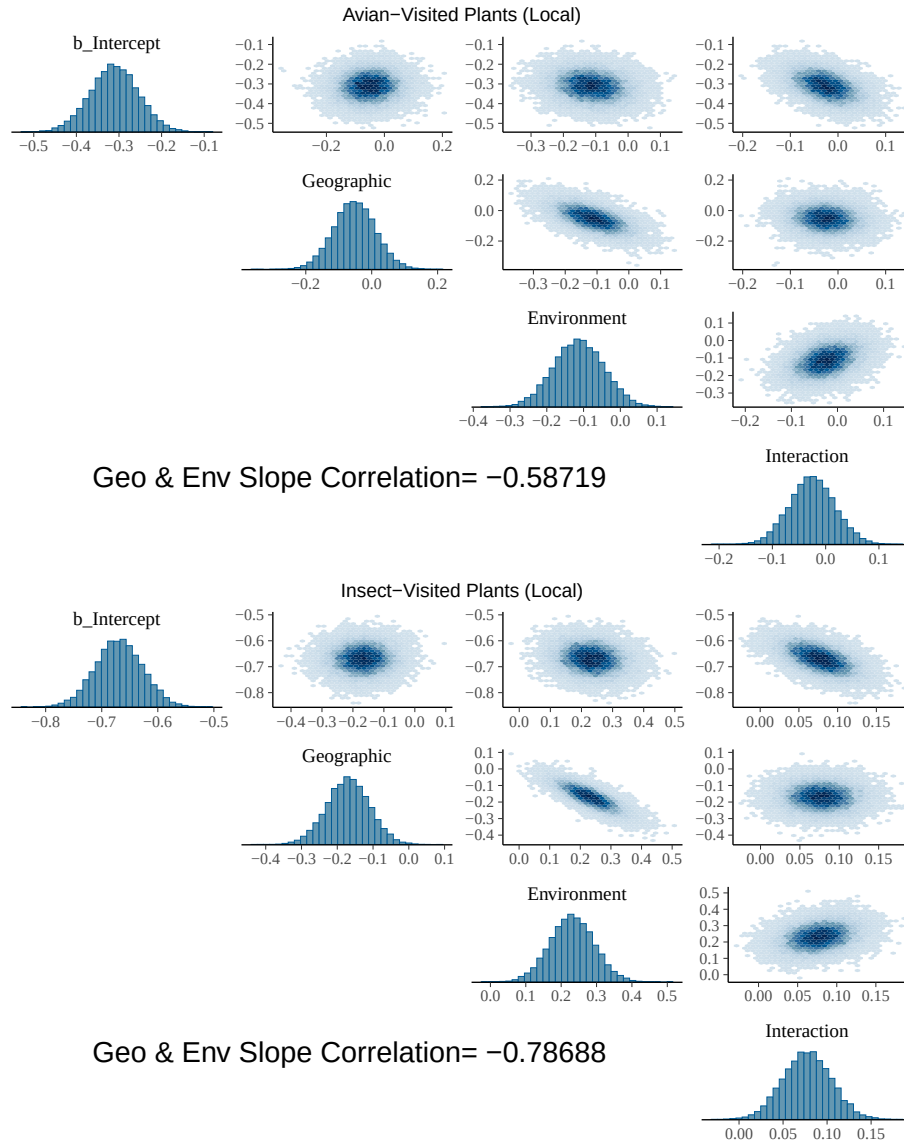


Figure B.11 Pairs plots of fit slope and intercept parameters for models of closeness centrality of **plants** in **local webs**. Hex plots represent the bivariate density of parameter estimates sampled from the posterior, while histograms represent univariate densities of each parameter. Top plots represent plants in avian-plant networks; lower plots represent plants in insect-plant networks. While there is some negative slope correlation between slope parameters for niche breadth and geographic range, observed $\hat{R}$ values < 1.01 indicate adequate chain convergence for inference.

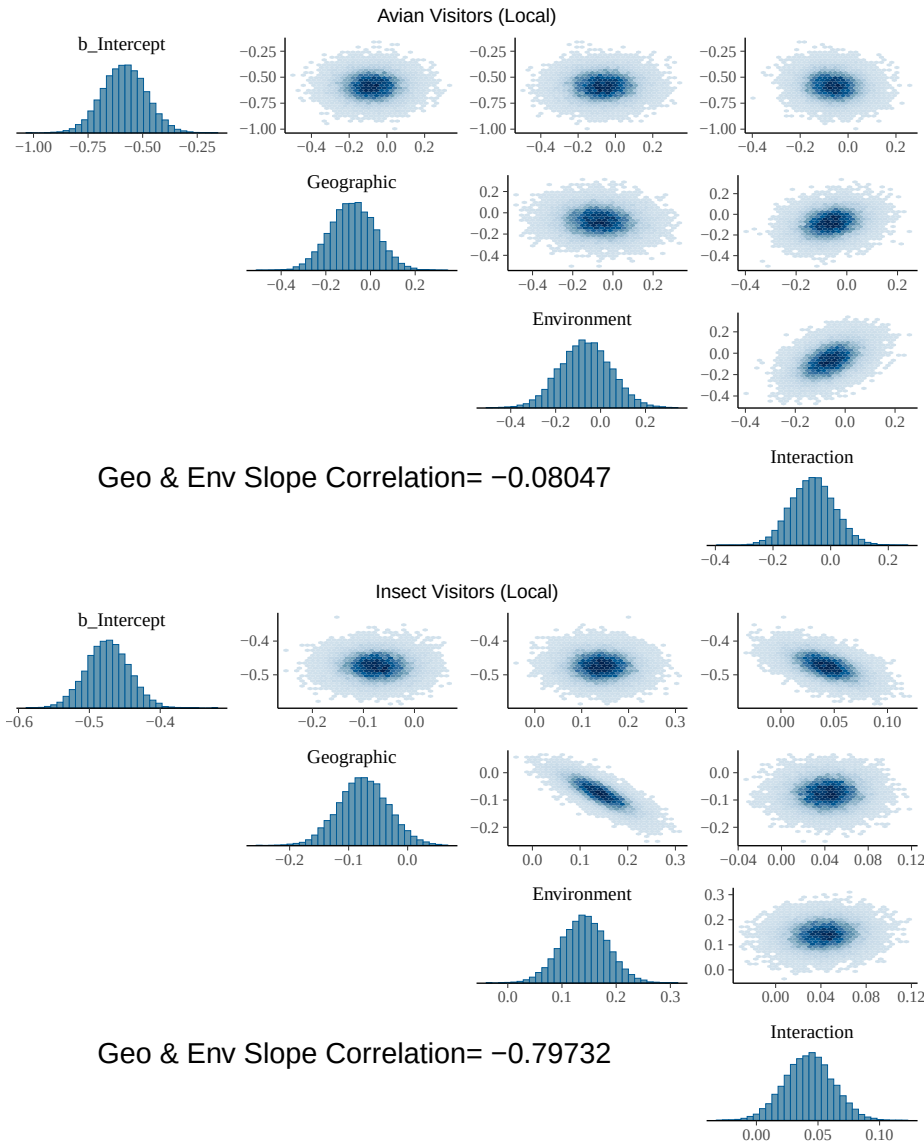


Figure B.12 Pairs plots of fit slope and intercept parameters for models of closeness centrality of **floral visitors** in **local webs**. Hex plots represent the bivariate density of parameter estimates sampled from the posterior, while histograms represent univariate densities of each parameter. Top plots represent birds in avian-plant networks; lower plots represent insects in insect-plant networks. While there is some negative slope correlation between slope parameters for niche breadth and geographic range in the model of local insect visitors, observed $\hat{R}$ values <1.01 indicate adequate chain convergence for inference.

APPENDIX C

SUPPLEMENTAL INFORMATION FOR CHAPTER 4

The relationship between network size and nestedness

Disentangling species richness (the sum of host and parasite richness) from nestedness is difficult, and many nestedness measures scale with species richness (Figure C.1). Comparisons of properties across networks of different sizes is a non-trivial problem, and one potential solution is the implementation of a null modeling approach. In our case, we repeatedly shuffled links between hosts and helminths in each network, while maintaining both parasite host range (i.e., the number of host species a given parasite infects), and parasite species richness (i.e., the number of parasite species a host is infected by). For each network, we produced 1000 randomized networks, creating a distribution of spectral radii. Using these null distributions, we calculated z -scores, which measure the degree of divergence of the empirical network from the randomized null networks. Below, we present the correlations between alternative measures of network nestedness (η (Jonhson et al., 2013), spectral radius, NODF, and $NODF_c$) as well as z -transformed versions of NODF and spectral radius. Nestedness as measured by η was not standardized, as null model randomization for this measure is difficult, as randomized networks maintaining aspects of the degree distribution of host and parasite species will result in the same η value as the empirical network.

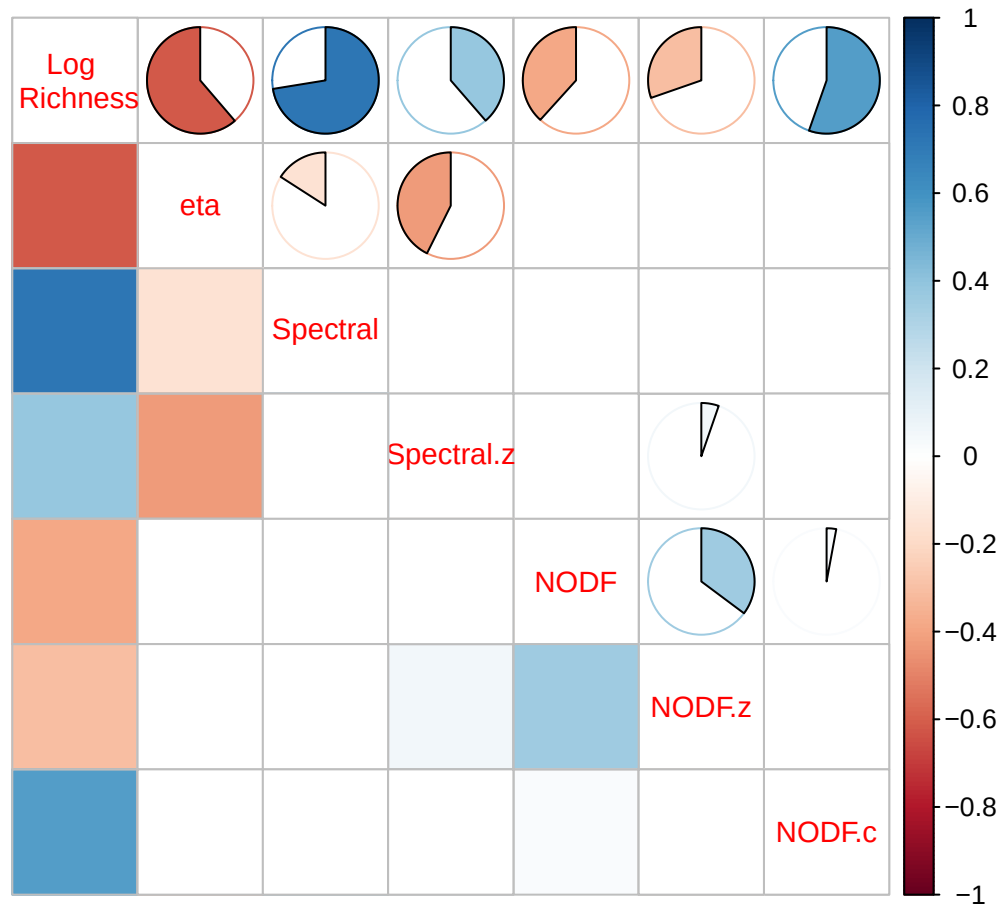


Figure C.1 Correlogram visualizing Pearson's correlation coefficients between network size (log species richness) and alternative measures of network nestedness in our data. Shaded squares represent significant correlations (p -value < 0.05). All measures of nestedness (η (Jonhson et al., 2013), spectral radius, and NODF, as well as z-scores of the latter two based on null randomizations) are significantly correlated with network size in our dataset

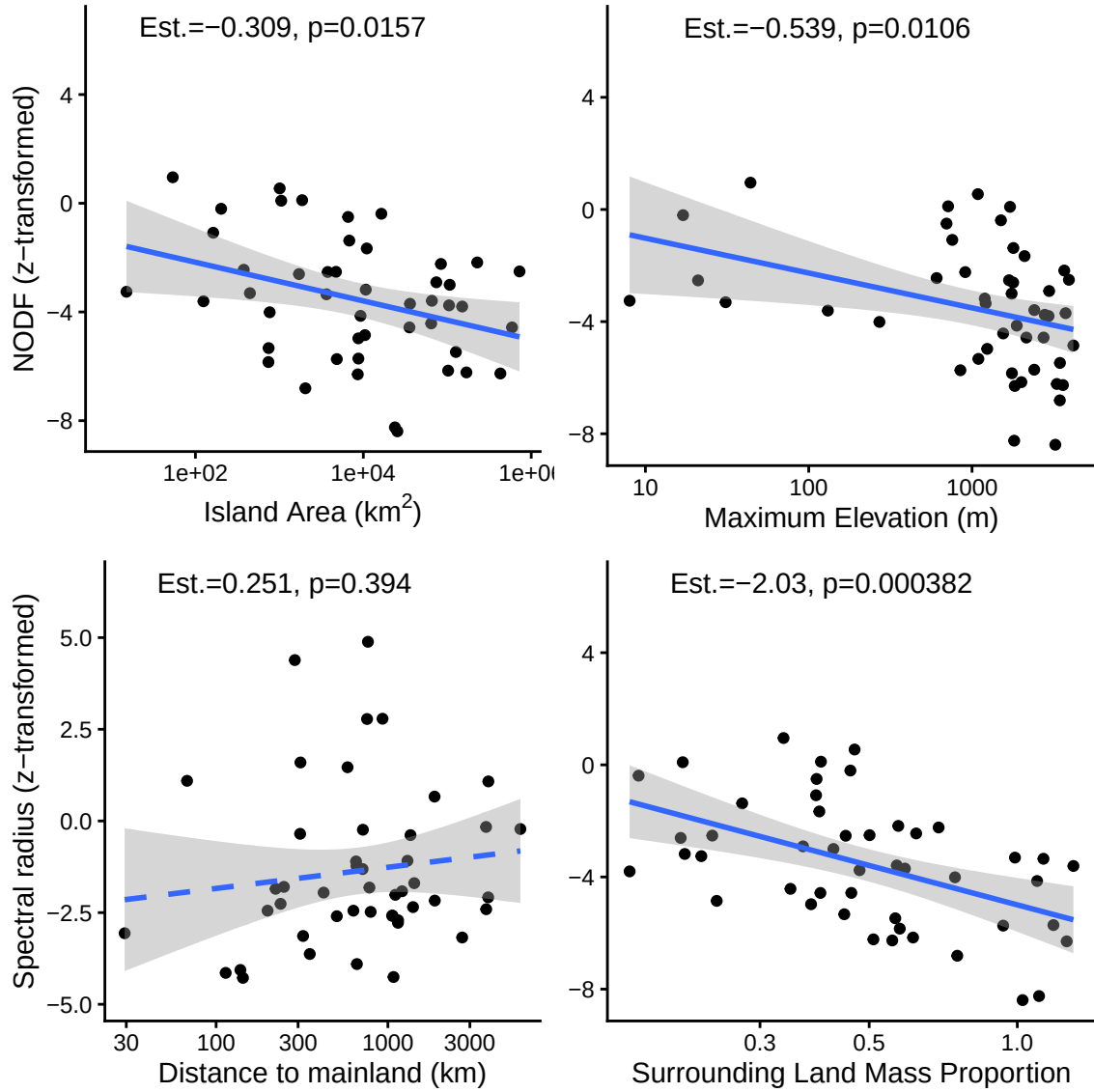


Figure C.2 Recreation of Figure 4.3 of the main text using z-transformed NODF as our nestedness metric of comparison. Despite null modeling approaches, this metric may still be sensitive to network size, and this gives conflicting results to those observed in $NODF_c$.

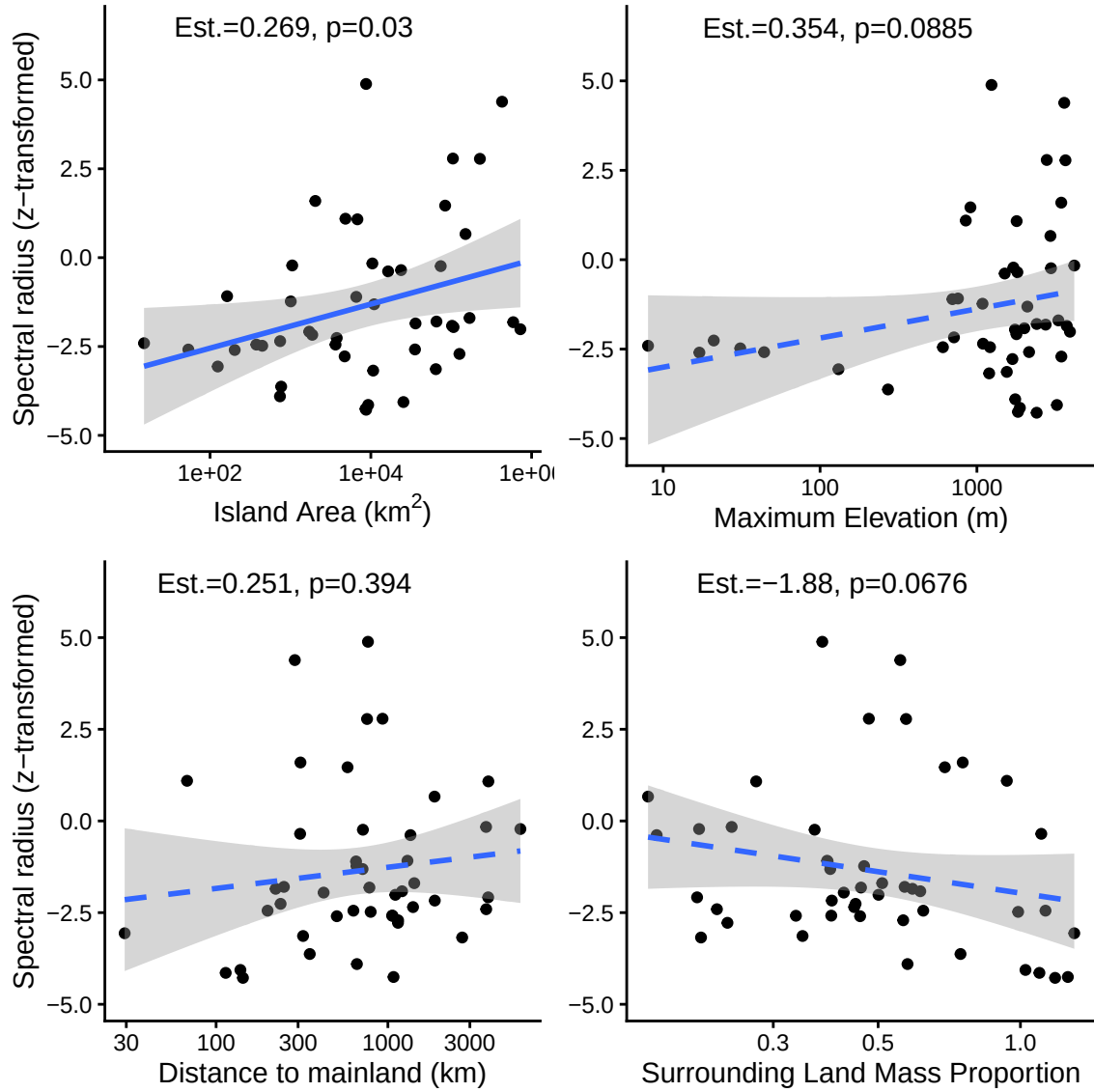


Figure C.3 Recreation of Figure 4.3 of the main text using z-transformed spectral radius as our nestedness metric of comparison. In contrast to z-standardized NODF, spectral radius yields qualitatively similar results to $NODF_c$ after z-transformation.

Statistical Analysis

In the main text, we used a negative binomial modeling multivariate regression approach to model the effects of island characteristics on species diversity. This approach greatly minimizes residual deviance compared to both Poisson and Quasi-Poisson

GLMs C.1. Models incorporating island area and isolation as measured by SLMP yielded lowest AIC values, and as such are presented in the main text.

Table C.1 Model deviance diagnostics for host diversity models.

Model	Residual deviance	Residual df	AIC	Deviance/df
Poisson: Area + SLMP	5621.91	55	5873.20	102.22
Quasi-Poisson: Area + SLMP	3497.06	55	–	63.58
Negative binomial: Area + Distance	65.10	55	488.86	1.18
Negative binomial: Area + SLMP	63.90	55	481.30	1.16
Negative binomial: Elevation + Distance	69.24	55	515.87	1.26
Negative binomial: Elevation + SLMP	68.08	55	509.48	1.24

Below, we report model diagnostic plots for the three models in the main text which show significant relationships between island characteristics and network properties.

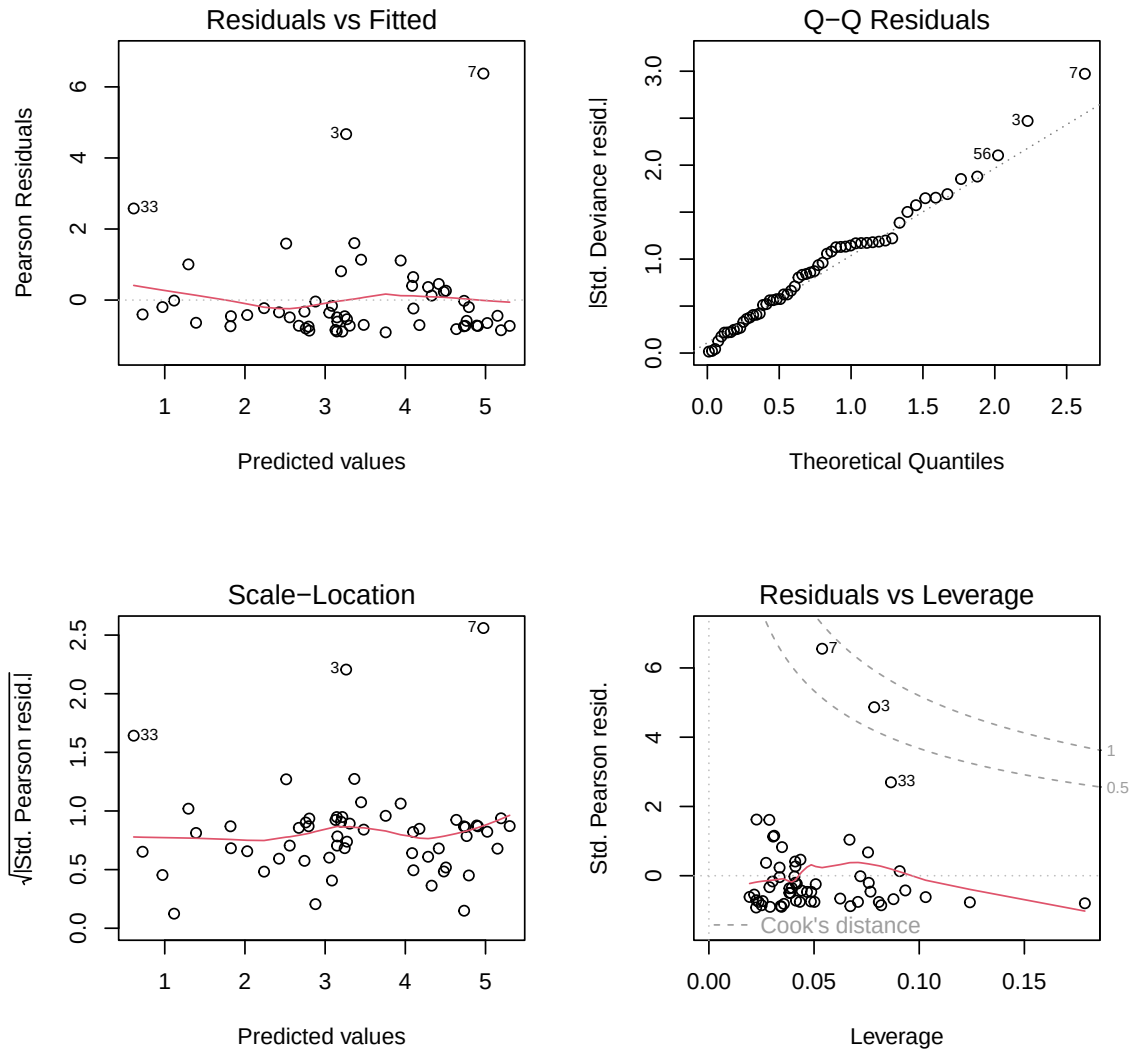


Figure C.4 Model diagnostics plots for our negative binomial model of **parasite richness** across islands, including residual vs fitted values (top left), Q-Q plot (top right), scale-location (bottom left), and residual vs leverage plot (bottom right), with dotted lines representing Cook's distance contours of 0.5 and 1, respectively.

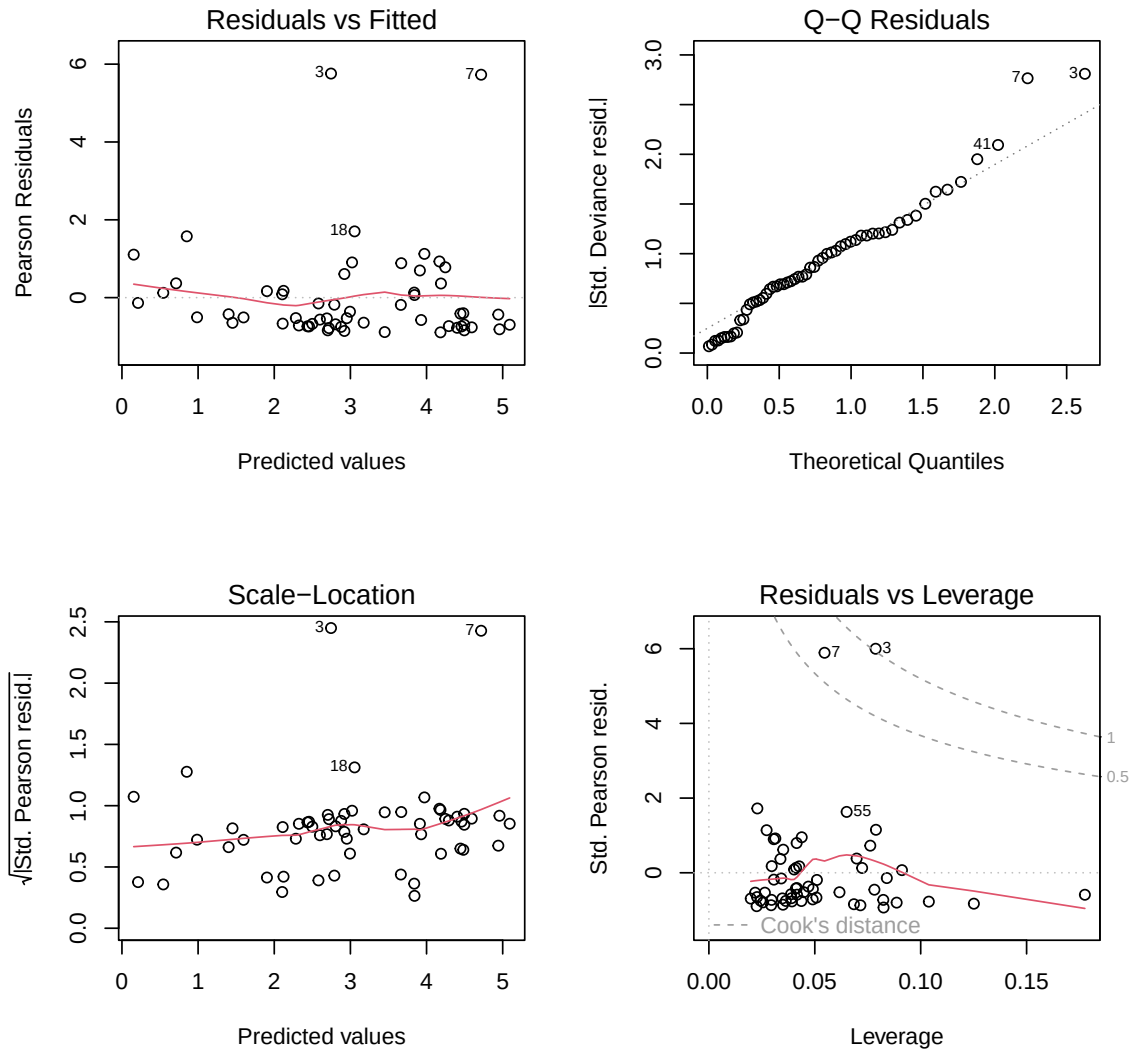


Figure C.5 Model diagnostics plots for our negative binomial model of **host richness** across islands, including residual vs fitted values (top left), Q-Q plot (top right), scale-location (bottom left), and residual vs leverage plot (bottom right), with dotted lines representing Cook's distance contours of 0.5 and 1, respectively.

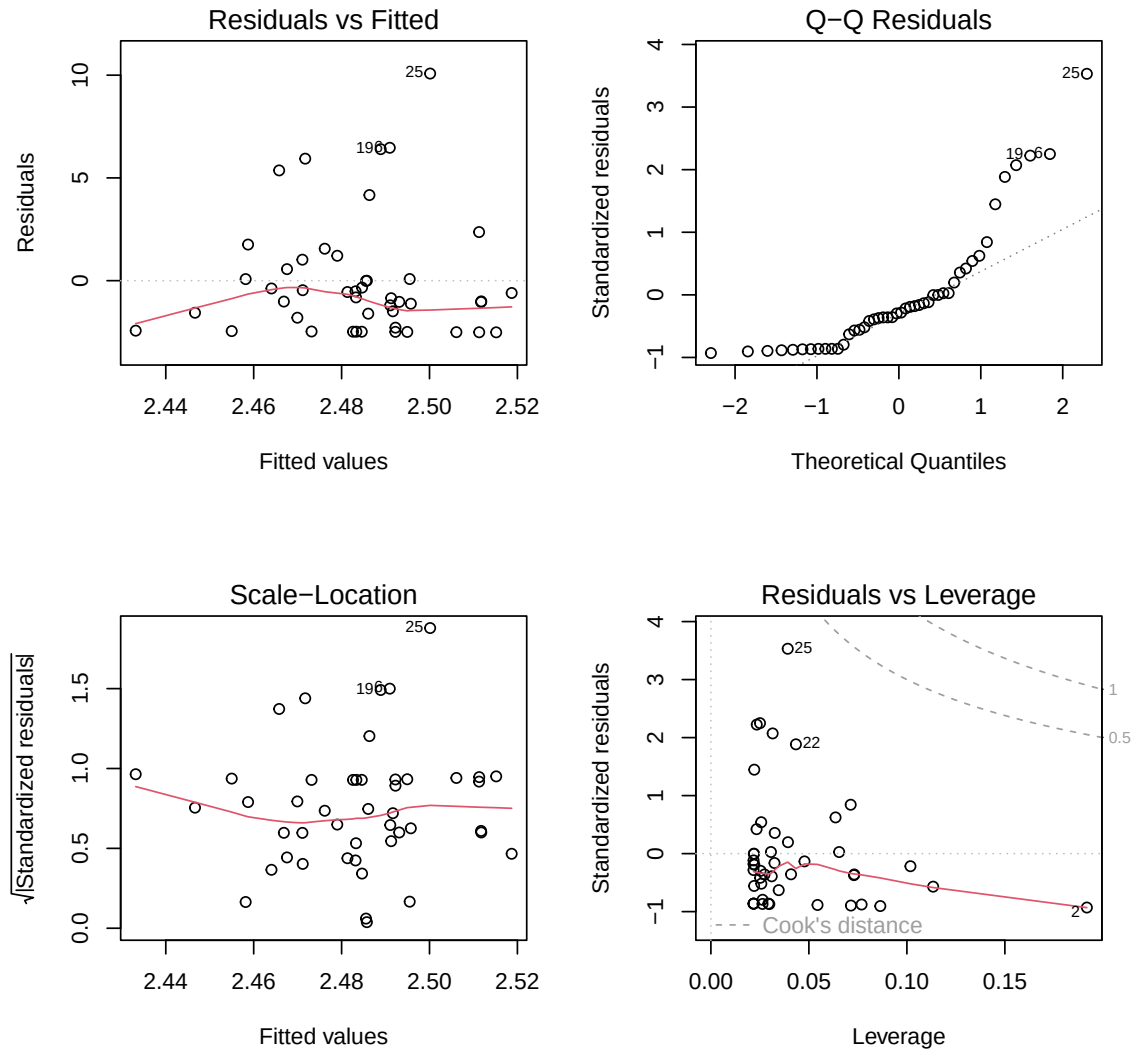


Figure C.6 Model diagnostics plots for our generalized linear model of **network nestedness** ($NODF_c$) across islands, including residual vs fitted values (top left), Q-Q plot (top right), scale-location (bottom left), and residual vs leverage plot (bottom right), with dotted lines representing Cook's distance contours of 0.5 and 1, respectively.

APPENDIX D

SUPPLEMENTAL INFORMATION FOR CHAPTER 6

Quantifying Multistrain Communities

For each community spread plate, the number of colonies of each composite phenotype were counted from photos. Ambient lighting environment slightly differed between treatments; we applied a small RGB scaling factor to low-nutrient environment photos to normalize differentiability between strains across treatments ($R=1.11$, $G=1.09$, $B=1.05$). Total number of colonies per plates was highly variable, a median number of colonies per plate of 161, but the overall range spanning from 5 colonies to 545. Figure D.1. Plates were excluded from analysis if contamination obscured a large amount ($<10\%$) of the plate, or had low plate abundance (<5 colonies); D.1 shows the distribution of removed plates across treatments.

Although experimental communities were inoculated only with colored strains, some colonies did not express visible phenotypes and could instead appear wild-type appearing. Because wild-type-appearing colonies in multispecies communities could originate from any strain present in the community, we used a Bayesian hidden-state model informed by monoculture expression rates and observed phenotype counts to estimate underlying strain relative abundances in each final community.

We model the relative abundances within each community-treatment combination as a singular grand mean, with individual replicates deviating from this mean according to a replicate-level variance parameter. Individual plates are modeled as multinomial draws from replicate-level distributions, modified by the independently modeled strain-specific phenotype expression rates. As such, we are able to model

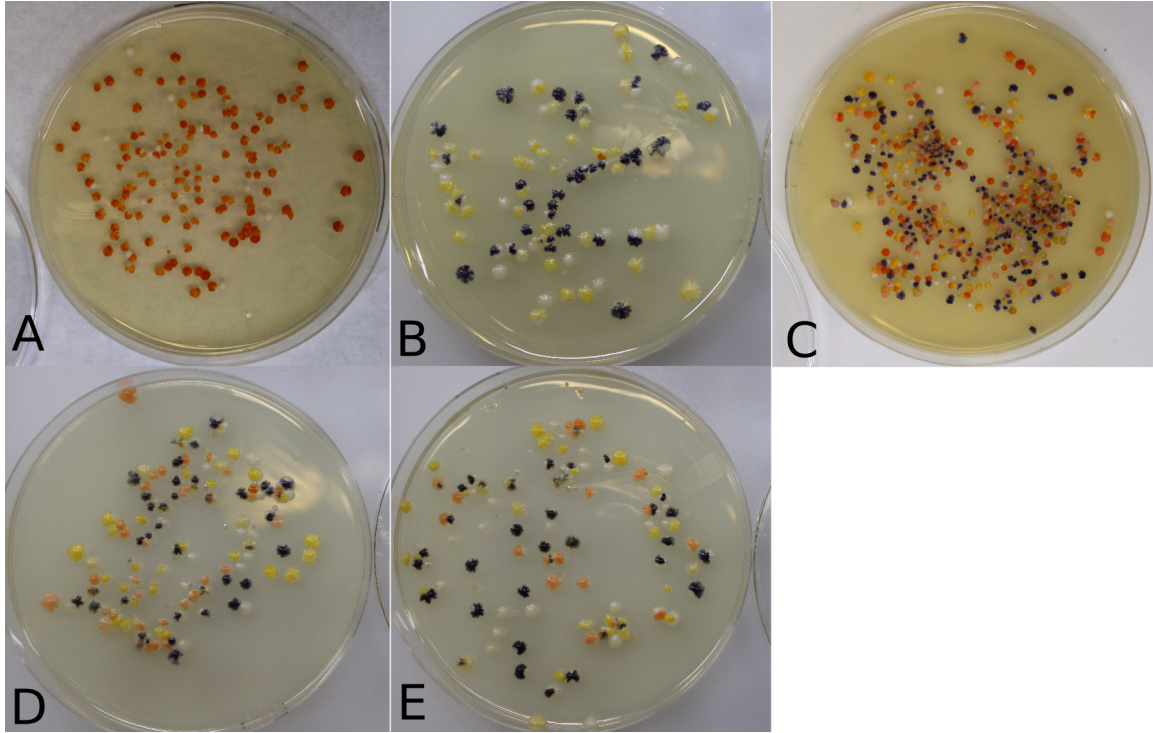


Figure D.1 Example plates of community samples from a variety of replicates. Panels A, B, and C, represent samples of one-, two-, and four-strain communities, respectively. Panels D and E represent two samples of the same replicate three-strain community. Although experimental communities were inoculated only with colored strains, a minority of colonies did not express visible phenotypes and could instead appear wild-type appearing (mean = 15.2% across strain-nutrient combinations), see Figure D.2.

the best-fit relative abundance distribution of each replicate given our observed phenotypic counts, along with our estimates uncertainty around those relative abundances. We present the results of these modeled relative abundances in main text Figure 6.3. Visual comparisons of raw and modeled relative abundances are presented in Figure D.2.

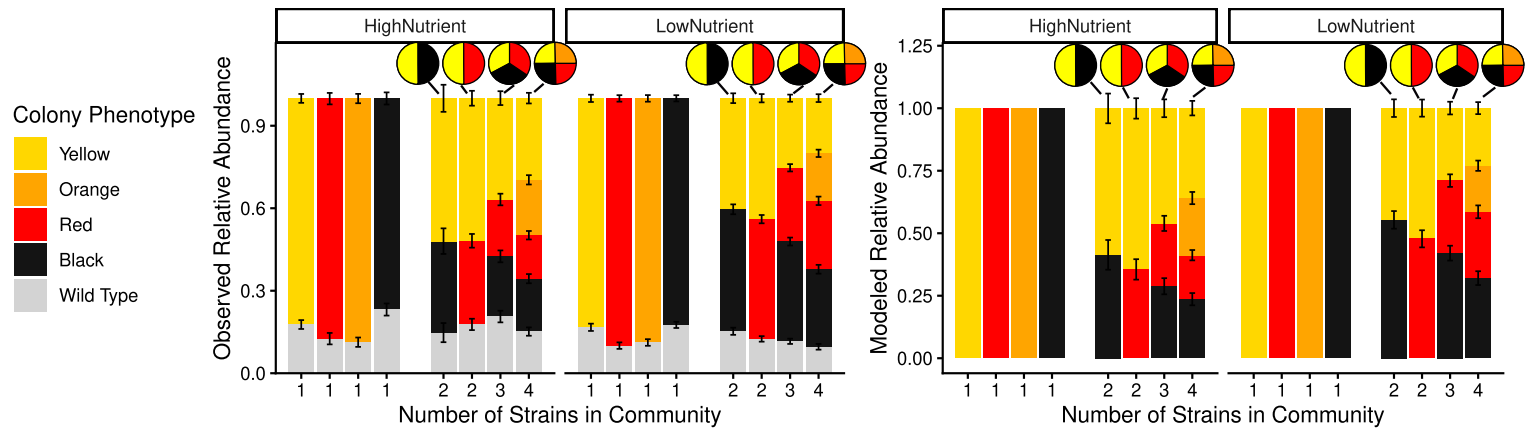


Figure D.2 Relative abundance of strains across community compositions. Although experimental communities were inoculated only with colored strains, some colonies did not express visible phenotypes and could instead appear wild-type appearing. Because wild-type-appearing colonies in multispecies communities could originate from any strain present in the community, we used a Bayesian hidden-state model informed by monoculture expression rates and observed phenotype counts to estimate underlying strain relative abundances in each final community. The left panel shows observed colony phenotypes across community compositions, and the right panel shows inferred strain relative abundances.

Table D.1 Summary of successful replicates across community compositions and nutrient environments. The experiment included eight community types grown in two nutrient environments, with eight replicates per community-environment combination. Each replicate was subsampled and spread-plated twice, yielding 256 total community samples. Samples were excluded from analysis if contamination obscured a large amount ($<10\%$) of the plate, or had low plate abundance (<5 colonies). For each community-environment combination, the table reports the number of biological replicates with two, one, or zero usable samples remaining after filtering. Across 128 community-environment replicates, 11 were fully excluded and 24 retained only one usable sample. These were generally spread relatively evenly community-environment combinations, with most excluding either one or zero replicates. The two exceptions are the high-nutrient yellow-black communities (3 replicates excluded) and low-nutrient yellow-red-orange-black communities (2 replicates excluded).

Community	High Nutrient			Low nutrient		
	2 samples	1 sample	0 samples	2 samples	1 sample	0 samples
Yellow	5	2	1	6	2	0
Red	4	3	1	6	1	1
Orange	5	3	0	6	2	0
Black	7	1	0	7	1	0
Yellow-Black	4	1	3	8	0	0
Yellow-Red	5	2	1	6	1	1
Yellow-Red-Black	6	2	0	7	0	1
All strains	7	1	0	4	2	2

Monoculture Growth Curves

We estimated strain-specific vital by growing each strain in liquid monocultures. Figure D.3 shows the example demographic curves fit to equation 6.1. The finite rate of increase (R) and carrying capacity (K) were estimated separately for each strain-by-nutrient combination; each timestep of our discrete model represented a 10 minute interval between OD600 measurements. Replicate trajectories were allowed to vary in R around their corresponding strain-by-nutrient mean, with a single replicate-level variance parameter shared across all groups. Observation error was modeled with a shared lognormal error term, and weakly informative priors were placed on log-transformed vital-rate parameters and variances. Models were fit using four MCMC chains with 1000 timesteps following a 1000-timestep burn-in period. Models were checked for convergence by analyzing ESS and $\hat{R}$ for each parameter estimate, as well as through posterior predictive visualizations. Though with high variability, we observed a potential negative trade-off between R and K across strains in both high- and low-nutrient treatments (Figure D.4).

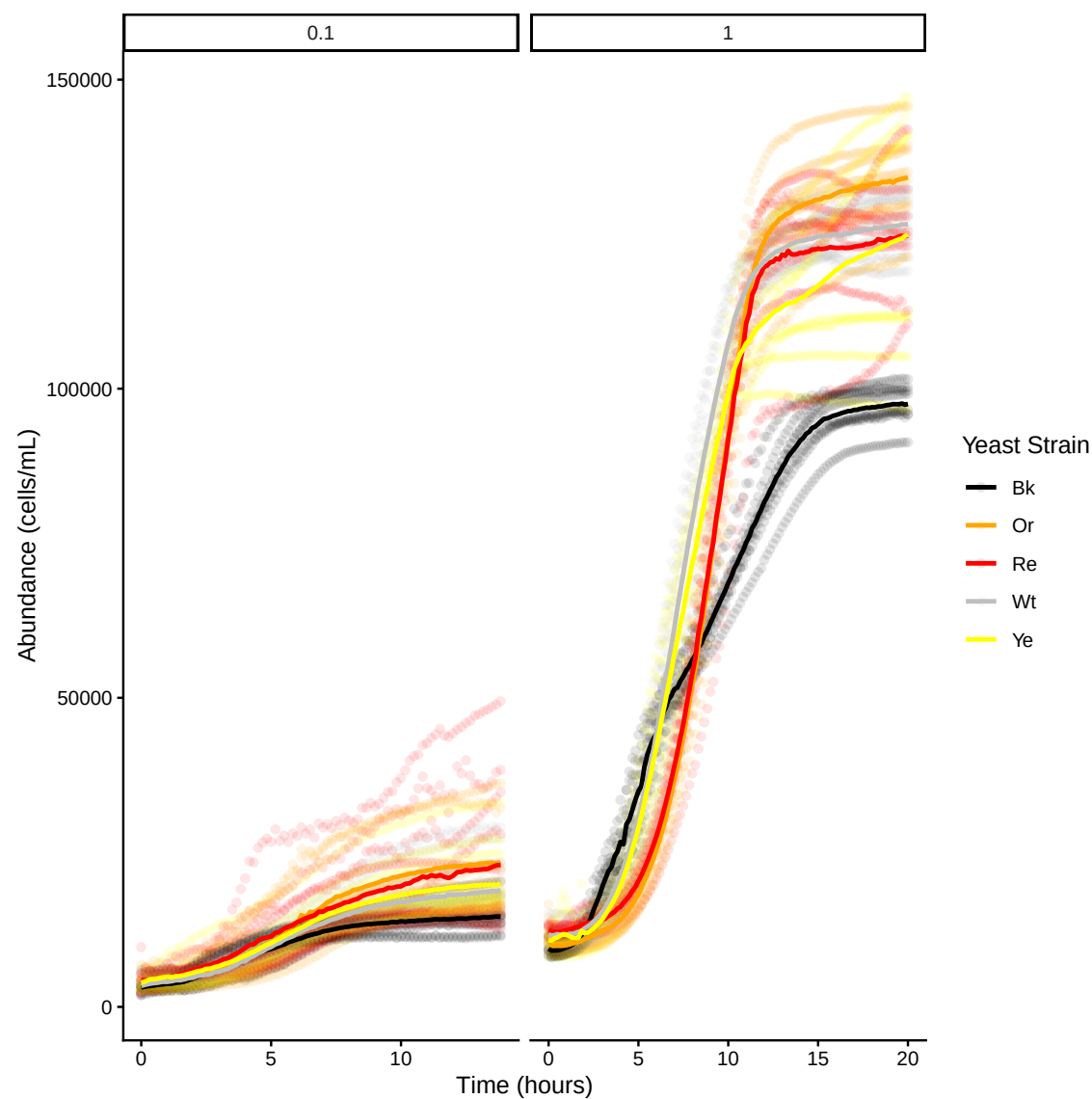


Figure D.3 Monoculture growth curves used to inform vital rate estimates. Density is measured through transforming optical density measurements to cellular density via standard curves. Left and right panels represent low and high-nutrient conditions. Points represent individual culture trajectories, while lines represent mean values across all replicates.

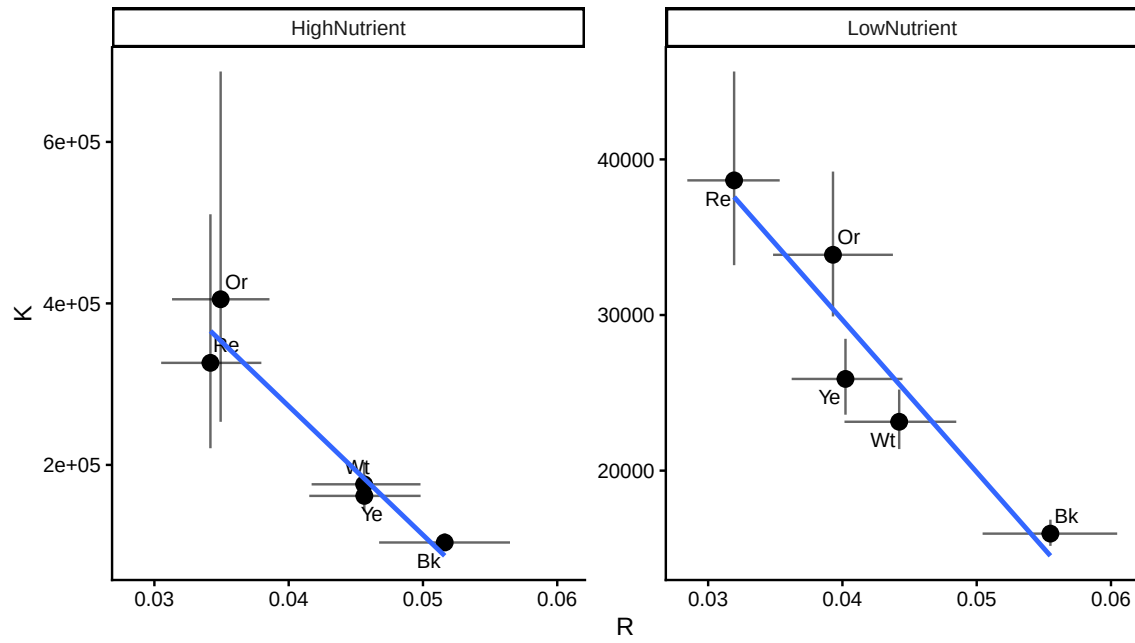


Figure D.4 Comparison of best fit carrying capacity (K) and finite rate of increase (R) values across nutrient environment. Error-bars represent 95% credible intervals. We observe a significant negative relationship between the two parameters across both nutrient environments.

APPENDIX E

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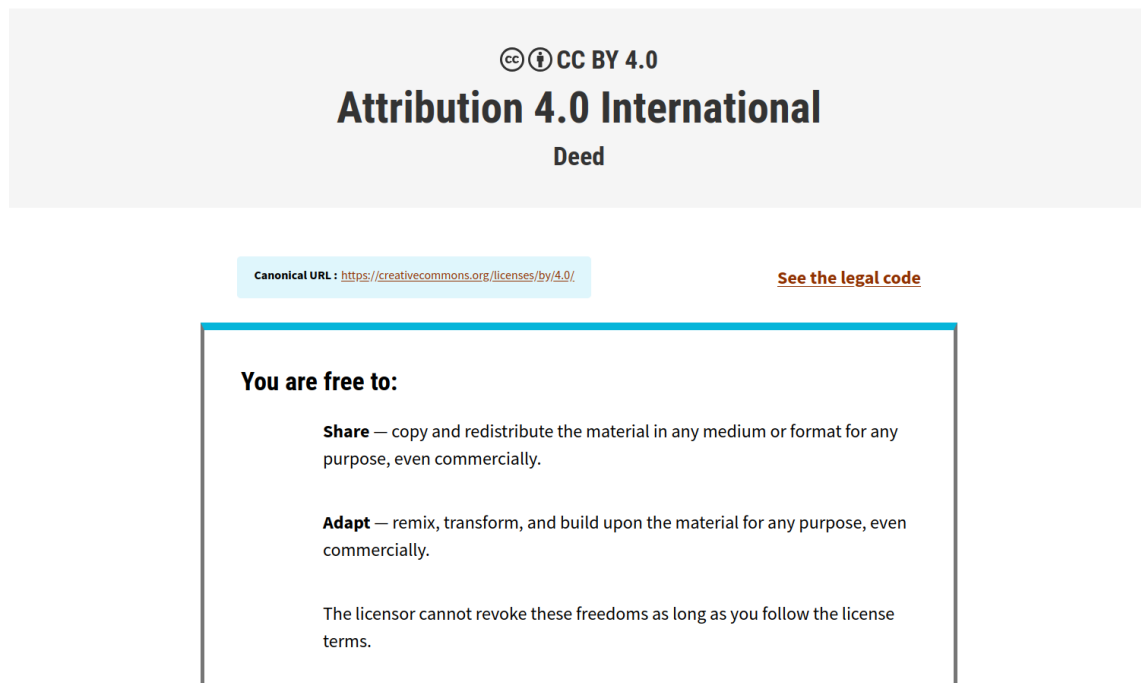


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